

EDO Board Prep Study Guide



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Contents

LIST OF FIGURES	VII
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LIST OF TABLES	IX
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1 ORGANIZATION	1
1.1 Command Chains and Governance	1
1.1.1 Operations Chain (Title 10)	2
1.1.2 Acquisition Governance (Department of the Navy)	4
1.1.3 Approving Authorities	6
1.2 Technical Authority and Engineering Agents	6
1.2.1 What Technical Authority Is	7
1.2.2 What EAs Do (and Do Not Do)	8
1.2.3 Where Engineering Agent Authority Comes From	8
1.2.4 Waterfront Triage: What Engages Whom	8
1.3 NAVSEA Organization and Warfare Centers	9
1.3.1 HQ Codes (must-know one-liners)	10
1.3.2 Warfare Centers (alphabetical)	12
1.4 NAVWAR Enterprise and Warfare Centers	16
1.4.1 HQ Directorates and FRD	17
1.4.2 Warfare Centers (alphabetical)	18
1.4.3 Project Overmatch	20
1.5 NAVAIR Warfare Centers	21
1.6 Navy Program Executive Offices (PEOs)	22
1.6.1 PEO/Warfare Center Interaction Model	26
2 PPBE	27
2.1 PPBE (Planning, Programming, Budgeting, and Execution)	27
2.1.1 What PPBE does	28
2.1.2 Timeline (annual rhythm)	28
2.1.3 Congressional Enactment (Regular Order)	29
2.1.4 Key Terms and Definitions	30
2.1.5 Programming v.s. Budgeting	31
2.1.6 PPBE Programming	32
2.1.7 Appropriations Life-cycle (Colors of Money)	32
2.1.8 IGT (Intragovernmental Transactions)	33
2.1.9 Common Ordering Instruments	33

2.1.10	<i>Fiscal Law (Know These Cold)</i>	33
2.1.11	<i>Reprogramming: Moving Resources After Enactment</i>	34
2.1.12	<i>From Enactment to Field Authority</i>	34
2.2	NWCF Essentials: Rates, Results, and Execution	34
2.2.1	<i>Key Definitions (Execution)</i>	36
2.2.2	<i>Flow: PMO Purchase Request to Treasury Payment</i>	36
2.2.3	<i>NWCF Metrics and Equations</i>	36
2.2.4	<i>NWCF Corpus v.s. Appropriations</i>	36
2.2.5	<i>Stabilized Rates and SLR</i>	37
2.2.6	<i>NWCF v.s. Mission-Funded (Appropriation) Commands</i>	37
2.2.7	<i>Standing Up or Modifying a NWCF Business Area</i>	37
2.2.8	<i>NWCF Cycle of Operations (Order to Cash)</i>	38
2.2.9	<i>Operating During a Continuing Resolution</i>	38
2.2.10	<i>Why Use a Warfare Center (Organic)</i>	38
2.3	Congressional Enactment	39
2.3.1	<i>Foundations and core definitions</i>	40
2.3.2	<i>Who drafts what</i>	40
2.3.3	<i>Regular-order timeline</i>	40
2.3.4	<i>The 12 regular appropriations bills</i>	41
2.3.5	<i>Continuing Resolution</i>	42
2.3.6	<i>Sequestration</i>	42
2.4	Program Funding and Execution	42
2.4.1	<i>BA, commitment, obligation, expenditure, outlay</i>	43
2.4.2	<i>Appropriation categories, scope, obligation windows</i>	44
2.4.3	<i>Funding policies</i>	44
2.4.4	<i>Navy Sustainment Funding: TYCOM v.s. Fleet (O&S)</i>	45
2.4.5	<i>Which money when (by acquisition phase)</i>	46
2.4.6	<i>MILCON v.s. O&M sustainment v.s. OPN modernization</i>	46
2.4.7	<i>Field Activity Financial Management (NWCF quick hits)</i>	46
2.5	Cost Elements, Estimates, and Learning Curves	47
2.5.1	<i>Direct v.s. Indirect Costs</i>	48
2.5.2	<i>Program Cost Terms (Know the Stack)</i>	48
2.5.3	<i>Why the LCCE Matters</i>	49
2.5.4	<i>Estimating Methods (Types, When, and Why)</i>	50
2.5.5	<i>Products and Players</i>	50
2.5.6	<i>Learning Curve (Production Efficiency)</i>	51
2.5.7	<i>Cost Escalation (Inflation Indices)</i>	51
2.5.8	<i>Will-Cost / Should-Cost</i>	51
3	CONTRACTING AND SOLICITATION	52
3.1	Intro to Contracting Fundamentals	52
3.1.1	<i>Why the Navy Uses Contracts</i>	53
3.1.2	<i>Essential Elements of a Binding Contract</i>	53
3.1.3	<i>Government Relationship to Contractors</i>	53
3.1.4	<i>FAR System and NAVSEA Overlays</i>	53

3.1.5	<i>Types of Contracting Officers (Know All Three)</i>	54
3.1.6	<i>PM and KO Partnership</i>	54
3.1.7	<i>Competition in Contracting Act Requirements</i>	54
3.1.8	<i>Responsiveness, Responsibility, and Key Determinations</i>	55
3.1.9	<i>J&A v.s. D&F</i>	55
3.1.10	<i>Who Signs D&Fs (and When)</i>	55
3.2	Solicitation Preparation	56
3.2.1	<i>Summary</i>	57
3.2.2	<i>Practitioner Steps</i>	57
3.2.3	<i>Checklist</i>	57
3.2.4	<i>Navy Overlays and Tailoring</i>	58
3.2.5	<i>Industry Engagement & Communications</i>	58
3.2.6	<i>Documentation & Approvals to Capture</i>	58
3.2.7	<i>Competitive Method Selection: Sealed Bidding v.s. Negotiated Procurement</i>	59
3.2.8	<i>Contract Type Landscape and Risk Allocation</i>	59
3.2.9	<i>Fixed-Price Incentive (Firm Target) Mechanics</i>	62
3.2.10	<i>Indefinite & Flexible Contract Forms</i>	62
3.2.11	<i>Debriefings and Protests (Forums and Timelines)</i>	63
3.2.12	<i>Why Cost-Plus After Low-Rate Initial Production? (PD and O&S Exceptions)</i>	63
3.2.13	<i>Simplified Acquisition and Small Business Programs</i>	63
3.2.14	<i>Alternative Acquisition Authorities</i>	64
3.2.15	<i>Technical Data Rights Overview</i>	64
3.2.16	<i>Uniform Contract Format Essentials</i>	64
3.3	Cost and Price Evaluation	64
3.3.1	<i>Summary</i>	65
3.3.2	<i>Practitioner Steps</i>	66
3.3.3	<i>Proposed Price, Cost, Profit, and Fee</i>	66
3.3.4	<i>Direct and Indirect Cost Structure</i>	67
3.3.5	<i>Other Direct Costs and Cost Allowability</i>	67
3.3.6	<i>Cost and Price Analysis</i>	67
3.3.7	<i>Forward Pricing Rates and Fully Burdened Labor</i>	67
3.3.8	<i>Indirect Rate Math and Profitability Checks</i>	68
3.3.9	<i>Oversight and Field Pricing Support</i>	68
3.3.10	<i>Checklist</i>	68
3.4	Earned Value Management (EVM)	69
3.4.1	<i>What EVM is</i>	70
3.4.2	<i>Key Components of EVM</i>	70
3.4.3	<i>When to Use EVM</i>	70
3.4.4	<i>EVM Compliance</i>	71
3.4.5	<i>EVM Principal Players</i>	71
3.4.6	<i>PMB</i>	72
3.4.7	<i>EVM Reviews and Reports</i>	73
3.4.8	<i>EVM Data Analysis</i>	75

4	DEFENSE ACQUISITIONS	77
4.1	Acquisition Policy	77
4.1.1	Summary	79
4.1.2	Practitioner Steps (Board Prep Focus)	80
4.1.3	Policy Stack and Authorities	81
4.1.4	Acquisition Players and Decision Forums	81
4.1.5	Adaptive Acquisition Framework Pathways	82
4.1.6	Warfighting Acquisition System Imperatives (2025)	85
4.1.7	Urgent Operational Needs Governance	85
4.1.8	Security Cooperation and Capital Alignment (2025 Reforms)	86
4.1.9	Acquisition Categories and Delegated Decision Authority	87
4.1.10	Project Overmatch Status	87
4.1.11	Navy Overlays and Touchpoints	88
4.2	Milestones and Phases	88
4.2.1	Defense Acquisition System Overview	91
4.2.2	Adaptive Acquisition Framework Pathways (Board Snapshot)	91
4.2.3	Baseline Discipline: Acquisition Program Baselines, Deviations, and Nunn-McCurdy	91
4.2.4	JCIDS Flow into the Defense Acquisition System	92
4.2.5	KPPs v.s. KSAs (Know the Difference)	92
4.2.6	Acquisition Strategy Essentials	92
4.2.7	International Acquisition Objectives	92
4.2.8	Navy Two-Pass / Seven-Gate Review	93
4.2.9	Technology Readiness Levels	93
4.2.10	T&E Organizations and Roles	94
4.2.11	Major Capability Acquisition Lifecycle Overview	94
4.2.12	Materiel Development Decision	97
4.2.13	Materiel Solution Analysis Phase Essentials	97
4.2.14	Milestone A Focus Areas	97
4.2.15	Technology Maturation and Risk Reduction Phase Essentials	97
4.2.16	Milestone B Focus Areas	98
4.2.17	Engineering and Manufacturing Development Phase Essentials	98
4.2.18	Milestone C Focus Areas	98
4.2.19	Production and Deployment Phase Essentials	98
4.2.20	Operations and Support Phase Essentials	99
5	REQUIREMENTS	100
5.1	Joint Capabilities Integration & Development System	100
5.1.1	Requirements Demand Versus Acquisition Supply	101
5.1.2	Capability Requirements and the Capability-Based Assessment Cadence	102
5.1.3	Strategy-to-Requirement Synchronization	103
5.1.4	Major Capability Acquisition Requirement Set	103
5.1.5	How KPPs Stay Integrated with DAS	104
5.1.6	Prioritizing Materiel and Non-Materiel Solutions	104
5.1.7	Governance Roles and Validation Chain	105
5.1.8	IT, C4I, and Coalition Interoperability Implications	105

5.1.9	Warfighting Acquisition System (JCIDS Replacement)	106
5.1.10	What EDOs Must Master Under WAS	107
5.1.11	Current Events: SECWAR Acquisition Transformation Strategy	107
5.2	Test & Evaluation	108
5.2.1	Board Prep Summary	109
5.2.2	Definition and Relationship to Verification and Validation	109
5.2.3	Statutory Requirements and Key Roles	110
5.2.4	Developmental Test Activities	111
5.2.5	Live Fire Test and Evaluation	112
5.2.6	Operational Test Phases	112
5.2.7	Fundamental Test and Evaluation Terms	112
5.2.8	Critical Technical Parameters and Technical Performance Measures	113
5.2.9	Test and Evaluation Master Plan	113
5.2.10	Modeling and Simulation	113
5.2.11	Navy Overlays and Integration	114
5.2.12	Fiscal and PPBE Touchpoints	114
5.2.13	Assumptions and Uncertainties	114
6	MAINTENANCE	115
6.1	Fleet Maintenance Assets	115
6.1.1	Echeloned Maintenance Levels	117
6.1.2	Shore Maintenance Organizations and Roles	118
6.1.3	Regional Maintenance Command Structure	120
6.1.4	Command, Funding, and Technical Authority Chains	121
6.1.5	Workforce Readiness and Assessments	121
6.2	Navy Modernization Program	121
6.2.1	Modernization vs. Maintenance	124
6.2.2	Governance, Roles, and Documentation	124
6.2.3	Process Flow and Funding Integration	125
6.2.4	Execution Enablers and Lessons Learned	125
6.3	Fleet Maintenance Policy and Implementation	126
6.3.1	Policy Baseline	129
6.3.2	Strategy Frameworks	130
6.3.3	Availability Approval and Execution Discipline	130
6.3.4	Trends, Risks, and Talking Points	130
6.4	AIT Fundamentals and Lessons Learned	131
6.4.1	Purpose and Advantages	133
6.4.2	Roles and Responsibilities	134
6.4.3	Planning Controls and MOA Discipline	134
6.4.4	Lessons Learned for Board Prep	135
6.5	Waterfront Organizations, Work Controls, and Safety	135
6.5.1	Naval Propulsion Program Oversight	138
6.5.2	SUBSAFE and Submergence Systems Controls	138
6.5.3	Work Authorization and Tag-out Discipline	138
6.5.4	Waterfront Organization Insights	139

6.6	Battle Damage Assessment and Repair	139
6.6.1	Strategic Context	142
6.6.2	BDAR Continuum	142
6.6.3	Command, Control, and Technical Authority	142
6.6.4	Logistics, Contracting, and Fiscal Readiness	143
6.6.5	Maritime Support Plan Force Packages	143
6.7	CNO Availability Planning	143
6.7.1	Balancing Requirements, Resources, and Readiness	146
6.7.2	Availability Work Package Development	146
6.7.3	Planning Teams and Integration Forums	147
6.7.4	Risk Controls and Board-Ready Talking Points	147
6.8	CNO Availability Execution	148
6.8.1	Project Teams and Responsibilities	150
6.8.2	Execution Phases	151
6.8.3	Governance, Quality, and Work Control	152
6.8.4	Lessons Learned	152
7	CIVILIAN PERSONNEL	154
7.1	Civilian Personnel Fundamentals	154
7.1.1	Why CIVPERS management matters	154
7.1.2	Workforce shaping and classification	154
7.1.3	Equal Employment Opportunity anchors	155
7.1.4	Employee relations essentials	155
7.1.5	Labor relations touchpoints	155
	Appendices	157
A	PLATINUM CARD	157
B	KEY ROLES IN ACQUISITION AND MODERNIZATION	162
B.1	EDO Flag Officers (snapshot)	162
B.2	Key Roles in Acquisition and Modernization	163
B.3	Combatant Commands (Reference)	170
B.4	Current Events Relevant to EDOs and Acquisition	170
B.4.1	Project Overmatch and Digital Modernization	173
B.4.2	EDO Community Updates	173
B.4.3	Acquisition Policy and Industrial Base Highlights	174
	REFERENCES	175
	LIST OF ACRONYMS	184

List of Figures

1.1	Title 10 Chain of Command	3
1.2	Organization Command Structure	4
1.3	Acquisition Governance Chain of Command	5
1.4	Technical Authority v.s. Program Authority	6
1.5	NAVSEA Organization Chart	13
1.6	Geographic locations of Navy Warfare Center	15
1.7	NAVWAR enterprise structure	17
1.8	NIWC LANT Locations	19
1.9	NIWC PAC Locations	20
2.1	PPBE Timeline for multiple fiscal years	28
2.2	Platinum Card: PPBE Planning Phase	29
2.3	Congressional Enactment Timeline	30
2.4	Platinum Card: Concurrent Program/Budget Review	32
2.5	Colors of Money Timeline	33
2.6	Congressional Enactment Timeline	41
2.7	Life Cycle Costs	49
2.8	Lif-Cycle Cost	49
3.1	Fixed Price Contracts	60
3.2	Cost Reimbursement Contracts	61
3.3	Contract-type emphasis across the Major Capability Acquisition phases.	62
3.4	Direct v.s. Indirect Costs	66
3.5	PMB “S”-Curve	72
3.6	PMB Development Flow	72
3.7	EVM Variances	75
4.1	Cost, Schedule, Performance	80
4.2	Trade Space, Performance v.s. Cost	80
4.3	Adaptive Acquisition Pathways	82
4.4	Major Capability Acquisition	83
4.5	MCA Lifecycle View	96
5.1	JCIDS to DAS Relationship	102

5.2	Concept to Requirement Flow	103
5.3	JCIDS Artifacts within MCA	104
5.4	MCA T&E Touchpoints	110
5.5	DoW T&E organization	111
6.1	Maintenance echelon coverage	118
6.2	AIT swim lane	134
6.3	BDAR continuum	142
6.4	CNO availability planning timeline	147
6.5	Nuclear waterfront team	151
A.1	Platinum Card: Life Cycle Cost Composition	157
A.2	Platinum Card: Cost Estimating Requirements	158
A.3	Platinum Card: PPBE Planning Phase	158
A.4	Platinum Card: Concurrent Program/Budget Review	159
A.5	Platinum Card: Below Threshold Reprogramming	159
A.6	Platinum Card: Resource Allocation Process	160
A.7	Platinum Card: Congressional Enactment Timeline	160
A.8	Platinum Card: Funding for Product Improvements	161
A.9	Platinum Card: Appropriation Life Cycle	161

List of Tables

NAVSEA Warfare Centers	12
NAVWAR HQ Directorates and FRDs	17
NAVWAR Warfare Centers	18
NAVAIR Warfare Centers	22
Key PPBE Terms	30
Programming v.s. Budgeting	31
Colors of Money	44
Exceptions to Full Funding	44
Appropriations per Phase	46
Comparison of Estimating Methods	50
TRL reference	93
Modernization document ownership and funding tie points	125
Maintenance policy map at a glance	129
CNO availability execution phases	152

1 Organization

1.1 Command Chains and Governance

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1.1.1 Operations Chain (Title 10)

Chain flow. President → Secretary of War (SECWAR) → Combatant Commander (CCDR). The Chairman of the Joint Chiefs of Staff (CJCS) is the principal military adviser; service chiefs are *not* in the operational chain of Title 10.

Board cue. Boards love the contrast with acquisition authority; avoid conflating operational and acquisition chains.

Echelons. Counted from the Chief of Naval Operations (CNO) (e.g., CNO is Echelon I, Systems Commands (SYSCOMs) are Echelon II, etc.).

Figure 1.1 shows the Title 10 chain of command. Figure 1.2 shows the Chain of Command from the President to Echelon III commands.

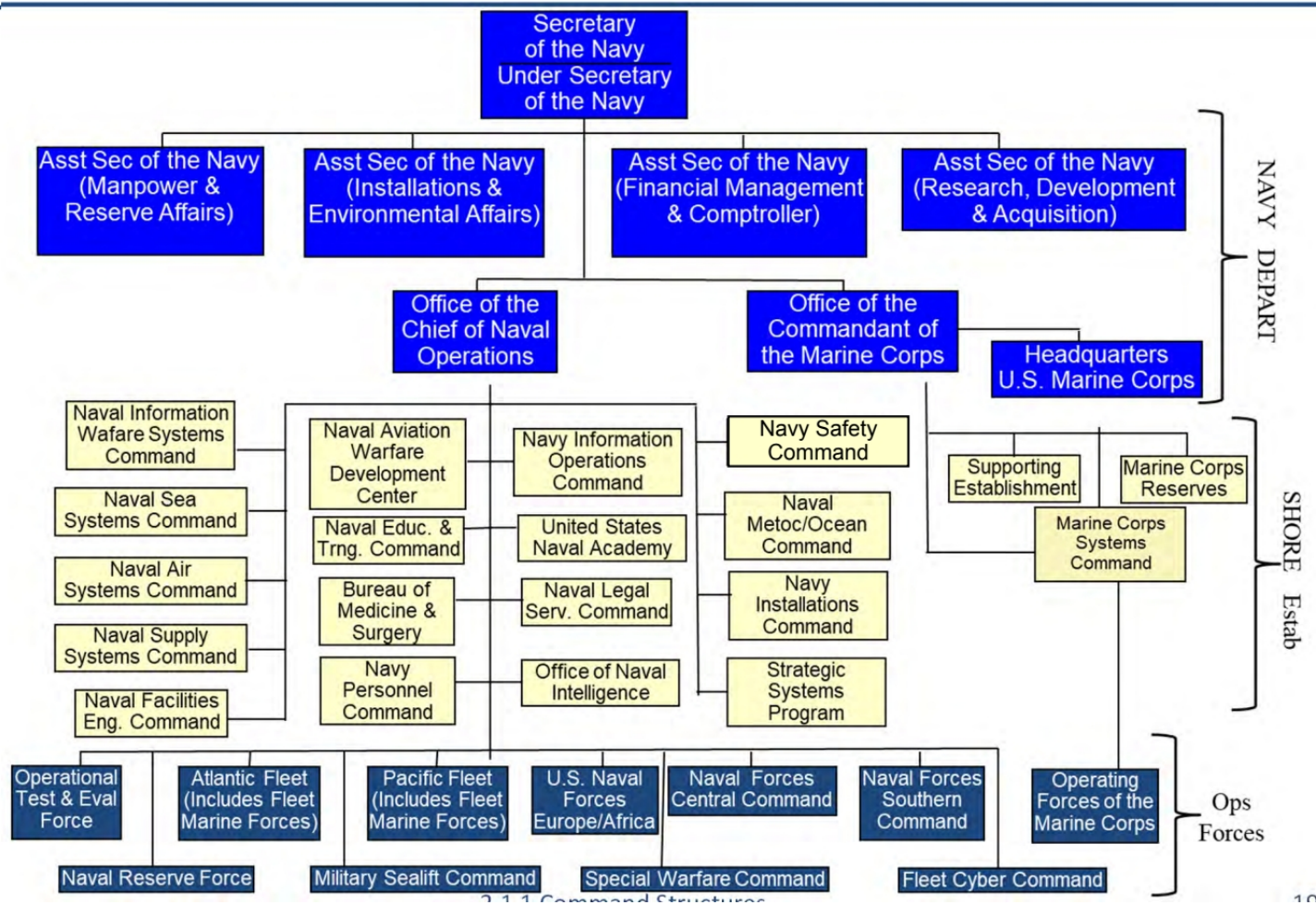


Figure 1.1. Title 10 Chain of Command. Source: 2.1.1. Command Structures, 2025 [1].

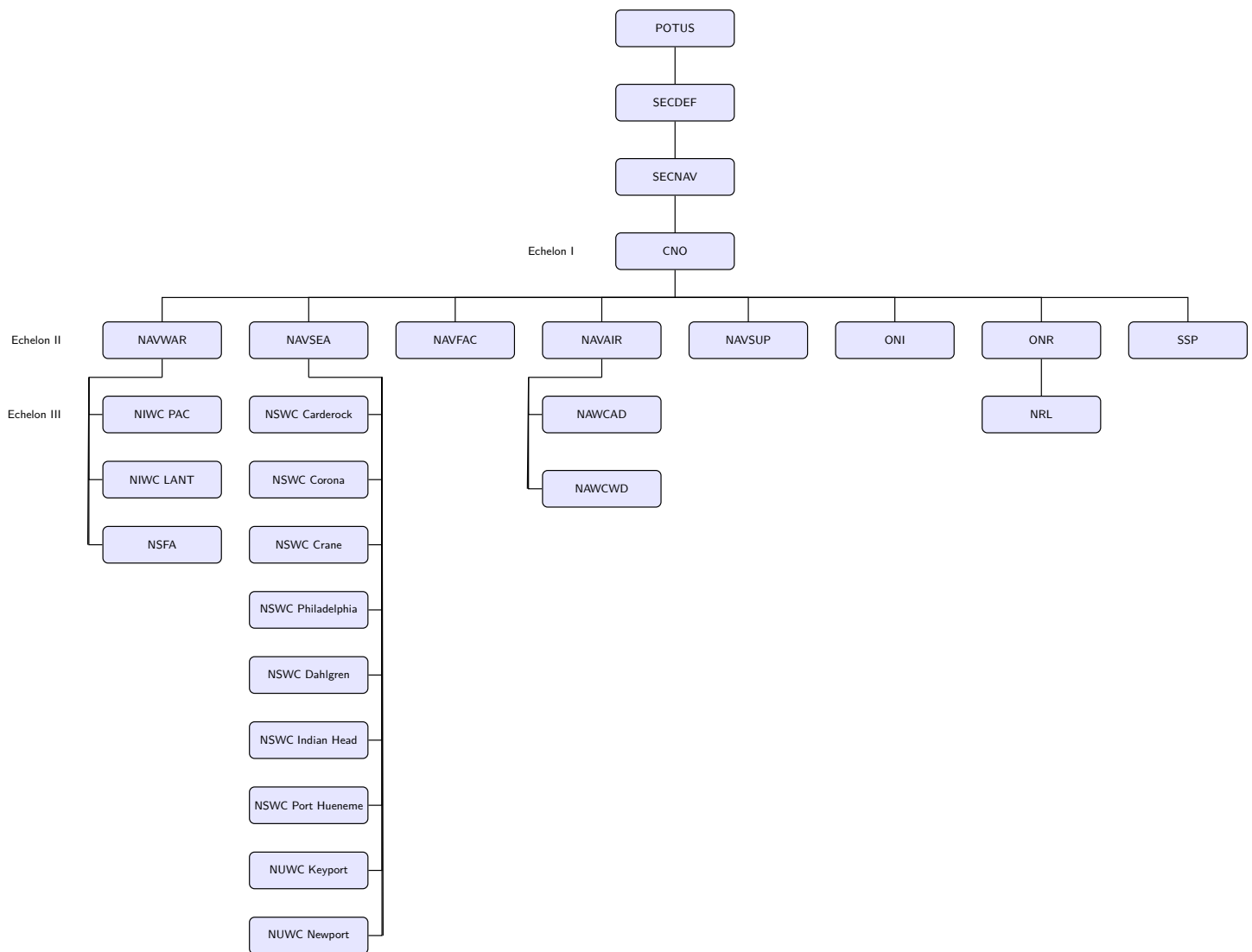


Figure 1.2. Organization Command Structure

1.1.2 Acquisition Governance (Department of the Navy)

Policy authority. Assistant Secretary of the Navy for Research, Development and Acquisition (ASN(RD&A)) is the Service Acquisition Executive (SAE) for the Department of the Navy (DON) and sets policy for Program Executive Offices (PEOs) and Direct Reporting Program Managers (DRPMs) [2, 3].

Matrix execution. PEOs hold program authority; SYSCOMs (Naval Sea Systems Command (NAVSEA), Naval Information Warfare Systems Command (NAVWAR), Naval Air Systems Command (NAVAIR)) provide the Technical Authority (TA) warrants and workforce; Naval Warfare Centers (NWCs) deliver research, development, test, and engineering support [2].

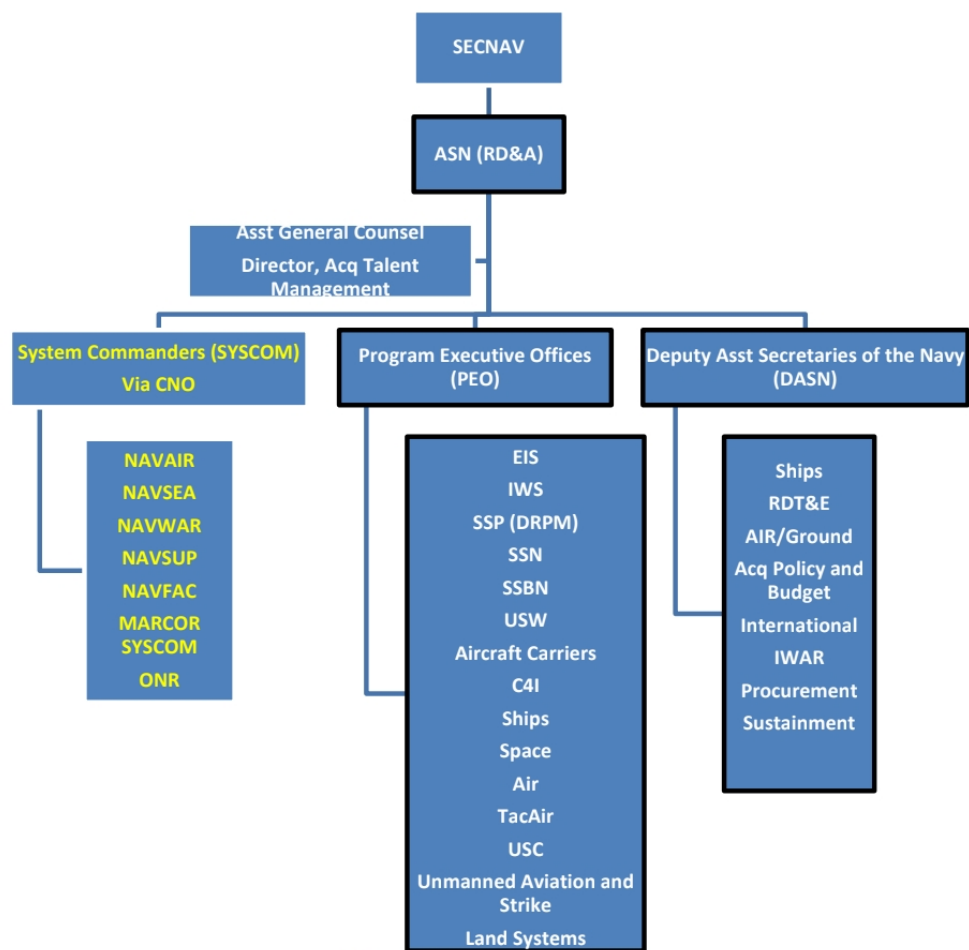
Flag/PEO billets (current). EDO-led today: ASN(RD&A)'s Principal Military Deputy (PMILDEP) (Vice Admiral (VADM) Seiko Okano), Commander, Naval Information Warfare Systems Command (COMNAVWAR) (Rear Admiral (Upper Half) (RADM) J. Kurt Rothenhaus), Program Executive Office, Ships (PEO Ships) (RADM Brian Metcalf), Program Executive

Office, Aircraft Carriers (PEO CVN) (RADM Casey Moton), and Program Executive Office, Attack Submarines (PEO SSN) (RADM Jonathan Rucker) anchor TA and acquisition governance for the community [4, 5, 6, 7, 8].

Non-Engineering Duty Officer (EDO) leaders (awareness): NAVSEA 08 is currently ADM William J. Houston (Submarine officer), Program Executive Office, Integrated Warfare Systems (PEO IWS) is RDML Thomas J. Dickinson (Surface Warfare Acquisition Professional), Program Executive Office, Manpower, Logistics and Business IT (PEO MLB) is RDML Kevin P. Biehn (Surface Warfare Acquisition Professional), NAVSEA 05 is RDML Jay A. Seif (Submarine Acquisition Professional), and Program Executive Office, Unmanned and Small Combatants (PEO USC) is filled by an SES (formerly RDML Kevin R. Smith) [9].

(Appendix B.1: See Appendix B.1 for the current flag billet roster and non-EDO billets.)

Figure 1.3 shows the Navy acquisition governance chain of command.



2.1.1 Command Structures

Figure 1.3. Acquisition Governance Chain of Command. Source: 2.1.1. Command Structures, 2025 [1].

1.1.3 Approving Authorities

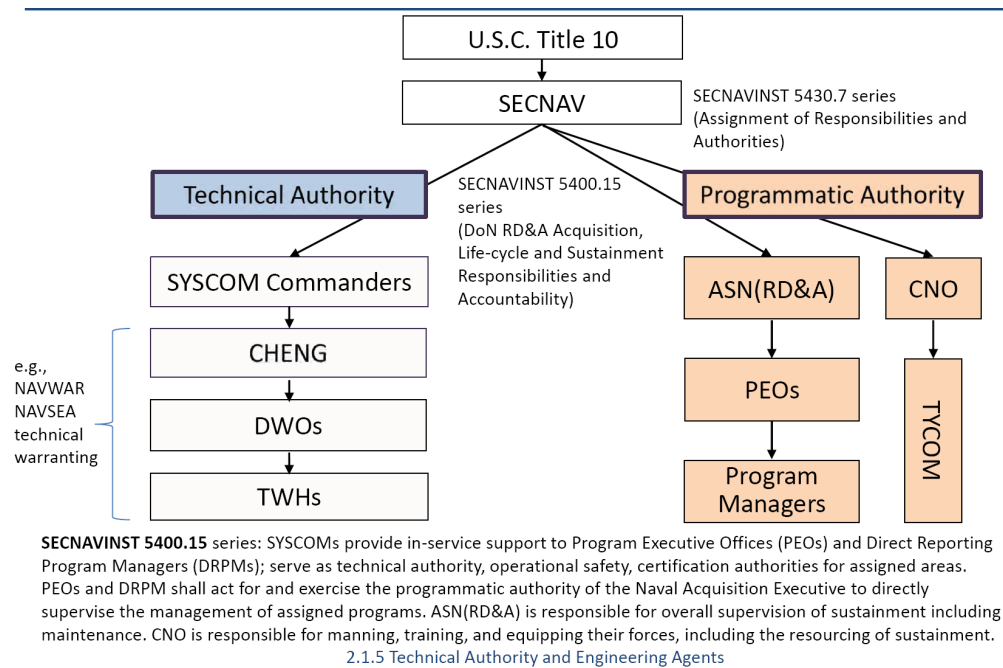


Figure 1.4. Technical Authority v.s. Program Authority alignment. Source: 2.1.5. Technical Authority and Engineering Agents, 2025 [10]

Figure 1.4 contrasts the roles of TA and Program Authority (PA) in Navy acquisition.

1.2 Technical Authority and Engineering Agents

- [1] “2.1.1. Command Structures”Engineering Duty Officer School, “2.1.1. Command Structures,” Engineering Duty Officer (EDO) School, Coursebook Module 2.1.1, version 5.2, Mar. 11, 2025, CIN A-4N-0021/-0034; coursebook PDF provided by author.
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1.2.1 What Technical Authority Is

TA. Independent engineering authority to set, maintain, and certify conformance to technical standards and baselines across the lifecycle [2].

Ultimate Technical Authority (UTA). Executed at the SYSCOM level (e.g., NAVSEA); resides with the Commander and flows through the Chief Engineer via a formal warranting system [10].

Delegated TA. Technical Warrant Holders (TWHs) (*Lead/Local/Warranted* TA) receive domain-specific authority across hull, mechanical, and electrical systems, combat systems, cybersecurity, and airworthiness or seaworthiness [10].

Core TA duties. Must-know responsibilities:

1. Establish and approve technical requirements, standards, and certification criteria;
2. Control the authoritative technical baseline (drawings, specs, interface control);

3. Adjudicate departures and waivers from specification and safety-critical changes; and
4. Certify readiness, readiness for service, and technical acceptability.

Independence. TA remains separate from cost, schedule, and programmatic authority to protect warfighter safety and mission assurance [2].

1.2.2 What EAs Do (and Do Not Do)

Engineering Agents (EAs) perform lifecycle engineering under TA governance. They are *not* TA unless they hold a TA warrant [10]. Common EA roles include:

In-Service Engineering Agent (ISEA). Lifecycle systems engineering for *fielded* systems (e.g., fleet introduction, distance support, troubleshooting/Casualty Report (CASREP) support, maintenance planning, obsolescence, technical manuals/data, configuration of the in-service baseline).

Design Agent (DA). Develops detailed design and configuration documentation for a system or platform (e.g., ship detail design, Initial Capabilities Documents (ICDs), drawings, Technical Data Packages (TDPs)) to TA-approved standards.

Alteration Engineering Agent (AEA). Produces alteration packages (e.g., installation drawings, test procedures, logistics updates) and supports fleet introduction for ship or system changes.

Systems Integration Agent (SIA). Orchestrates system-of-systems integration (e.g., interfaces, interoperability, cybersecurity in the integration space) across combat system elements and platforms.

Technical Direction Agent (TDA). Issues and maintains technical work direction for installations and maintenance (e.g., Technical Work Document (TWD), local instructions) consistent with the TA-approved baseline.

1.2.3 Where Engineering Agent Authority Comes From

Statutory basis. [2] assigns DON acquisition responsibilities, directing SYSCOMs to execute TA and to designate Warfare Centers as EAs that support PEOs and Program Managers (PMs).

Delegations from TA. NAVSEA, NAVAIR, and NAVWAR Chief Engineers issue written warrants or designation letters (Lead/Lab/Local TA) that flow requirements to specific Warfare Center codes; EA charters reference those warrants [10].

Execution orders. PEOs and PMs provide technical direction letters, project orders, and statements of work that scope the EA's tasks while preserving independence for specification compliance and certification [11].

Accountability. EAs report to the SYSCOM Chief Engineer for technical rigor and to the sponsoring PM for cost and schedule; loss of a warrant or charter terminates their authority to issue technical documentation [10].

1.2.4 Waterfront Triage: What Engages Whom

Spec/drawing nonconformance (build or repair). Generate a departure or waiver request for the proper TA warrant holder; minor deviations may be approved by Local or Lead TA, while major or safety-critical issues go to the Chief Engineer/UTA.

In-service failure (CASREP/technical assist). ISEA opens a support case, provides immediate workarounds and troubleshooting, coordinates root-cause analysis, updates technical data, and recommends permanent fixes (engineering change or alteration).

Change to configuration. If a permanent change is needed, the AEA develops the alteration package; SIA validates interfaces; DA updates drawings and TDPs; TA approves and certifies.

Principle to remember. EAs execute engineering; TA sets the rules and certifies compliance.

1.3 NAVSEA Organization and Warfare Centers

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1.3.1 HQ Codes (must-know one-liners)

NAVSEA headquarters staff code (SEA) 01 – Comptroller provides financial policy, advice and quality services to ensure NAVSEA’s customers’ budgets are efficiently and effectively executed. SEA 01 manages appropriation areas as well as providing cost engineering, and industrial analysis.

SEA 02 – Contracts and its field contracting offices under the Contracts Competency award approximately \$36 billion in contracts annually for new construction ships and submarines, ship repair, major weapon systems and services.

SEA 03 – Cyber Engineering & Digital Transformation has the responsibility for providing the infrastructure and support services for the programs and developers in the NAVSEA community. SEA 03 delivers combat power to the fleet through enterprise digital capabilities, infrastructure for cyber-secure digital work and innovation, and enhanced enterprise user’s experience. (*Appendix B.1: Rear Admiral (Lower Half) (RDML) Huan Nguyen serves as NAVSEA Deputy Commander for Cyber Engineering*)

- **Cyber Security:** SEA 03 provides the NAVSEA Enterprise with guidance on cyber issues and ensures that all ships are able and ready to detect, defend and recover from cybersecurity attacks.
- **Information Technology (IT) Services:** SEA 03 maintains and facilitates Network Operations and IT Service Delivery for Headquarters NAVSEA end users to enable maximum productivity for NAVSEA civilians and military personnel in the support of the warfighter.
- **Digital Transformation:** SEA 03 transforms NAVSEA into a digital capability adopting new digital technologies, capitalizing on data, increasing digital skills, and modernizing business processes.

SEA 04 – Industrial Operations has the important mission of getting ships to sea and keeping them ready. SEA 04 is the preferred integrator of maintenance and industrial operations for its Enterprise customers. SEA 04 manages the four Naval Shipyards and the four Supervisor of Shipbuildings (SUPSHIPS) (*Appendix B.1: RDML Scott M. Brown is the NAVSEA Deputy Commander for Industrial Operations*)

SEA 05 – Naval Systems Engineering & Logistics Directorate (TA) is responsible for providing the engineering and scientific expertise, knowledge, and technical authority necessary to design, build, maintain, repair, modernize, certify, and dispose of the Navy's ships, submarines, and associated warfare systems. SEA 05 is organized into 16 groups. (*Appendix B.1: RDML Peter D. Small is the NAVSEA Chief Engineer and Deputy Commander for Naval Systems Engineering (also Commander, Naval Surface Warfare Center (NSWC)/Naval Undersea Warfare Center (NUWC))*)

- Office of the Chief Engineer (SEA 05B)
- Cost Engineering and Industrial Analysis (SEA 05C)
- Explosive Ordnance Engineering (SEA 05E)
- Integrated Warfare Systems Engineering (SEA 05H)
- Logistics and Maintenance Warfare Systems Engineering (SEA 05M)
- Undersea Warfare Systems Engineering (SEA 05N)
- Ship Integrity and Performance Engineering (SEA 05P)
- Industrial Engineering, Technical Policy and Standards (SEA 05S)
- Technology Office (SEA 05T)
- Submarine/Submersible Design and Systems Engineering (SEA 05U)
- Aircraft Carrier Design and Systems Engineering (SEA 05V)
- Surface Warfare Systems Engineering (SEA 05W)
- Marine Engineering (SEA 05Z)

SEA 06 – Sustainment. Life-cycle product support.

SEA 07 – Undersea Warfare. Dual-hatted as Program Executive Office, SSBN (PEO SSBN); provides a full spectrum of research, development, test and evaluation, Hull, Mechanical, and Electrical (HM&E) systems engineering and fleet support services to the in-service submarine and undersea forces. Submarine/Undersea Warfare Technology (SUBTECH) coordinates the development of technologies to fulfill undersea warfare capability requirements.

SEA 08 – Nuclear Propulsion. Manages all technical matters pertaining to naval reactors. Triple-hat duties:

SEA 08 Deputy Commander. Leads nuclear propulsion activities within NAVSEA.

Director, Naval Nuclear Propulsion (Office of the Chief of Naval Operations (OPNAV) 00N). Serves as the Navy's principal nuclear propulsion authority.

Deputy Administrator for Naval Reactors. Interfaces with the National Nuclear Security Administration and the Department of Energy.

- SEA 09 – Safety and Regulatory Compliance.** Strengthens and aligns safety oversight and reporting.
- SEA 10 – Total Force and Corporate Operations** performs all operations support for NAVSEA directorates and field activities as well as PEOs. Support includes administrative products and services, career planning, employee development, facilities, foreign military sales coordination, human resources, security, and university research assistance.
- SEA 21 – Surface Ship Maintenance, Modernization and Sustainment** manages the complete lifecycle support for all non-nuclear surface ships and is the principal interface with the Surface Warfare Enterprise. The directorate is responsible for the maintenance and modernization of non-nuclear surface ships currently operating in the Fleet. Through planned modernization and upgrade programs, SEA 21 will equip today’s surface ships with the latest technologies and systems to keep them in the Fleet through their service lives. Additionally, SEA 21 oversees the ship inactivation process, including ship transfers or sales to friendly foreign navies, inactivation and/or disposal.
- Program Manager, Ships (PMS) 321.** Unmanned small combatants and amphibious ships.
- PMS 326.** International fleet support.
- PMS 339.** Surface training systems.
- PMS 421.** Large surface combatant modernization and sustainment.
- PMS 443.** Bridge integration and HM&E sustainment.
- PMS 451.** Destroyer Modernization 2.0 portfolio.
- SEA 21I.** Inactive Ships Directorate.
- Surface Maintenance Engineering Planning Program (SURFMEPP) Activity.** Surface Maintenance Engineering Planning Program.

1.3.2 Warfare Centers (alphabetical)

Table lists NAVSEA Warfare Centers with locations and descriptions. Figure 1.6 shows the geographic locations of NWCs. For the EDO board, most recommend drawing the US to show where the NWCs are located. The other option is to list them out, but a map is more visual.

TABLE I: NAVSEA WARFARE CENTERS WITH LOCATIONS AND DESCRIPTIONS.

Center	Location	Description
NSWC Carderock	West Bethesda, MD	Carderock Division is the U.S. Navy’s state-of-the-art research, engineering, modeling, and test center for ships and ship systems; the Navy and maritime communities rely on it for advanced platforms, analytics, and cost-conscious performance improvements.

Continued on next page

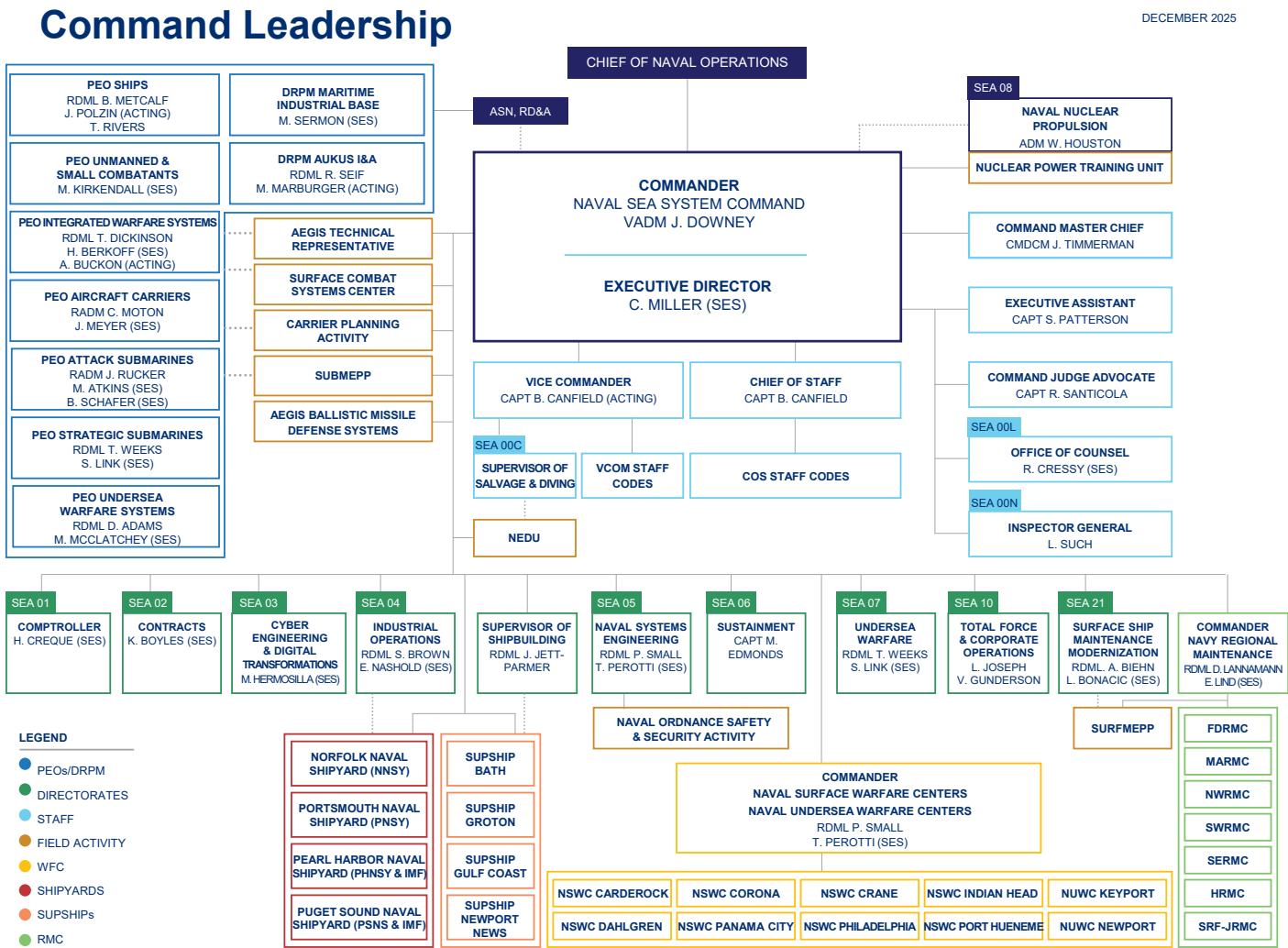


Figure 1.5. NAVSEA Organization Chart, December 2025.Source: NAVSEA Organization Chart (December 2025), 2025 [9]

TABLE I: NAVSEA WARFARE CENTERS WITH LOCATIONS AND DESCRIPTIONS. (Continued)

Center	Location	Description
NSWC Corona	Corona, CA	NSWC Corona Division is the Navy's independent assessment agent, leading NAVSEA data analytics and providing transparency for warfighting readiness, fleet LVC training integration, and metrology/calibration accuracy.
NSWC Crane	Crane, IN	NSWC Crane provides technical engineering solutions and total lifecycle leadership for many of the systems that protect and enable the Warfighter. Every decision we make, technology we create or business relationship we foster is performed with the fighting man and woman in mind. NSWC Crane has concentrated its resources and core competencies in the 3 mission areas which best support the Warfighter. The mission areas are Strategic Mission, Electromagnetic Warfare, and Expeditionary Warfare.
NSWC Dahlgren	Dahlgren, VA	The Naval Surface Warfare Center Dahlgren Division originally established itself as the major testing area for naval guns and ammunition. Today, it continues to provide the military with the development and integration of warfare systems for the warfighter, warfighting and the future fleet.
NSWC Indian Head	Indian Head, MD	As a field activity of the Naval Sea Systems Command, and part of the Navy's Science and Engineering Establishment, NSWC IHD is the Navy's premier facility for ordnance, energetics and explosive ordnance disposal solutions. We support the warfighter of today and tomorrow through discoveries that anticipate the next generation's future needs.
NSWC Panama City	Panama City, FL	The mission of NSWC, Panama City Division is to conduct research, development, test and evaluation, and in-service support of mine warfare systems, mines, naval special warfare systems, diving and life support systems, amphibious and expeditionary maneuver warfare systems, and other missions that occur primarily in coastal (littoral) regions.
NSWC Philadelphia	Philadelphia, PA	Provide research, development, test and evaluation, acquisition support, engineering, systems integration, in-service engineering and fleet support with cyber-security, comprehensive logistics, and life-cycle savings through commonality for surface and undersea vehicle machinery, ship systems, equipment and material.
NSWC Port Hueneme	Port Hueneme, CA	Integrate, Test, Evaluate, and Provide Life-Cycle Engineering and Product Support for Warfare Systems.
NUWC Keyport	Keyport, WA	Keyport's mission is focused on developing and applying advanced technical capabilities to test, evaluate, field, and maintain undersea warfare systems and related defense assets. These advanced technical capabilities directly support the full spectrum of Navy undersea programs.
NUWC Newport	Newport, RI	The Naval Undersea Warfare Center Division Newport provides research, development, test and evaluation, engineering, analysis, and assessment, and fleet support capabilities for submarines, autonomous underwater systems, and offensive and defensive undersea weapon systems, and stewards existing and emerging technologies in support of undersea warfare.

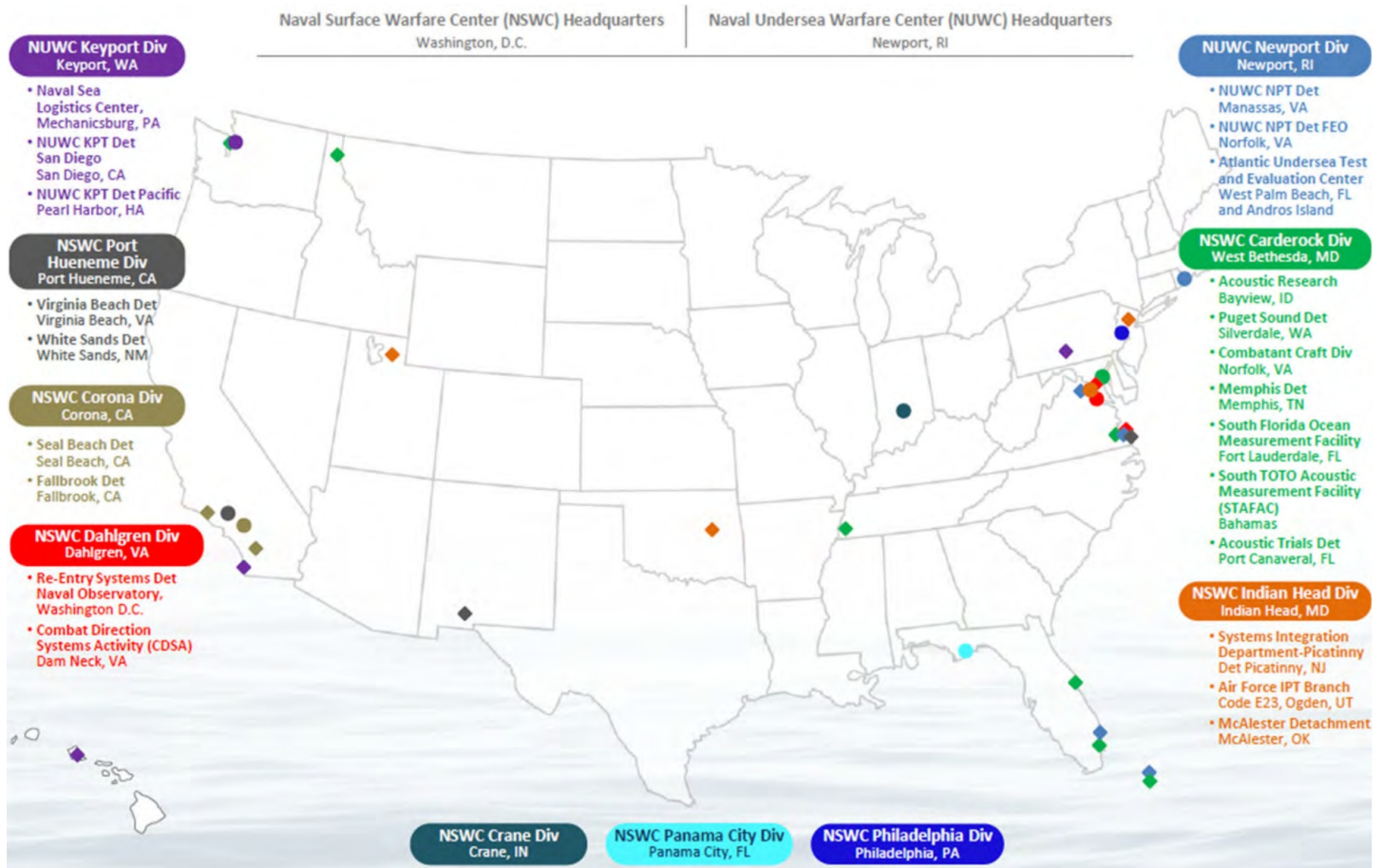


Figure 1.6. Geographic locations of Navy Warfare Centers. Source: NAVSEA Organization Chart (December 2025), 2.1.4. Naval Warfare Centers, 2025, 2024 [9, 11].

1.4 NAVWAR Enterprise and Warfare Centers

- [1] “2.1.1. Command Structures”Engineering Duty Officer School, “2.1.1. Command Structures,” Engineering Duty Officer (EDO) School, Coursebook Module 2.1.1, version 5.2, Mar. 11, 2025, CIN A-4N-0021/-0034; coursebook PDF provided by author.
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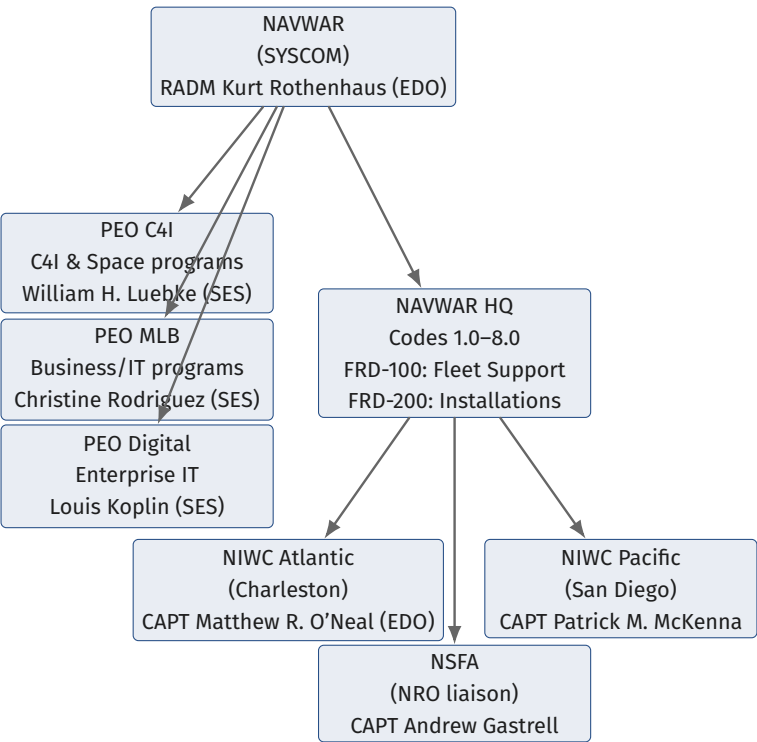


Figure 1.7. NAVWAR enterprise structure showing the SYSCOM commander with PEO portfolios, headquarters directorates, Field Readiness Directorates, and warfare centers.

1.4.1 HQ Directorates and FRD

Table I lists NAVWAR HQ directorates and Fleet Readiness Directorates (FRDs) with one-line roles.

TABLE II: NAVWARHQ DIRECTORATES AND FIELD READINESS DIRECTORATES – ONE-LINE ROLES.

Code	Office	One-liner
1.0	Comptroller	Budget formulation/execution; BSO duties; funds control and financial reporting.
2.0	Contracts	Contracting policy/oversight; acquisition strategy; award/administration of instruments.
3.0	Counsel	Legal counsel for acquisition/ethics; protests, claims, agreements, and compliance.
4.0	Logistics & Fleet Support	Life-cycle logistics; sustainment planning; supply/tech data; fleet support integration.
5.0	Chief Engineer / (TA for IT & Cyber)	Technical standards/architecture; interoperability; cyber TA; certification/air-gapping where required.
6.0	Program Management	Program governance; milestone readiness; portfolio integration; interface with PEOs.
7.0	Science & Technology	S&T portfolio; experimentation/prototyping; tech transition to programs of record.

Continued on next page

TABLE II: NAVWARHQ DIRECTORATES AND FIELD READINESS DIRECTORATES – ONE-LINE ROLES. (Continued)

Code	Office	One-liner
8.0	Corporate Operations	Workforce, security, facilities, enterprise information-technology/chief-information-officer services, PAO, and other corporate enablers.
FRD-100	Fleet Support	Sustainment services and distance support; field engineering and readiness assistance.
FRD-200	Installations	C4I installation planning/execution; shore/afloat integration and cutover support.

Source: 2.1.3. NAVWAR Enterprise, 2024 [12]

1.4.2 Warfare Centers (alphabetical)

Table II lists NAVWAR Warfare Centers with locations and descriptions.

TABLE III: NAVWAR WARFARE CENTERS WITH LOCATIONS AND DESCRIPTIONS.

Center	Location	Description
NIWC LANT	Charleston, SC	Conduct research, development, prototyping, engineering, test and evaluation, installation, and sustainment of integrated information warfare capabilities and services across all warfighting domains with an emphasis on Expeditionary Tactical Capabilities & Enterprise IT and Business Systems in order to drive innovation and warfighter information advantage
NIWC PAC	San Diego, CA	Conduct research, development, prototyping, engineering, test and evaluation, installation, and sustainment of integrated information warfare capabilities and services across all warfighting domains with emphasis on Basic and Applied Research and Tactical Systems Afloat and Ashore in order to drive innovation and warfighter information advantage.
NSFA	Chantilly, VA	Navy link to the NRO; Provide naval warfare and acquisition expertise to national reconnaissance programs; Coordinate naval space research, development, and acquisition activities with national reconnaissance programs; Provide and coordinate training and tools to enable the fleet to use national space capabilities; Conducts intelligence-related activities essential for US national security.

Source: 2.1.3. NAVWAR Enterprise, 2024 [12]

NIWC LANT DETACHMENTS

- HQ: Charleston, SC.
- New Orleans, LA.

- Norfolk, VA.
- Tampa, FL.
- Washington, DC.
- Stuttgart, Germany.
- Naples, Italy.
- Rota, Spain.
- Manama, Bahrain.



Figure 1.8. NIWC LANT Locations

NIWC PAC DETACHMENTS

- **HQ:** San Diego, CA.
- Everett, WA.
- Philadelphia, PA.
- Pearl City, HI.
- Santa Rita, Guam.
- Yokosuka, Japan.



Figure 1.9. NIWC PAC Locations

1.4.3 Project Overmatch

Mission.

Deliver a resilient Naval Operational Architecture that links platforms, weapons, sensors, and data fabrics for Distributed Maritime Operations and Combined Joint All-Domain Command and Control (CJADC2); NAVWAR provides the digital architecture, data transport, and cyber TA that underpin the hybrid fleet.

Five Eyes coalition network.

Under an Oct 2024 Project Arrangement with Australia, Canada, New Zealand, and the United Kingdom, Cooperative Project Personnel embed in San Diego to co-develop kill-chain networks; Capt. Remil Capili (PM), VADM Seiko Okano (steering chair/PMILDEP), Shayna Bond (international deputy), and David Flowers (international lead) oversee the coalition lab and Rim of the Pacific Exercise (RIMPAC) 26 demonstrations (*Appendix B.1: VADM Seiko Okano is the PMILDEP flag billet holder in Appendix B.1*) [13].

Allied exercises.

Overmatch Five Eyes (Australia, Canada, New Zealand, United Kingdom, United States) (FVEY) solutions are being exercised at Trident Warrior 24, Talisman Sabre 25, and a capstone during RIMPAC 26 to prove resilient comms in contested electromagnetic environments[13].

Rapid onboarding.

Open Data and Applications Government-owned Interoperable Repositories (DAGIR) and Maven Smart Systems pipelines let NAVWAR ingest commercial Artificial Intelligence and Machine Learning (AI/ML) tools in weeks, accelerating fleet readiness visualizations and data-driven decision aids across the hybrid force [14].

Governance.

As PMILDEP and DRPM Overmatch, VADM Okano keeps digital architecture tied to Program Decision Meetings and PEO portfolio trades, ensuring Overmatch data standards propagate into new ship, sub, and aircraft programs [3, 13].

1.5 NAVAIR Warfare Centers

- [1] “2.1.1. Command Structures”Engineering Duty Officer School, “2.1.1. Command Structures,” Engineering Duty Officer (EDO) School, Coursebook Module 2.1.1, version 5.2, Mar. 11, 2025, CIN A-4N-0021/-0034; coursebook PDF provided by author.
- [2] *SECNAVINST 5400.15D*: Department of the Navy Research, Development and Acquisition—Organizational Relationships and Responsibilities*SECNAVINST 5400.15D, Department of the Navy Research, Development and Acquisition—Organizational Relationships and Responsibilities*, SECNAVINST 5400.15D, Instruction, Department of the Navy, Jan. 19, 2021. Accessed: 2025-09-29. [Online]. Available: <https://www.secnav.navy.mil/doni/SECNAVInstructions/05000-05999/5400.15D.pdf>
- [3] *SECNAVINST 5000.2G*: Defense Acquisition System and JCIDS Implementation in the Department of the Navy*SECNAVINST 5000.2G, Defense Acquisition System and JCIDS Implementation in the Department of the Navy*, SECNAVINST 5000.2G, Instruction, Department of the Navy, Apr. 8, 2022. Accessed: 2025-09-29. [Online]. Available: <https://www.secnav.navy.mil/doni/SECNAVInstructions/05000-05999/5000.2G.pdf>
- [4] “Vice Admiral Seiko Okano”U.S. Navy. “Vice Admiral Seiko Okano.” Principal Military Deputy, ASN(RD&A), Accessed: 2025-12-05. [Online]. Available: <https://www.navy.mil/Leadership/Flag-Officer-Biographies/BioDisplay/Article/2283053/vice-admiral-seiko-okano/>
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- [9] “NAVSEA Organization Chart (December 2025)”NAVSEA. “NAVSEA Organization Chart (December 2025).” NAVSEA Organization Chart (December 2025); verify current fills, Accessed: 2025-12-13. [Online]. Available: https://www.navsea.navy.mil/Portals/103/Documents/Command%20Org%20Charts/NAVSEA_Org_Chart_Dec2025.pdf
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- [11] “2.1.4. Naval Warfare Centers”Engineering Duty Officer School, “2.1.4. Naval Warfare Centers,” Engineering Duty Officer (EDO) School, Coursebook Module 2.1.4, version 5.0, Dec. 18, 2024, CIN A-4N-0021/-0034.
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- [13] “Project Overmatch and Five Eyes Coalition Partners Strengthen Ties” Naval Information Warfare Systems Command. “Project Overmatch and Five Eyes Coalition Partners Strengthen Ties.” Story ID 551112, Accessed: 2025-12-05. [Online]. Available: <https://www.dvidshub.net/news/551112/project-overmatch-and-five-eyes-coalition-partners-strengthen-ties>
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Table III lists NAVAIR Warfare Centers with locations and descriptions. This is an optional item to know for the EDO board, not as critical as NAVSEA/NAVWAR.

TABLE IV: NAVAIR WARFARE CENTERS WITH LOCATIONS AND DESCRIPTIONS.

Center	Location	Description
NAWCAD	Patuxent River, MD	Aircraft/engines/avionics RDT&E; Test Pilot School.
NAWCWD	China Lake / Point Mugu, CA	Weapons systems, ranges, guided-missile integration.

Source: 2.1.4. Naval Warfare Centers, 2024 [11]

1.6 Navy Program Executive Offices (PEOs)

- [1] “2.1.1. Command Structures” Engineering Duty Officer School, “2.1.1. Command Structures,” Engineering Duty Officer (EDO) School, Coursebook Module 2.1.1, version 5.2, Mar. 11, 2025, CIN A-4N-0021/-0034; coursebook PDF provided by author.
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- [7] “Rear Admiral Casey Moton”U.S. Navy. “Rear Admiral Casey Moton.” Program Executive Officer, Aircraft Carriers, Accessed: 2025-12-05. [Online]. Available: <https://www.navy.mil/Leadership/Flag-Officer-Biographies/BioDisplay/Article/2236471/rear-admiral-casey-moton/>
- [8] “Rear Admiral Jonathan Rucker”U.S. Navy. “Rear Admiral Jonathan Rucker.” Program Executive Officer, Attack Submarines, Accessed: 2025-12-05. [Online]. Available: <https://www.navy.mil/Leadership/Flag-Officer-Biographies/BioDisplay/Article/3052230/rear-admiral-jonathan-rucker/>
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- [11] “2.1.4. Naval Warfare Centers”Engineering Duty Officer School, “2.1.4. Naval Warfare Centers,” Engineering Duty Officer (EDO) School, Coursebook Module 2.1.4, version 5.0, Dec. 18, 2024, CIN A-4N-0021/-0034.
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- [15] “3.1.5. Field Activity Financial Management”Engineering Duty Officer School, “3.1.5. Field Activity Financial Management,” Engineering Duty Officer (EDO) School, Coursebook Module 3.1.5, version 5.2, Mar. 26, 2025, CIN A-4N-0021/-0034.

PEO CVN. Designs, builds, delivers, and sustains nuclear-powered aircraft carriers. (*Appendix B.1: RADM Casey J. Moton holds the PEO CVN billet*)

PMS 312. In-service aircraft carrier program management.

PMS 378. Nuclear-Powered Aircraft Carrier (CVN)-78 class program management.

PMS 379. CVN-79/80 program management.

PEO IWS. Develops and sustains ship and submarine combat systems.

Note

RDML T. Dickinson is currently PEO IWS. He is a Surface Warfare Acquisition Professional, not a EDO.

Integrated Warfare Systems (IWS) 1.0. Aegis combat system lead.

IWS 1.0F. Aegis fleet readiness.

IWS 2.0. Above-water sensors portfolio.

IWS 3.0. Surface ship weapons integration.

IWS 4.0. International and foreign military sales.

IWS 5.0. Undersea systems.

IWS 6.0. Command-and-control systems.

IWS 9.0. Guided Missile Destroyer (DDG)-1000, littoral combat ship, and patrol craft combat systems.

IWS 11.0. Terminal defense systems.

IWS 12.0. North Atlantic Treaty Organization (NATO) Seasparrow programs.

IWS 80.0. Atalanta combat systems.

IWS X. Integrated combat-system document center.

PEO Ships. Oversees surface combatant and amphibious ship construction and modernization. (*Appendix B.1: RADM Brian A. Metcalf serves as PEO Ships*)

PEO USC. Manages littoral combat ships, frigates, expeditionary platforms, and unmanned surface/undersea portfolios. (*Appendix B.1: RDML Kevin R. Smith serves as PEO USC*)

Team Submarines. Integrates strategic and attack submarine acquisition and sustainment.

PEO SSN. Attack submarine programs. (*Appendix B.1: RADM Jonathan E. Rucker serves as PEO SSN*)

PMS 351. New attack submarine acquisition.

PMS 390. Undersea special mission systems.

PMS 391. In-service submarine sustainment.

PMS 394. Advanced undersea system's development.

PMS 450. Virginia class program management.

PEO SSBN. Strategic deterrent submarine portfolio.

PMS 396. In-service Ballistic Missile Submarine, Nuclear (SSBN) sustainment.

PMS 397. Columbia class program management.

Submarine Maintenance Engineering Planning Program (SUBMEPP). Surface maintenance engineering planning program support to ballistic submarine availabilities.

Program Executive Office, Undersea Warfare Systems (PEO UWS). Submarine combat, cyber, and sensor systems.

PMS 401. Submarine acoustic systems.

PMS 404. Undersea weapons.

PMS 415. Undersea defensive warfare systems.

PMS 425. Combat and weapon control systems.

PMS 435. Electromagnetic systems.

PMS 485. Maritime surveillance systems.

Program Executive Office, Command, Control, Communications, Computers, and Intelligence (PEO C4I). Delivers fleet Command, Control, Communications, Computers, and Intelligence (C4I) capabilities.

Program Management Warfare (NAVWAR PEO C4I/NAVAIR PMS series) (PMW) 120. Battlespace awareness and information operations.

PMW 130. Information Assurance (IA) and cybersecurity programs.

PMW 150. Navy command-and-control systems.

PMW 160. Tactical networks.

PMW 170. Communications and Global Positioning System (GPS) navigation.

PMW 740. International C4I integration.

PMW 750. Carrier strike and air integration.

PMW 760. Ship integration.

PMW 770. Undersea integration.

PMW 790. Shore and expeditionary integration.

PMW series focus. Two PMW series within PEO C4I have distinct roles:

PMW 1XX. Major capability development portfolios (new platforms and end-to-end networks) where PEO C4I owns the baseline capabilities document, acquisition strategy, and milestone execution; Warfare Centers supply lead systems integration, test ranges, and engineering agents via tailored project orders [2, 11].

PMW 7XX. Fleet integration portfolios synchronizing new capabilities into existing hulls, aircraft, and shore nodes; emphasizes installation planning, logistics, and interoperability packages coordinated through regional Warfare Centers and type commanders [2, 15].

Program Executive Office, Digital and Enterprise Services (PEO Digital). Provides enterprise digital services (e.g., Flank Speed collaboration environment).

PEO MLB. Modernizes manpower, logistics, and business information-technology systems.

1.6.1 PEO/Warfare Center Interaction Model

Role Designation. PEOs charter Warfare Centers as EAs (ISEA, DA, SIA) via technical direction letters aligned with [2]; the Warfare Center executes under Navy Working Capital Fund (NWCF) funding while reporting performance to the PM [11].

Requirements Translation. PM Integrated Product Teams (IPTs) decompose Capabilities Development Documents (CDDs) into technical requirements that Warfare Center TA holders validate and flow down through alteration packages, interface control documents, and test plans [11].

Funding Mechanisms. PEO Program Management Offices (PMOs) issue project orders or Economy Act orders using their appropriations; Warfare Centers accept into NWCF, schedule Work Breakdown Structures (WBSes), and recover rates through Stabilized Labor Rates (SLRs) while keeping the PM apprised of burn rates and Net Operating Result (NOR) impacts [15].

Governance Rhythm. Monthly technical reviews focus on requirements churn, test results, and configuration control; quarterly business reviews reconcile execution to SLR assumptions and assess workforce mix, drawing on Warfare Center cost visibility [2, 15].

Acquisition Decision Events. Warfare Centers provide independent readiness assessments (test status, technical risk) for PEO milestone decisions, often serving as certifying authorities for safety releases or flight clearances prior to fielding [2, 11].

2 PPBE

2.1 PPBE (Planning, Programming, Budgeting, and Execution)

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- [18] “DoW Financial Management Regulation 7000.14—R, Volume 3” “DoW Financial Management Regulation 7000.14—R, Volume 3,” Office of the Under Secretary of War (Comptroller), Regulation Vol. 3, Jul. 1, 2025. Accessed: 2025-09-29. [Online]. Available: <https://comptroller.defense.gov/fmr/>
- [19] “3.1.3. Program Funding” Engineering Duty Officer School, “3.1.3. Program Funding,” Engineering Duty Officer (EDO) School, Coursebook Module 3.1.3, version 5.3, Sep. 8, 2025, CIN A-4N-0021/-0034.
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- [21] 41 U.S.C. § 6307: Project Order statute 41 U.S.C. § 6307, Project Order statute, 2011. [Online]. Available: <https://uscode.house.gov/>
- [22] 31 U.S.C. § 1535: Economy Act 31 U.S.C. § 1535, Economy Act, 2018. [Online]. Available: <https://uscode.house.gov/>
- [23] 31 U.S.C. § 1301(a): Application — Purpose Statute 31 U.S.C. § 1301(a), Application — Purpose Statute, 2018. [Online]. Available: <https://uscode.house.gov/>
- [24] 31 U.S.C. § 1502(a): Bona Fide Needs Rule 31 U.S.C. § 1502(a), Bona Fide Needs Rule, 2018. [Online]. Available: <https://uscode.house.gov/>
- [25] 31 U.S.C. § 1341; § 1517: Anti-Deficiency Act 31 U.S.C. § 1341; § 1517, Anti-Deficiency Act, 2018. [Online]. Available: <https://uscode.house.gov/>
- [26] 10 U.S.C. § 2208: Working-capital funds 10 U.S.C. § 2208, Working-capital funds, 2023. Accessed: 2025-10-12. [Online]. Available: <https://uscode.house.gov/>
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2.1.1 What PPBE does

Aligns strategy to resources across the Future Years Defense Program (FYDP). *Planning* (strategy → guidance), *Programming* (balanced force/program → Program Objective Memorandum (POM)), *Budgeting* (validated request → President’s Budget (PB)), *Execution* (obligations/outlays → performance feedback) [16].

2.1.2 Timeline (annual rhythm)

Figure 2.1 shows the Planning, Programming, Budgeting and Execution (PPBE) timeline across multiple fiscal years with key milestones during each calendar year.

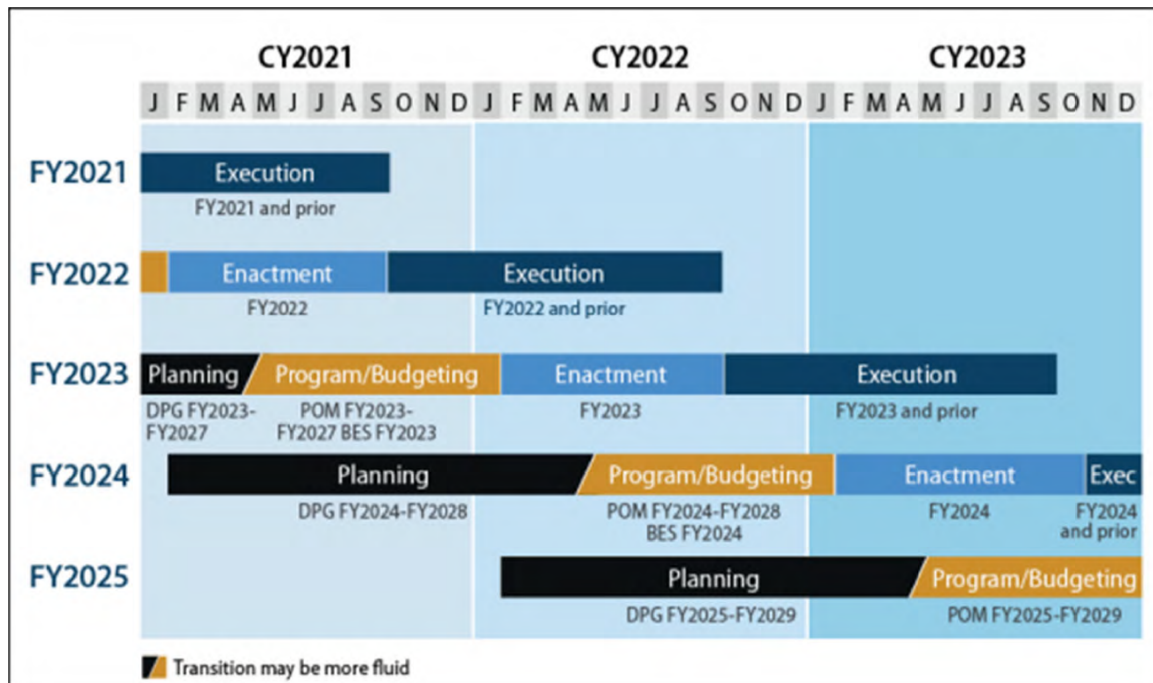


Figure 2.1. PPBE Timeline for multiple fiscal years. Source: 3.1.1. PPBE (Planning, Programming, Budgeting, and Execution), 2025 [16].

The Platinum Card view in Figure 2.2 is a quick board sketch for the Planning phase (War aims → strategy → guidance) before the POM/PB forks.

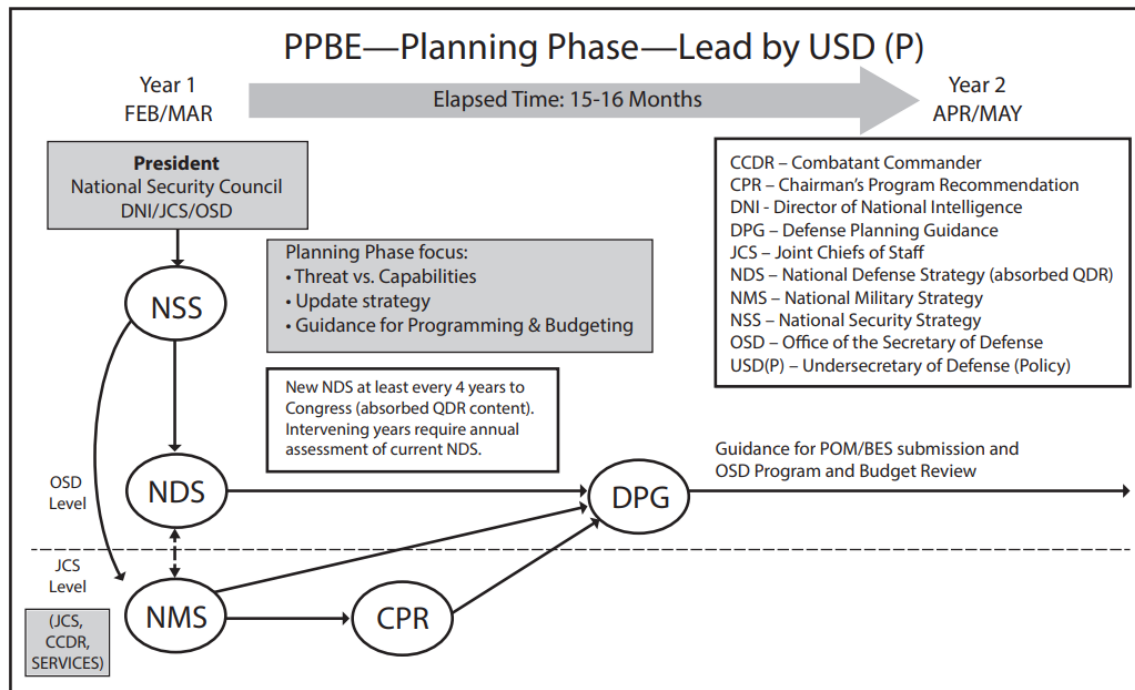


Figure 2.2. Platinum Card view of the PPBE Planning phase (USDP-led strategic guidance path). Source: 3.1.1. PPBE (Planning, Programming, Budgeting, and Execution), 2025 [16].

2.1.3 Congressional Enactment (Regular Order)

Figure 2.3 highlights how Congress moves from the PB submission through authorization and appropriations when operating on-time without a Continuing Resolution (CR). Being able to walk this chart helps bridge PPBE milestones with Hill activity.

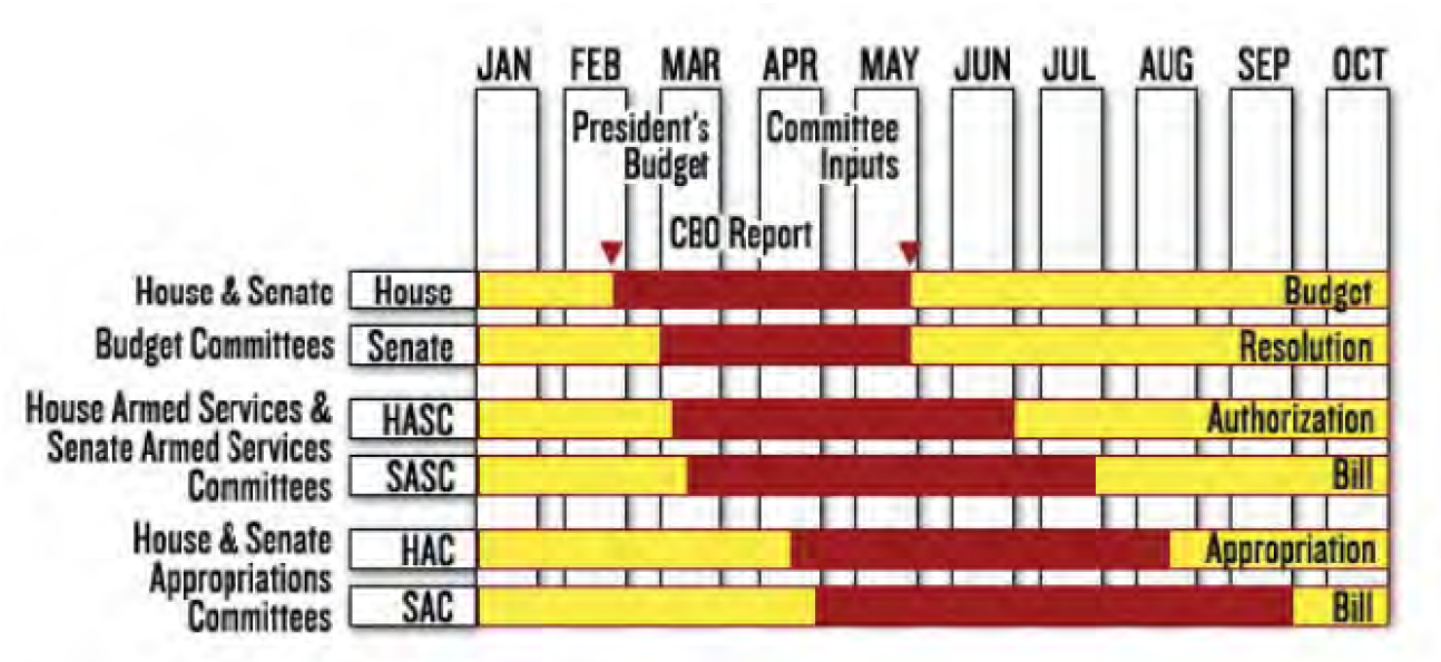


Figure 2.3. Congressional Enactment Timeline. Source: 3.1.2. Congressional Enactment, 2025 [17].

2.1.4 Key Terms and Definitions

Table IV lists key terms and definitions.

TABLE V: KEY TERMS IN PPBE

Term	One-liner
PY	Last completed FY: execution look-back and CR baselines.
CY	Ongoing FY: execution and mid-year reviews.
BY	Next FY in submission.
POM	Services balanced force/program across the FYDP.
FYDP	5-year program structure and resources.

Source: DoW Financial Management Regulation 7000.14—R, Volume 3, 2025 [18]

Other terms and definitions from [16].

Cost Assessment and Program Evaluation (CAPE). CAPE supports the Secretary and other Department of War (DoW) senior leaders through non-advocate, fact-based decision support on the size, shape, affordability, disposition, and readiness of the future joint force to achieve the National Defense Strategy. They do this by: (1) Supporting the Secretary in Analyzing and Adjusting the Defense Program with the Strategic Portfolio Review and by Leading the Department’s Program Review; (2) Facilitating the Acquisition Process with Independent Cost Estimates and Guidance and sufficiency review for Analysis of

Alternatives; and (3) Leading DoW’s Analytics and Cost Communities by Building the skills of the Department’s analytical workforce and Providing deeper insights on DoW programs.

Program Decision Memorandum (PDM). Records SECWAR program decisions at the end of Program Review.

OPNAV Integration of Capabilities and Resources (N8). Builds the Navy POM; trades across portfolios with cost/risk realism.

OPNAV Warfare Systems (N9). Validates/advocates warfare requirements; integrates by mission area.

Warfare Integration Directorate (N91). Manages cross-domain mission integration and architecture; orchestrates POM issue papers and mission engineering.

Warning

Office designators can shift; verify current subcodes week-of.

2.1.5 Programming v.s. Budgeting

Table V compares key differences between Programming and Budgeting. One way to think about this: (1) Budgeting is for the money for this year and (2) Programming is for the 5-year FYDP.

TABLE VI: PROGRAMMING V.S. BUDGETING		
	Programming	Budgeting
Purpose	Build a balanced force across FYDP	Price/validate an executable BY request
Lead	N8 with supporting codes/PEOs/SYSCOMs	FMB/Office of the Secretary of War (Comptroller)/Office of Management and Budget
Key products	POM, PDM	Budget estimate, reclama, President’s Budget

Source: 3.1.1. PPBE (Planning, Programming, Budgeting, and Execution), 2025 [16]

Figure 2.4 from the Platinum Card captures how the Service-level Program/Budget reviews run in parallel after Planning guidance issues, keeping POM trades and the Comptroller’s reclaims synchronized.

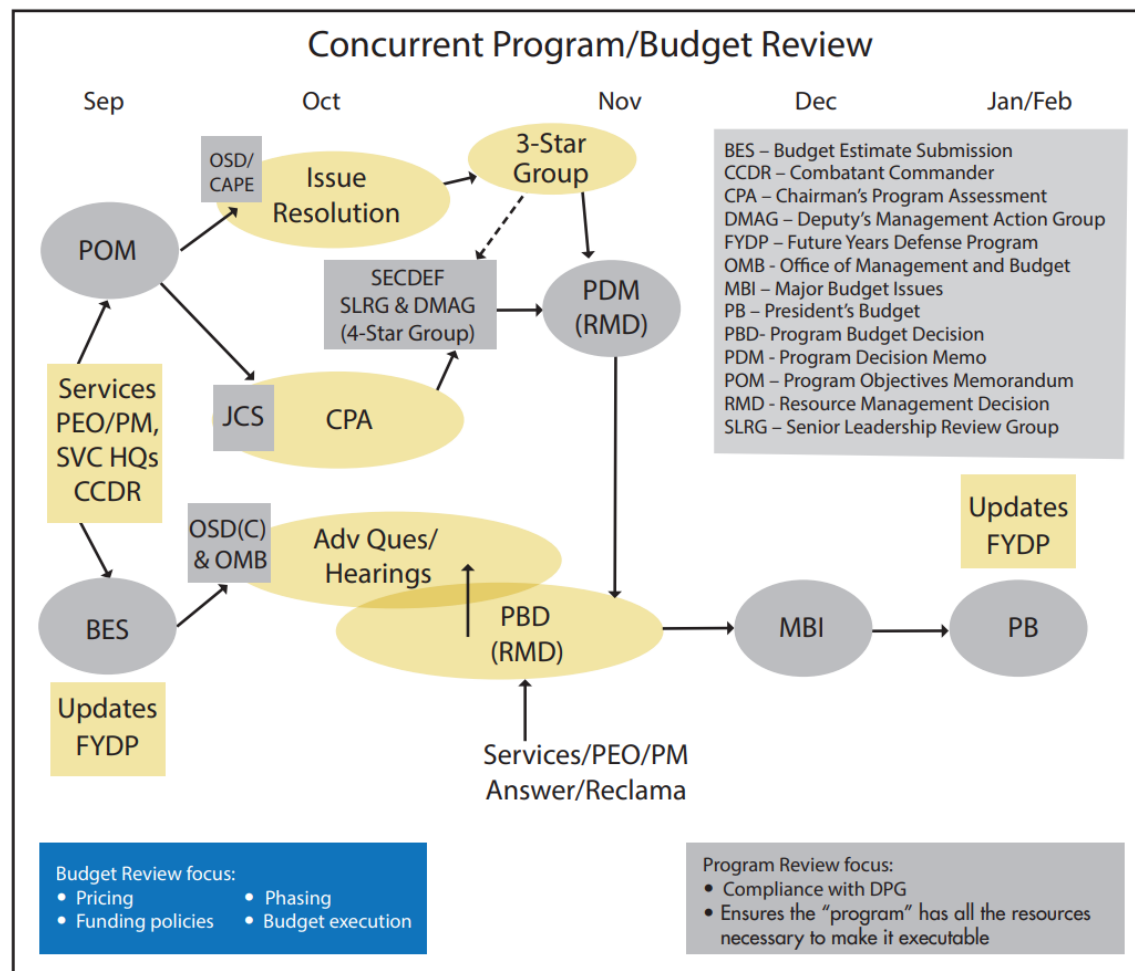


Figure 2.4. Platinum Card view of the concurrent Program and Budget reviews. Source: 3.1.1. PPBE (Planning, Programming, Budgeting, and Execution), 2025 [16].

2.1.6 PPBE Programming

3-STAR V.S. 4-STAR REVIEWS: During Office of the Secretary of War (OSD)'s Program Review, issues raised on the Services' POMs move first to a 3-Star Programmers Panel (staff-level, chaired by Director, CAPE (DCAPE)) that vets issue papers, builds options (with offsets), and forwards recommendations. Unresolved or strategy-level trades go to the 4-Star forum, the Deputy's Management Action Group (DMAG) co-led by the Deputy SECWAR (with the Vice-Chairman of the Joint Chiefs of Staff (VCJCS)), for senior adjudication. Decisions at this stage are documented as a PDM or (in recent cycles) a programmatic Resource Management Decision (RMD), which updates the FYDP and hands off to the Comptroller's Budget Review that culminates in budgetary RMDs.

2.1.7 Appropriations Life-cycle (Colors of Money)

Figure 2.5 visualizes when each major appropriation category is Current, Expired, or Cancelled. Pair this with the obligation windows discussion so you can sketch the chart quickly at the board.

YEARS											
Appropriation Categories	1	2	3	4	5	6	7	8	9	10	11
O&M											
RDT&E											
Procurement											
Ships											
MILCON											
MILPERS											

LEGEND		
Current		Available for new obligations, obligation adjustments, expenditures, and outlays.
Expired		Available for obligation adjustments, expenditures, and outlays.
Cancelled		Unavailable for obligations, obligation adjustments, expenditures, and outlays.

Figure 2.5. Colors of Money Timeline. Source: 3.1.3. Program Funding, 2025 [19].

2.1.8 IGT (Intragovernmental Transactions)

Definition. Orders and collections between federal entities using Treasury’s G-Invoicing: 7600A (order), 7600B (agreement/performance), Intra-Governmental Payment and Collection (IPAC) for collections [20].

Why it matters. Intragovernmental Transaction (IGT) acceptance *is an obligation* on the customer side and the *revenue recognition driver* for the performing Working Capital Fund (WCF) activity.

2.1.9 Common Ordering Instruments

Work Order. Internal directive used within a command to control scope/cost/schedule.

Project Order. Statutory order for a *definite, specific, and entire* project; obligational on acceptance; performs like a contract between DoW activities. (41 U.S.C. § 6307) [21]

Military Interdepartmental Purchase Request (MIPR) (DD Form 448). Economy Act order between federal entities; obligational when accepted; performed by the servicing agency. (31 U.S.C. § 1535) [20, 22]

2.1.10 Fiscal Law (Know These Cold)

Purpose Statute (“Misappropriation”). Funds must be used only for their appropriated purpose (31 U.S.C. § 1301(a)) [23].

Bona Fide Needs Rule. Use current-year appropriations only for legitimate needs of that fiscal year (31 U.S.C. § 1502(a)) [24].

Anti-Deficiency Act. Do not obligate/expend in excess of available amounts or before funds are available (31 U.S.C. § 1341, § 1517) [25].

2.1.11 Reprogramming: Moving Resources After Enactment

FOUR MECHANISMS (HIGH LEVEL).

1. **Congressional Prior Approval Reprogramming (PA Reprog):** Actions above statutory/committee thresholds or affecting congressional special-interest items or new starts. Requires approval from all four defense committees; timing driven by committee cycles [18].
2. **Internal Reprogramming (IR):** Realignment within an appropriation that do not cross thresholds or trigger congressional interest; used for proper purpose alignment without changing totals. Notice sent to committees [18].
3. **Below-Threshold Reprogramming (BTR):** Component-level authority to move funds below set dollar/percent limits within the same appropriation/fiscal year; limits reset annually by appropriations acts/committee guidance [18].
4. **Letter Transfer (LT):** Treasury non-expenditure transfer moving budgetary resources between appropriations/treasury symbols when authorized (e.g., statute or enacted language) [18].

Hint

General rules of thumb—cannot change color-of-money or extend time; must complete before funds' obligation period ends; expired-year accounts limited to valid upward/downward adjustments only [18].

2.1.12 From Enactment to Field Authority

1. Treasury *warrants* the appropriation; Office of Management and Budget (OMB) *apportions* on Standard Form 132.
2. OSD (Comptroller) issues allocations to components such as the DON.
3. OPNAV (Financial Management Budget (OPNAV) (FMB) / OPNAV Programming Division (N82)) allocates to SYSCOM Budget Submitting Offices (BSOs) (Echelon II), which suballocate to Warfare Centers (Echelon III), followed by *allowances/allotments* to the Enterprise Resource Planning (ERP) system [18].
4. Commands record commitments, obligations, and expenses, and request outlays via Defense Finance and Accounting Service (DFAS).

2.2 NWCF Essentials: Rates, Results, and Execution

- [3] *SECNAVINST 5000.2G: Defense Acquisition System and JCIDS Implementation in the Department of the Navy* SECNAVINST 5000.2G, *Defense Acquisition System and JCIDS Implementation in the Department of the Navy*, SECNAVINST 5000.2G, Instruction, Department of the Navy, Apr. 8, 2022. Accessed: 2025-09-29. [Online]. Available: <https://www.secnav.navy.mil/doni/SECNAVInstructions/05000-05999/5000.2G.pdf>

- [16] “3.1.1. PPBE (Planning, Programming, Budgeting, and Execution)”Engineering Duty Officer School, “3.1.1. PPBE (Planning, Programming, Budgeting, and Execution),” Engineering Duty Officer (EDO) School, Coursebook Module 3.1.1, version 5.2, Mar. 26, 2025, CIN A-4N-0021/-0034.
- [17] “3.1.2. Congressional Enactment”Engineering Duty Officer School, “3.1.2. Congressional Enactment,” Engineering Duty Officer (EDO) School, Coursebook Module 3.1.2, version 5.2, Mar. 26, 2025, CIN A-4N-0021/-0034.
- [18] “DoW Financial Management Regulation 7000.14—R, Volume 3”“DoW Financial Management Regulation 7000.14—R, Volume 3,” Office of the Under Secretary of War (Comptroller), Regulation Vol. 3, Jul. 1, 2025. Accessed: 2025-09-29. [Online]. Available: <https://comptroller.defense.gov/fmr/>
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- [20] “G-Invoicing Program Guide and Resources”“G-Invoicing Program Guide and Resources,” U.S. Department of the Treasury, Bureau of the Fiscal Service, Accessed: 2025-09-29. [Online]. Available: <https://www.fiscal.treasury.gov/g-invoicing/>
- [21] 41 U.S.C. § 6307: Project Order statute41 U.S.C. § 6307, Project Order statute, 2011. [Online]. Available: <https://uscode.house.gov/>
- [22] 31 U.S.C. § 1535: Economy Act31 U.S.C. § 1535, Economy Act, 2018. [Online]. Available: <https://uscode.house.gov/>
- [23] 31 U.S.C. § 1301(a): Application — Purpose Statute31 U.S.C. § 1301(a), Application — Purpose Statute, 2018. [Online]. Available: <https://uscode.house.gov/>
- [24] 31 U.S.C. § 1502(a): Bona Fide Needs Rule31 U.S.C. § 1502(a), Bona Fide Needs Rule, 2018. [Online]. Available: <https://uscode.house.gov/>
- [25] 31 U.S.C. § 1341; § 1517: Anti-Deficiency Act31 U.S.C. § 1341; § 1517, Anti-Deficiency Act, 2018. [Online]. Available: <https://uscode.house.gov/>
- [26] 10 U.S.C. § 2208: Working-capital funds10 U.S.C. § 2208, Working-capital funds, 2023. Accessed: 2025-10-12. [Online]. Available: <https://uscode.house.gov/>
- [27] “DoW FMR 7000.14-R, Volume 3, Chapter 19: Defense Working Capital Funds”“DoW FMR 7000.14-R, Volume 3, Chapter 19: Defense Working Capital Funds,” Office of the Under Secretary of War (Comptroller), Jul. 1, 2025. Accessed: 2025-09-29. [Online]. Available: <https://comptroller.defense.gov/fmr/>
- [28] “Continuing Resolutions: Overview of Components and Recent Practices”J. V. Saturno et al., “Continuing Resolutions: Overview of Components and Recent Practices,” Congressional Research Service, R42647, Mar. 27, 2025. Accessed: 2025-09-29. [Online]. Available: <https://crsreports.congress.gov/>
- [29] 10 U.S.C. § 3501: Multiyear Procurement10 U.S.C. § 3501, Multiyear Procurement, 2023, Title 10 recodification; formerly Title 10 U.S.C. § 2306b. Accessed: 2025-09-29. [Online]. Available: <https://uscode.house.gov/>
- [30] “DFARS 217.172: Multiyear Contracts”“DFARS 217.172: Multiyear Contracts,” Regulation, 2025. Accessed: 2025-09-29. [Online]. Available: <https://www.acquisition.gov/dfars/part-217>
- [31] “Federal Acquisition Regulation (FAR)”“Federal Acquisition Regulation (FAR),” Regulation, 2025. Accessed: 2025-09-29. [Online]. Available: <https://www.acquisition.gov/far>
- [32] DoDI 5000.85: Major Capability AcquisitionDoDI 5000.85, *Major Capability Acquisition*, DoDI 5000.85, Instruction, Change(s) as applicable, Department of War, Aug. 6, 2020. Accessed: 2025-09-29. [Online]. Available: <https://www.esd.whs.mil/Directives/issuances/dodissuances/>
- [33] “3.1.6. Cost Estimating”Engineering Duty Officer School, “3.1.6. Cost Estimating,” Engineering Duty Officer (EDO) School, Coursebook Module 3.1.6, version 5.0, Dec. 18, 2024, CIN A-4N-0021/-0034.
- [34] “Adaptive Acquisition Framework”Defense Acquisition University. “Adaptive Acquisition Framework,” Accessed: 2025-09-29. [Online]. Available: <https://aaf.dau.edu/>

2.2.1 Key Definitions (Execution)

Commitment. Administrative reservation of funds in anticipation of an obligation; reduces available authority but is not yet a legal liability [18] (e.g., National Reconnaissance Office (NRO)’s Request for Contract Action (RCA) process).

Obligation. Legal liability incurred that encumbers funds [18] (e.g., award of a contract, acceptance of an IGT order).

Expenditure/Expense. Recording of cost when goods/services are received/accepted (accrual basis) [18] (e.g., sending check to pay an invoice).

Outlay/Disbursement. Treasury cash payment to a vendor or performing activity [18] (e.g., payment check is cashed in).

2.2.2 Flow: PMO Purchase Request to Treasury Payment

1. PMO *commits* funds (approved in the ERP system).
2. Contract award or IGT order acceptance creates the *obligation* [18, 20].
3. Performance/acceptance posts *expense* (accrual).
4. DFAS schedules the invoice for payment; Treasury disburses (*outlay*) via IPAC (for IGT) or commercial Electronic Funds Transfer (EFT) [20].

2.2.3 NWCF Metrics and Equations

NOR (annual profit/loss).

$$\text{NOR} = \text{Revenue} - \text{Total Expenses}$$

Target performance remains near break-even over time (small NOR) [18].

Accumulated Operating Result (AOR) (retained earnings/equity over time).

$$\text{AOR}_t = \text{AOR}_{t-1} + \text{NOR}_t \pm \text{Other Adjustments}$$

Represents the *corpus* (net position) of the fund [18].

2.2.4 NWCF Corpus v.s. Appropriations

AOR (working capital). Revolving cash and retained earnings authorized under 10 U.S.C. § 2208 [26]; remains available without fiscal year limitation to finance operations until recovered through stabilized rates [27].

Customer Appropriations. Mission-funded (e.g., Operation and Maintenance, Navy (OMN), Research, Development, Test, and Evaluation (RDT&E), Shipbuilding and Conversion, Navy (SCN)) dollars obligated by the customer when a project order or inter/intra-governmental order is accepted; purpose, time, and amount statutes still apply to the customer [18].

Capital Investment Program (CIP). Financed by NWCF corpus but budgeted in the capital budget exhibit; used for plant/equipment modernization and amortized back through rates, not through a separate appropriation [27].

No Augmentation. NWCF activities cannot augment customer appropriations; corpus only bridges cash timing between expense recognition and reimbursement [27].

2.2.5 Stabilized Rates and SLR

Stabilized Rates. Customer prices set in the budget build to recover expected full costs (labor, material, overhead, depreciation/C CIP) with a near-zero NOR goal [18].

SLR. Published labor \$/hr for a shop/code; recovers direct labor + fringe + overhead + G&A + capital depreciation recovery [18]. The rate is:

$$\text{SLR} = \frac{\text{Direct Labor} + \text{Fringe} + \text{OH} + \text{G\&A} + \text{Depreciation Recovery}}{\text{Direct Labor Hours}}$$

Adjustment Battle Rhythm. Rates are established two years ahead during the PPBE budget build and held constant throughout budget year execution; mid-year changes require Office of the Under Secretary of War (Comptroller) (OUSW(C)) approval when earned rates diverge materially from planned costs, and Navy BSOs typically review SLR accuracy monthly/quarterly to recommend any out-of-cycle adjustments [27].

2.2.6 NWCF v.s. Mission-Funded (Appropriation) Commands

WHICH BILLETS AT NWCF ARE MISSION-FUNDED. Military administration and leadership billets (CO/XO/Admin) are funded by **Military Personnel (MILPERS)** and treated as mission-funded within NWCF; certain command/HQ oversight billets may also be OMN funded by policy, so verify locally. These costs are not recovered in stabilized rates. Others that are assigned to NWCF will bill hours towards work and the NWCF will reimburse MILPERS [18].

2.2.7 Standing Up or Modifying a NWCF Business Area

1. **Business Case Development:** Sponsor (e.g., ASN(RD&A) or SYSCOM) prepares analytical justification showing workload, demand signal, and ability to operate on a revolving basis without violating purpose/time/amount [27].
2. **DON Approval Chain:** Secretary of the Navy (SECNAV) (delegated to Assistant Secretary of the Navy (Financial Management and Comptroller) (ASN(FM&C))) endorses the concept and forwards to OUSW(C) while coordinating with Navy Comptroller to align PPBE exhibits [27].
3. **OUSW(C)/OMB Review:** DoW Chief Financial Officer (CFO) validates cash requirements, rate methodology, and capital plan, then seeks OMB alignment for the PB [27].
4. **Congressional Notification:** Congress must be notified (and, when required, explicitly authorize in appropriations or authorizations) before execution; 10 U.S.C. § 2208 [26] restricts creation of new WCFs without legislative awareness [27].

5. **Implementation:** Once approved, the new business area is issued a NWCF business unit code, begins PPBE rate build two years out, and transitions legacy appropriated accounts via opening balance adjustments [27].

2.2.8 NWCF Cycle of Operations (Order to Cash)

1. **Customer order.** IGT 7600A or project order is received and *accepted*, creating the customer's obligation.
2. **Work in process.** Labor and material are applied; costs accumulate; billing events are scheduled per percent complete or delivery.
3. **Revenue recognition and billing.** The NWCF recognizes revenue and bills via IPAC (IGT) or a commercial invoice if authorized.
4. **Collection (cash).** Treasury IPAC/electronic funds transfer posts; NWCF cash increases; NOR/AOR update through the period close [18, 20].

2.2.9 Operating During a Continuing Resolution

Customer Funding Limits. Appropriated customers remain bound by prior-year obligation rates and anti-deficiency constraints; NWCF orders cannot exceed apportioned CR amounts until an appropriations act is passed [28].

Cash Cushion. Existing NWCF corpus allows Warfare Centers to keep executing accepted orders (labor continues, suppliers paid) even if reimbursements lag, provided cash balances stay within the Financial Management Regulation (FMR)'s upper/lower operating limits [27].

No New Starts. CR guidance prohibits new start projects, major capital investments, or rate changes absent explicit exception; NWCF managers defer new workloads that would obligate customer funds beyond CR allowances [28].

Rate Discipline. Stabilized rates remain frozen; only emergency OUSW(C)-approved rate adjustments may occur, so BSOs focus on expense control to avoid large NOR swings during the CR period [27].

2.2.10 Why Use a Warfare Center (Organic)

Things PMOs/industry can't do. Inherently governmental TA warrants, certification authority, certain safety releases; highly classified or nuclear workspaces.

Things they shouldn't do. Independent test/assessment, spec adjudication, tech baseline control, blue-&-gold separation to avoid Organizational Conflict of Interest (OCI).

Things they won't do. Sustainment engineering at scale, depot-level organic repair, fleet distance support.

2.3 Congressional Enactment

- [3] *SECNAVINST 5000.2G: Defense Acquisition System and JCIDS Implementation in the Department of the Navy* SECNAVINST 5000.2G, *Defense Acquisition System and JCIDS Implementation in the Department of the Navy*, SECNAVINST 5000.2G, Instruction, Department of the Navy, Apr. 8, 2022. Accessed: 2025-09-29. [Online]. Available: <https://www.secnav.navy.mil/doni/SECNAVInstructions/05000-05999/5000.2G.pdf>
- [16] “3.1.1. PPBE (Planning, Programming, Budgeting, and Execution)” Engineering Duty Officer School, “3.1.1. PPBE (Planning, Programming, Budgeting, and Execution),” Engineering Duty Officer (EDO) School, Coursebook Module 3.1.1, version 5.2, Mar. 26, 2025, CIN A-4N-0021/-0034.
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- [22] 31 U.S.C. § 1535: Economy Act 31 U.S.C. § 1535, Economy Act, 2018. [Online]. Available: <https://uscode.house.gov/>
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- [24] 31 U.S.C. § 1502(a): Bona Fide Needs Rule 31 U.S.C. § 1502(a), Bona Fide Needs Rule, 2018. [Online]. Available: <https://uscode.house.gov/>
- [25] 31 U.S.C. § 1341; § 1517: Anti-Deficiency Act 31 U.S.C. § 1341; § 1517, Anti-Deficiency Act, 2018. [Online]. Available: <https://uscode.house.gov/>
- [26] 10 U.S.C. § 2208: Working-capital funds 10 U.S.C. § 2208, Working-capital funds, 2023. Accessed: 2025-10-12. [Online]. Available: <https://uscode.house.gov/>
- [27] “DoW FMR 7000.14-R, Volume 3, Chapter 19: Defense Working Capital Funds” “DoW FMR 7000.14-R, Volume 3, Chapter 19: Defense Working Capital Funds,” Office of the Under Secretary of War (Comptroller), Jul. 1, 2025. Accessed: 2025-09-29. [Online]. Available: <https://comptroller.defense.gov/fmr/>
- [28] “Continuing Resolutions: Overview of Components and Recent Practices” J. V. Saturno et al., “Continuing Resolutions: Overview of Components and Recent Practices,” Congressional Research Service, R42647, Mar. 27, 2025. Accessed: 2025-09-29. [Online]. Available: <https://crsreports.congress.gov/>
- [29] 10 U.S.C. § 3501: Multiyear Procurement 10 U.S.C. § 3501, Multiyear Procurement, 2023, Title 10 recodification; formerly Title 10 U.S.C. § 2306b. Accessed: 2025-09-29. [Online]. Available: <https://uscode.house.gov/>
- [30] “DFARS 217.172: Multiyear Contracts” “DFARS 217.172: Multiyear Contracts,” Regulation, 2025. Accessed: 2025-09-29. [Online]. Available: <https://www.acquisition.gov/dfars/part-217>
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- [34] “Adaptive Acquisition Framework” Defense Acquisition University. “Adaptive Acquisition Framework,” Accessed: 2025-09-29. [Online]. Available: <https://aaf.dau.edu/>

2.3.1 Foundations and core definitions

Power of the purse. U.S. Constitution, Article I.

Section 8.

“The Congress shall have power to... provide for the Common Defense... and general welfare...”

Section 9.

“No money shall be drawn from Treasury, but in consequence of Appropriations made by law...”

Key terms. Budget Authority (BA) (legal authority to incur obligations), Total Obligational Authority (TOA), obligations, outlays, Legislative Proposal (LEGPROP).

Authorization v.s. Appropriation. National Defense Authorization Act (NDAA) authorizes programs/policy; appropriations provide BA.

2.3.2 Who drafts what

Authorization. House Armed Services Committee (HASC) /Senate Armed Services Committee (SASC).

Appropriations. House Appropriations Committee (HAC) /Senate Appropriations Committee (SAC).

Budget Resolution. House Budget Committee (HBC) /Senate Budget Committee (SBC) (sets topline aggregates and 302 allocations; not a law).

Independent analysis/oversight. Congressional Budget Office (CBO) (scoring), Congressional Research Service (CRS) (research), Government Accountability Office (GAO) (audits/oversight).

2.3.3 Regular-order timeline

Figure 2.6 shows the timeline from the PB → Appropriation Bill with approximate timelines assuming no CR.

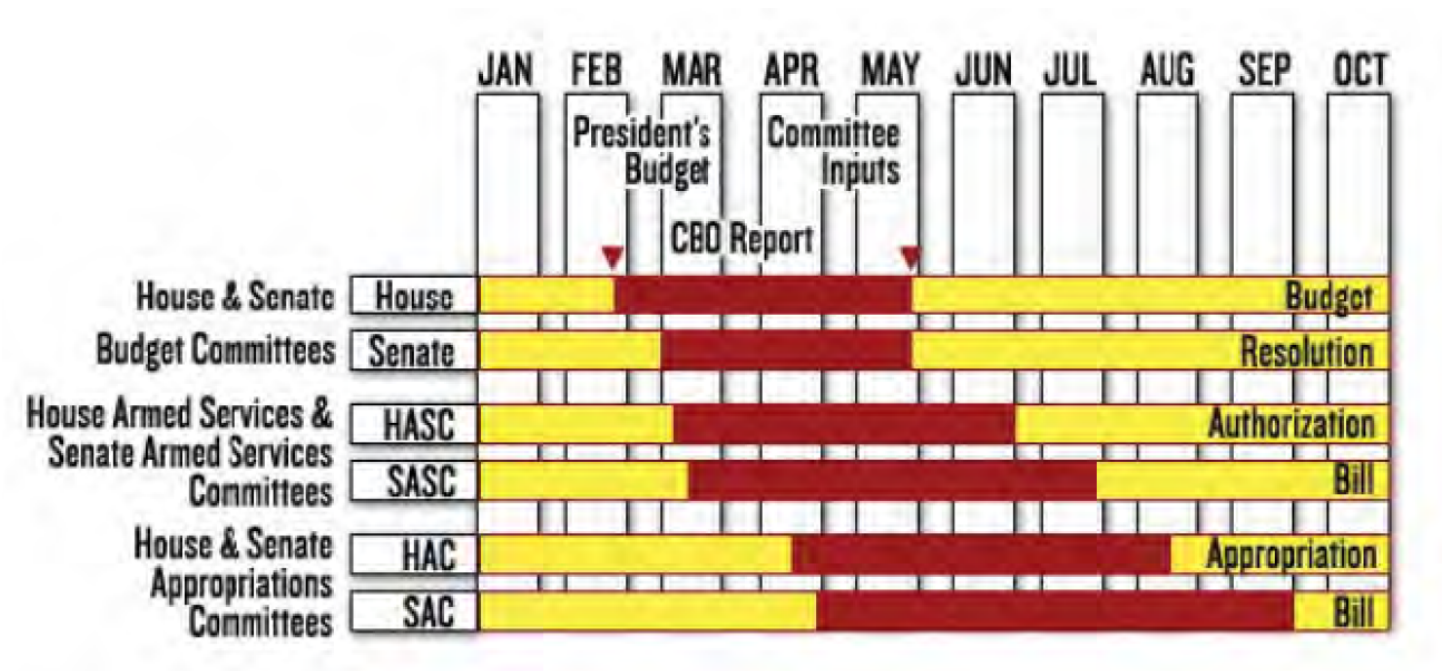


Figure 2.6. Congressional Enactment Timeline. Source: 3.1.2. Congressional Enactment, 2025 [17].

2.3.4 The 12 regular appropriations bills

- Defense.** Department of War appropriations.
- Military Construction and Veterans Affairs.** Military construction, housing, and Department of Veterans Affairs programs.
- Energy and Water.** Department of Energy, Army Corps of Engineers, and related infrastructure programs.
- Homeland Security.** Department of Homeland Security operations and components.
- Interior and Environment.** Department of the Interior and Environmental Protection Agency activities.
- Labor, Health and Human Services, Education.** Departments of Labor, Health and Human Services, and Education, plus related agencies.
- Legislative Branch.** U.S. Congress and supporting legislative branch agencies.
- Financial Services and General Government.** Treasury, judiciary, Small Business Administration, and other independent agencies.
- Transportation and Housing and Urban Development.** Departments of Transportation and Housing and Urban Development.
- Commerce, Justice, Science.** Departments of Commerce and Justice plus science agencies such as National Aeronautics and Space Administration (NASA) and the National Science Foundation.

State and Foreign Operations. Department of State, U.S. Agency for International Development, and foreign assistance programs.

Agriculture and Food and Drug Administration (FDA). Department of Agriculture and Food and Drug Administration activities.

Note

The bills that EDO's care about the most are Defense, Military Construction (MILCON) & Department of Veterans Affairs (VA) and Energy & Water. This because those bills directly affect funding for our acquisitions.

Hint

These bills are sometimes packaged into minibuses or a single omnibus appropriations act.

2.3.5 Continuing Resolution

CONTINUING RESOLUTION RULES (BOARD ONE-LINER). *No new starts*; rate/quantity changes generally constrained unless specified; execution limited by OMB apportionment, CR anomalies, and any authorities Congress adds in the joint explanatory statement [28].

2.3.6 Sequestration

Automatic, across-the-board reductions if statutory caps/triggers are breached; applied by account unless modified by law.

2.4 Program Funding and Execution

- [3] *SECNAVINST 5000.2G: Defense Acquisition System and JCIDS Implementation in the Department of the Navy* SECNAVINST 5000.2G, *Defense Acquisition System and JCIDS Implementation in the Department of the Navy*, SECNAVINST 5000.2G, Instruction, Department of the Navy, Apr. 8, 2022. Accessed: 2025-09-29. [Online]. Available: <https://www.secnav.navy.mil/doni/SECNAVInstructions/05000-05999/5000.2G.pdf>
- [16] "3.1.1. PPBE (Planning, Programming, Budgeting, and Execution)" Engineering Duty Officer School, "3.1.1. PPBE (Planning, Programming, Budgeting, and Execution)," Engineering Duty Officer (EDO) School, Coursebook Module 3.1.1, version 5.2, Mar. 26, 2025, CIN A-4N-0021/-0034.
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2.4.1 BA, commitment, obligation, expenditure, outlay

1. *BA* available (legal authority).
2. *Commitment* (administrative reservation).
3. *Obligation* (legal liability via contract/order).
4. *Expenditure* (payment recorded).
5. *Outlay* (cash disbursed/Treasury).

Source references [18, 20].

2.4.2 Appropriation categories, scope, obligation windows

Table VI shows the Colors of Money with their spending categories and obligation windows. Figure 2.5 from § 2.1.7 visually shows when each category is Current, Expired, or Cancelled.

TABLE VII: APPROPRIATION CATEGORIES, SCOPE, OBLIGATION WINDOWS

Category	Typical Scope	Obligation Window
RDT&E	Science & technology; development; test & evaluation	2 years
OPN ^a	End items/modernization and spares	3 years
OMN	Operations, maintenance, training, minor mods	1 year
MILCON	Facilities construction and real property	5 years
MILPERS	Personnel pay	1 year

^a SCN is a specific procurement with a 5-year obligation window.
Source: DoW Financial Management Regulation 7000.14—R, Volume 3, 2025 [18]

2.4.3 Funding policies

- Annual OMN/MILPERS.** Fund only bona fide needs of the fiscal year.
- RDT&E.** Fund work as it occurs by fiscal year.
- Full-funding. (Procurement/MILCON).** Fund the total usable end-item at award.¹

Common techniques/exceptions: Advance Procurement (long-lead items), Multiyear Procurement, Economic Order Quantity, Cost-to-Complete when provided in law.

EXCEPTIONS TO THE FULL FUNDING POLICY

Below is the list of the key exceptions to the Full Funding Policy, what they allow, and the must-know rules/approvals (Table VII shows full details).

TABLE VIII: KEY EXCEPTIONS TO THE FULL FUNDING POLICY: WHAT THEY ALLOW AND WHAT TO REMEMBER

Exception	Typical Appropriation	What it allows	Must-know rules / approvals
MYP	APN, OPN, SCN	Single contract covering multiple fiscal years of buys to achieve predictable demand and unit-cost savings; may include EOQ purchases common across years.	Requires specific congressional authorization; stable requirements/design; credible savings; low technical risk; term generally up to 5 years; plan/budget any cancellation liability.

Continued on next page

¹Exceptions to Full funding are select SCN programs (e.g., CVNs, SSBNs, and Landing Helicopter Docks (LHDs)/Landing Helicopter Assaults (LHAs)) with Congressional approval.

TABLE VIII: KEY EXCEPTIONS TO THE FULL FUNDING POLICY: WHAT THEY ALLOW AND WHAT TO REMEMBER (Continued)

Exception	Typical Appropriation	What it allows	Must-know rules / approvals
Advance Procurement (incl. LLTM/EOQ)	APN, OPN, SCN	Limited <i>early</i> funding (often 1–2 years before the main “buy” year) for long-lead components or EOQ lots to protect schedule or achieve price breaks.	Narrow and part-specific; must be justified in budget docs; EOQ commonly tied to MYP (or explicit block authority); does <i>not</i> equal full funding of the end item.
BB	Primarily SCN	A single contract buying multiple ships (or end items) across fiscal years to gain tooling/learning-curve efficiencies.	Requires explicit congressional authorization; funds still obligated year-by-year; not under the MYP statute, so oversight mechanics differ.
CTC	SCN	Additional funds in a later FY to finish an item (e.g., ship) when matured estimates exceed prior appropriations.	Completes original approved scope (not added capability); appears as a distinct request/justification; common in long-duration shipbuilding.
RCOH for CVN	SCN (with prior Advance Procurement)	Mid-life refueling and major overhaul funded incrementally over multiple years; early procurement of nuclear fuel/critical components.	Recognized exception due to size/complexity; incremental SCN with advance procurement years ahead of the principal execution.

Note:

- (i) Economic Order Quantity (EOQ) purchases are commonly approved within Multiyear Procurement (MYP) or other explicit multi-year constructs;
- (ii) Block Buy resembles MYP in intent (savings) but uses program-specific authorization;
- (iii) Plan for cancellation liability under MYP when required.

Source: DoW Financial Management Regulation 7000.14—R, Volume 3, 10 U.S.C. § 3501, DFARS 217.172: Multiyear Contracts, 2025, 2023, 2025 [18, 29, 30]

2.4.4 Navy Sustainment Funding: TYCOM v.s. Fleet (O&S)

Who holds O&M,N. Type Commander (TYCOM) leaders (*TYCOMs*; e.g., Commander, Naval Surface Forces (COMNAVSURFOR), Commander, Naval Air Forces (COMNAVAIRFOR), Commander, Submarine Forces (COMSUBFOR)) are the primary executors of OMN for unit readiness: flying hours, steaming days, intermediate-level maintenance, training, and afloat/ashore support.

Fleet Commanders. U.S. Pacific Fleet (PACFLT) and U.S. Fleet Forces (USFF) set operational priorities and readiness goals, adjudicate contingencies, and may centrally manage certain Fleet-wide initiatives; they influence TYCOM allocations but most execution occurs at the TYCOM.

Program Office (SYSCOM/PEO). Funds in-service engineering, tech data, Diminishing Manufacturing Sources, and modernization using Other Procurement, Navy (OPN)/Aircraft Procurement, Navy (APN) (end items/kits) and RDT&E/OMN (installation, trials) as appropriate; manages sustainment Indefinite-Delivery, Indefinite-Quantities (IDIQs)/Performance-Based Logistics (PBL) with color-of-money compliance.

Who pays what. OPN/APN: hardware/software upgrade kits, initial spares; OMN: installs, fleet introduction/training, minor mods, repair parts consumption; RDT&E: development/test of engineering changes. Naval Supply Systems Command (NAVSUP)/NWCF activities recover costs via rates for supply/repair.

Board cue. In operations and support, TYCOMs execute most OMN; Program Managers fund modernization and in-service support from OPN/APN/RDT&E. Align funding lines with the WBS and the Colors-of-Money rules to avoid Anti-Deficiency Act risk.

Quick information regarding exceptions to the full funding policy:

MYP. Up to 5 years; stable design; verifiable savings; cancellation liability planned.

Advance Procurement Long-Lead Time Material (LLTM)/EOQ. Focused early buys for schedule/savings; not full funding.

Block Buy (BB). Congressionally authorized multi-ship buys; obligate by Fiscal Year (FY); not the MYP statute.

Cost-to-Complete (CTC). Complete original scope when estimates mature upward; separate SCN request.

Refueling and Complex Overhaul (RCOH). Planned midlife overhaul; incremental SCN with advance procurement of cores/critical material.

2.4.5 Which money when (by acquisition phase)

Table VIII shows the nominal types of appropriation used for Materiel Solution Analysis (MSA), Technology Maturation and Risk Reduction (TMRR), Engineering and Manufacturing Development (EMD), Production and Deployment Phase (P&D), and Operations and Support Phase (O&S)

TABLE IX: APPROPRIATION TYPES PER PHASE

Phase	Typical appropriation(s)
MSA	RDT&E (analyses, prototyping); small OMN for studies
TMRR	RDT&E (risk reduction, development, test)
EMD	RDT&E (develop/build/test/qualify); selected long-lead/AP/EOQ in Procurement when authorized
PD	Procurement (e.g., OPN); limited RDT&E for fixes
OS	OMN sustainment; spares via Procurement as applicable

2.4.6 MILCON v.s. O&M sustainment v.s. OPN modernization

Rule of thumb: Real property construction ⇒ MILCON; keeping systems running (repair/overhaul, services) ⇒ OMN; new capability/end-item upgrades ⇒ Procurement (e.g., OPN).

2.4.7 Field Activity Financial Management (NWCF quick hits)

Goal. NWCF activities price to *break even over time*; in-year result is **NOR**, cumulative is **AOR**.

SLR. Recover direct labor + overhead + G&A; shocks flow to NOR/AOR and are managed over time.

Carryover. Unfilled orders at FY end; subject to limits/management.

IGT via G-Invoicing. Treasury-mandated platform for intra-gov orders/settlement; apportionment/availability rules still govern customer funds.

2.5 Cost Elements, Estimates, and Learning Curves

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2.5.1 Direct v.s. Indirect Costs

- Direct.** Traceable to a final cost objective (e.g., technician hours, end-item material).
- Indirect.** Pooled and allocated (overhead, General and Administrative (G&A), facilities, depreciation). See the Federal Acquisition Regulation (FAR) cost principles [31].

2.5.2 Program Cost Terms (Know the Stack)

- Flyaway Cost.** Recurring production cost of an aircraft/weapon delivered (excludes non-recurring tooling, spares, support).
- Weapon System Cost.** Flyaway plus support equipment, training, data, and initial spares.
- Procurement Cost.** Weapon system cost plus peculiar support and MILCON as applicable.
- Life-Cycle Cost (LCC).** All costs from concept through disposal (RDT&E, procurement, O&S, disposal).
- Total Ownership Cost (TOC).** LCC plus broader infrastructure/indirect enterprise costs.

Note

Use DON/DoW cost handbook definitions; be consistent across documents. [3, 32].

Figure 2.7 shows the LCC from above and how they relate to one another. Note how each cost builds to the next as different types of expenditures are added.

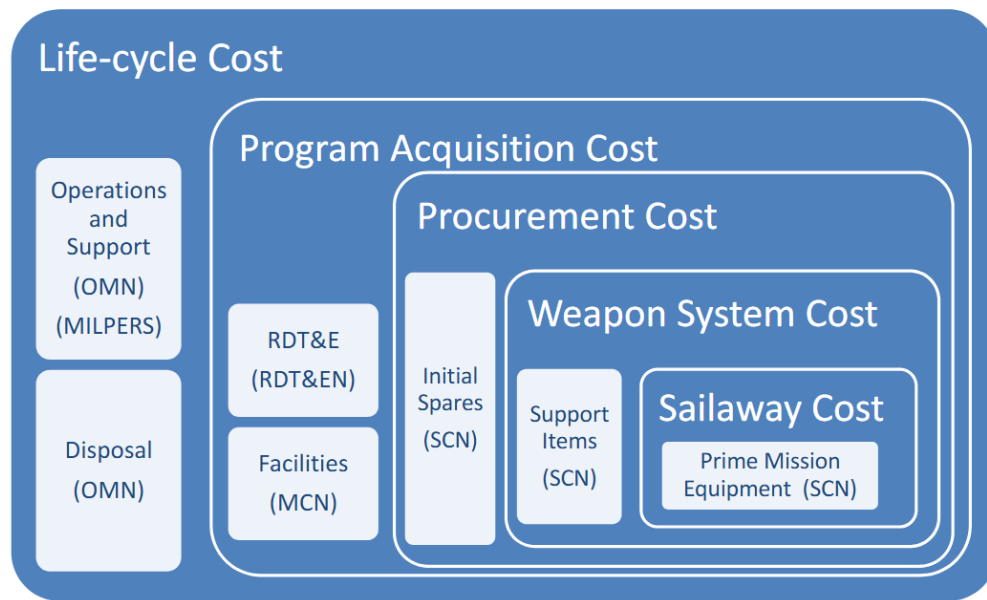


Figure 2.7. Composition of Life Cycle Costs. Source: 3.1.6. Cost Estimating, 2024 [33].

2.5.3 Why the LCCE Matters

Establishes the affordability baseline, informs Acquisition Program Baseline (APB), drives color-of-money phasing, and underpins “will-cost / should-cost” targets [32]. Figure 2.8 shows the different types of funding that will be used in different phases of the Major Capability Acquisition (MCA). Most of the LCC will be from the O&S phase.

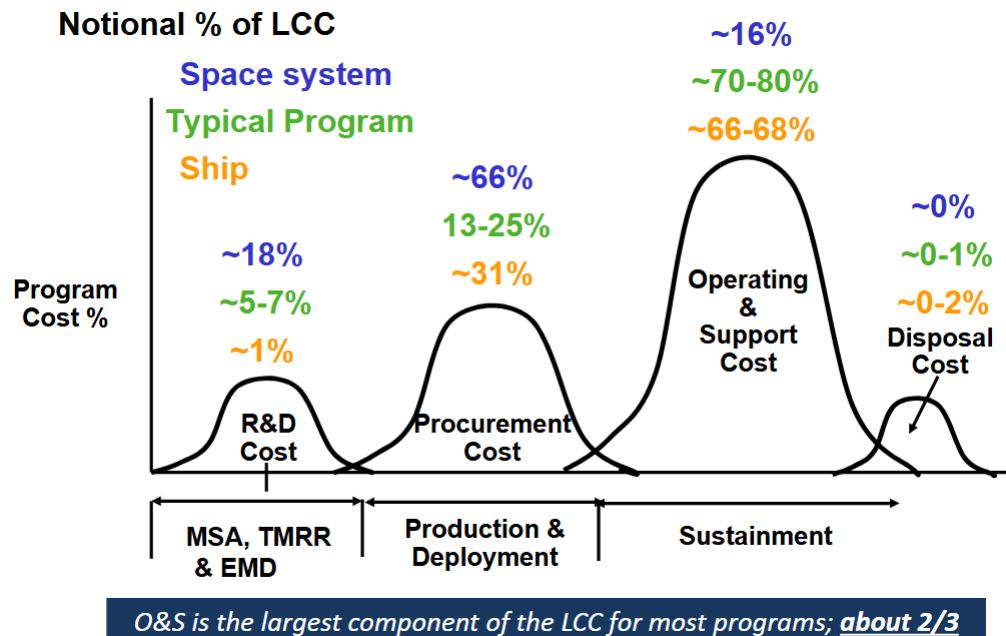


Figure 2.8. LCC over time. The colors of money used are shown for each phase including percent of total costs. Source: 3.1.6. Cost Estimating, 2024 [33]

2.5.4 Estimating Methods (Types, When, and Why)

Analogy (top-down). *When:* Earliest stages (ICD/Materiel Development Decision (MDD), pre- Analysis of Alternatives (AoA)) when design is immature and only analogous programs or components exist. *Why:* Fast, defensible scoping by scaling from a close comparator using adjustment factors (size, weight, speed, complexity). Good for bounding affordability and informing trades before detailed data exists.

Parametric (Cost Estimating Relationship (CER)-based). *When:* AoA through TMRR when historical data supports CERs (e.g., weight, Source Lines of Codes (SLOCs), thrust, kW). *Why:* Enables rapid sensitivity analysis and trade-space exploration with statistical confidence intervals; supports independent cross-checks and early budget phasing.

Engineering (bottom-up). *When:* EMD and production planning when Bill of Materials (BoMs), routing, labor standards, and quotes are available. *Why:* Most traceable and auditable; ties to technical baseline and schedule networks; supports Integrated Baseline Reviews (IBRs) and negotiation objectives.

Actuals / Extrapolation. *When:* From late TMRR into production when pilots/Low-Rate Initial Production (LRIP) actuals exist and learning effects are visible. *Why:* Highest fidelity; captures learning curves and process improvements; anchors estimates to realized performance and Cost and Software Data (CSDR) actuals [34].

TABLE X: COMPARISON OF ESTIMATING METHODS

Type	Description	When Used	Why Used
Analogy	Uses cost of a similar past system, adjusted for differences	Early concept/refinement (Pre-Milestone A/B)	Quick estimate when data limited
Parametric	Uses statistical cost estimating relationships (CERs) like cost vs. weight or power	Milestone B/C; early design	Captures cost drivers using quantifiable parameters
Engineering (Bottom-Up)	Builds from detailed WBS—materials, labor, overhead	Post-CDR, Low-Rate Initial Production	High accuracy for mature designs
Actual Cost / Extrapolation	Uses historical/production actuals (learning curves)	Full-Rate Production / O&S	Most accurate; supports should-cost and sustainment budgeting

2.5.5 Products and Players

Cost Analysis Requirements Description (CARD). Technical/schedule baseline for independent estimates.

Independent Cost Estimate (ICE). Component Acquisition Executive (CAE) or OSD CAPE for Acquisition Category (ACAT) I; Program Office Estimate (POE).

CSDR. Contractor cost & software data reporting for actuals/CERs.

APB/Selected Acquisition Report (SAR)/Defense Acquisition Executive Summary (DAES). Baselines and reporting [32].

2.5.6 Learning Curve (Production Efficiency)

If unit time/cost follows an $s\%$ slope, doubling quantity reduces unit cost by $s\%$. Typical aerospace slopes: 85–90% early, flattening as processes mature. Plan lots and spares buys accordingly [34].

2.5.7 Cost Escalation (Inflation Indices)

Escalate base-year costs to then-year using approved DoW inflation indices; separate real growth from price growth; align phasing to obligation windows [18].

2.5.8 Will-Cost / Should-Cost

Will-Cost. Conservative reference estimate used for budgeting and margin planning.

Should-Cost. Management targets that drive efficiencies (e.g., lean events, value engineering, rate negotiations) [32].

3 Contracting and Solicitation

3.1 Intro to Contracting Fundamentals

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3.1.1 Why the Navy Uses Contracts

- Contracts create a legally enforceable relationship between the Government and industry partners, defining rights and responsibilities for each party.
- Written terms provide structure for changes via agreed conditions, protecting both the Navy and the Contractor throughout acquisition actions.
- Binding agreements ensure accountability for cost, schedule, and performance outcomes that cannot be achieved through informal arrangements.

3.1.2 Essential Elements of a Binding Contract

Mutual Assent. Requires an offer and acceptance with a meeting of the minds on all material terms such as scope, price, quantity, and delivery.

Consideration. Both sides exchange value (e.g., payment, performance, schedule relief) that courts will evaluate for adequacy in Government contracting.

Capacity. Parties must be legally competent to enter the agreement; lack of capacity (e.g., incapacity, lack of authority) undermines enforceability.

Lawful Purpose. The contract must pursue a legal objective; agreements for unlawful acts are void.

3.1.3 Government Relationship to Contractors

Transactional. Each party enters with distinct incentives (contractor profit versus government outcomes); avoid unauthorized commitments.

Professional. Both teams must understand the contract and collaborate to meet requirements.

Collaborative. Contractors bring technical expertise while programmatic decision-making remains inherently governmental.

Constrained. Interactions are bound by ethics rules, conflicts-of-interest standards, and contract clauses.

3.1.4 FAR System and NAVSEA Overlays

FAR. Government-wide acquisition regulation establishing policy, procedures, and contract clauses [31].

Defense Federal Acquisition Regulation Supplement (DFARS). DoW-level supplement tailoring FAR provisions for defense-unique requirements, including competition, source selection, and industrial base policy [35].

Navy Marine Corps Acquisition Regulation Supplement (NMCARS). DON supplement that adds Navy-specific directives such as approval thresholds, peer reviews, and templates [36].

SUPSHIP Operation Manual (SOM) Chapter 3. NAVSEA guidance for SUPSHIP on contract formation, modification, and surveillance (*Appendix B.1: RDML Jonathan J. Jett-Parmer leads SUPSHIP as NAVSEA Deputy Commander; with RDML Robert J. Dodson (United States Navy Reserve (USNR)) as reserve deputy*) [37].

NAVSEA Source Selection Guide. Standardizes competitive source selection procedures, roles, and documentation for SYSCOM procurements [38].

NAVSEA Contracts Handbook. Practical desk reference covering policy interpretations, clause usage, and best practices for NAVSEA contracting professionals [39].

3.1.5 Types of Contracting Officers (Know All Three)

Procuring Contracting Officer (PCO). Leads acquisition planning through award; signs contracts and bilateral modifications on behalf of the Government.

Administrative Contracting Officer (ACO). Oversees post-award administration, surveillance, and contractor performance; typically assigned via Defense Contract Management Agency (DCMA) or NAVSEA field offices.

Termination Contracting Officer (TCO). Manages partial or complete contract terminations, settlement proposals, and equitable adjustments.

All contracting officers must hold a warrant that delineates dollar and authority limits; only a warranted Contracting Officer (KO) can obligate the United States [31].

3.1.6 PM and KO Partnership

PM. Accountable to the Milestone Decision Authority (MDA) for cost, schedule, and performance; integrates technical authority, requirements, and budget execution across the lifecycle.

KO. Provides acquisition strategy execution expertise, ensures compliance with statute/regulation, and is responsible for contract integrity and enforceability.

Board Cue. PMs lead program outcomes; KOs safeguard the contracting instrument. Neither can assume the other's authorities, so coordination before solicitations, negotiations, or modifications is mandatory.

3.1.7 Competition in Contracting Act Requirements

Full and Open Competition. Default posture: all responsible sources may compete under the FAR [31, 40].

Full and Open After Exclusion of Sources. Permits set-asides (e.g., small business, 8(a)) or alternate-source strategies when justified by the FAR [31].

Approval for Other than Full and Open. Requires documented justification and senior approval per FAR Subpart 6.3 [31, Subpart 6.3].

SEVEN EXCEPTIONS TO FULL AND OPEN COMPETITION (MEMORIZE)

1. Only one responsible source will satisfy agency requirements (see FAR [31, § 6.302-1]).
2. Unusual and compelling urgency (see FAR [31, § 6.302-2]).
3. Industrial mobilization; engineering, developmental, or research capability (see FAR [31, § 6.302-3]).
4. International agreement (see FAR [31, § 6.302-4]).
5. Authorized or required by statute (see FAR [31, § 6.302-5]).
6. National security (see FAR [31, § 6.302-6]).
7. Public interest (see FAR [31, § 6.302-7]).

Hint

Be ready to cite an example scenario for each exception and the approval level required.

3.1.8 Responsiveness, Responsibility, and Key Determinations

Responsiveness (sealed bidding). Bid must conform to all material terms of the Invitation for Bids; nonconforming bids are rejected without discussion per the FAR [31].

Responsibility. Prospective contractor must possess adequate resources, schedule compliance, performance record, integrity, and necessary systems to receive award (FAR [31, § 9.104]).

3.1.9 J&A v.s. D&F

Justification and Approval (J&A). Documents the rationale for other-than-full-and-open competition, identifies the chosen statutory exception, and records approval by the appropriate official. Must be posted to <https://sam.gov> after award with required redactions [31].

Determination and Findings (D&F). Formal determination that specific conditions are satisfied before taking an action (e.g., use of special contract types, multiyear contracting); states the findings that support the determination [31].

Note

At the NRO, documents and rationale are posted on the low and high-side Acquisition Research Center (ARC). We are required to post other-full-and-open competition for five days on the ARC to allow opportunity for other contractors to bid.

3.1.10 Who Signs D&Fs (and When)

General Rule. FAR [31, § 1.704] requires the contracting officer to sign D&Fs when the action is within their delegated authority, unless a higher approval level is specified elsewhere in the regulation or delegation memo.

Head Contracting Activity (HCA). FAR [31, §§ 16.603-3, 16.504(c)(1)(ii)(D)] reserve approval for actions such as issuing a letter contract or awarding a single-award task/delivery IDIQ expected to exceed \$100M; DON HCAs may redelegate no lower than a flag/Senior Executive [36, § 5201.707].

SAE. Multiyear contracting, extraordinary contractual relief, or other actions identified in FAR [31, § 17.105-1] and DFARS [35, § 217.172] require a D&F signed by the Service SAE (ASN(RD&A) for the Navy, who is also the DON Senior Procurement Executive) or a specifically delegated official.

Document Content. Every D&F must cite the specific statutory/regulatory authority, describe supporting facts, and state the determination in clear language; expiration dates and any required follow-on reviews must also be included per FAR [31, § 1.707] and NMCARS [36, § 5201.707] guidance.

3.2 Solicitation Preparation

- [31] “Federal Acquisition Regulation (FAR)” “Federal Acquisition Regulation (FAR),” Regulation, 2025. Accessed: 2025-09-29. [Online]. Available: <https://www.acquisition.gov/far>
- [32] *DoDI 5000.85: Major Capability Acquisition* DoDI 5000.85, Major Capability Acquisition, DoDI 5000.85, Instruction, Change(s) as applicable, Department of War, Aug. 6, 2020. Accessed: 2025-09-29. [Online]. Available: <https://www.esd.whs.mil/Directives/issuances/dodissuances/>
- [35] “Defense Federal Acquisition Regulation Supplement (DFARS)” “Defense Federal Acquisition Regulation Supplement (DFARS),” Regulation, 2025. Accessed: 2025-10-12. [Online]. Available: <https://www.acquisition.gov/dfars>
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- [40] 10 U.S.C. § 3201: Competition requirements 10 U.S.C. § 3201, Competition requirements, 2023. Accessed: 2025-10-12. [Online]. Available: <https://uscode.house.gov/>
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- [43] 10 U.S.C. § 4022: Follow-on production contracts or transactions 10 U.S.C. § 4022, Follow-on production contracts or transactions, 2023. Accessed: 2025-10-12. [Online]. Available: <https://uscode.house.gov/>
- [44] 15 U.S.C. § 638: Small Business Innovation Research program 15 U.S.C. § 638, Small Business Innovation Research program, 2023. Accessed: 2025-10-12. [Online]. Available: <https://uscode.house.gov/>

[45] “3.2.4. Cost and Price Evaluation”Engineering Duty Officer School, “3.2.4. Cost and Price Evaluation,” Engineering Duty Officer (EDO) School, Coursebook Module 3.2.4, version 5.0, Dec. 18, 2024, CIN A-4N-0021/-0034.

[46] “3.6.1. Introduction to EVM”Engineering Duty Officer School, “3.6.1. Introduction to EVM,” Engineering Duty Officer (EDO) School, Coursebook Module 3.6.1, version 5.2, Mar. 26, 2025, CIN A-4N-0021/-0034.

[47] “3.6.2. EVM Data Analysis”Engineering Duty Officer School, “3.6.2. EVM Data Analysis,” Engineering Duty Officer (EDO) School, Coursebook Module 3.6.2, version 5.2, Mar. 26, 2025, CIN A-4N-0021/-0034.

3.2.1 Summary

Uniform Contract Format (UCF) scaffolding. Tailor the UCF (Sections A through M) so Section C defines the requirement, Section L tells offerors how to respond, and Section M mirrors evaluation factors and the stated basis of award per the FAR [31, 41].

Release readiness. Do not release the Request for Proposal (RFP) until the acquisition plan/strategy is approved, funds are certified, legal review is complete, synopsis rules are satisfied, and the source-selection teams are chartered per FAR/NMCARS rules [31, 36, 41].

Communication discipline. Request for Informations (RFIs), draft RFPs, industry days, and Q&A all flow through the KO; once released, clarifications must be shared with all offerors through written amendments or controlled exchanges in accordance with the FAR [31, 41].

3.2.2 Practitioner Steps

1. Finalize the requirement package (spec /Statement of Work (SOW), Contract Data Requirements Lists (CDRLs), Independent Government Cost Estimate (IGCE), market research report) with IPT concurrence and Small Business coordination documentation per the FAR [31, 41].
2. Structure Section L instructions: volumes, page limits, submission medium, proposal due date, classification controls, and FAR [31, § 5.203] synopsis timing.
3. Align Section M evaluation factors, subfactors, and relative importance statements with the basis of award (Lowest Price Technically Acceptable (LPTA) v.s. trade-off) and rating methodology approved by the Source Selection Authority (SSA), KO, legal, and cost/price community consistent with the FAR [31, 38].
4. Verify pre-release approvals and postings: acquisition plan or strategy, Determination of Acquisition Strategy (as applicable), certified funds, and <https://www.SAM.gov> (System for Award Management (SAM)) synopsis [36, 41].
5. Manage amendments and records after release: capture bidder Q&A, issue conformed RFPs, adjust milestones when requirements shift, and file every action in the official contract record per the FAR and activity guidance [31, 39].

3.2.3 Checklist

- Acquisition plan/strategy signed; conformed requirements package appended per NMCARS [36].

- Funding document (Purchase Request (PR), project order, or MIPR) signed and funds certified for the planned obligation per the FAR [31].
- Market research report and Small Business Coordination Record approved per the FAR [31].
- Source Selection Plan endorsed by SSA, legal, and cost/price; nondisclosure/OCI statements executed for SSA, Source Selection Advisory Council (SSAC), Source Selection Evaluation Board (SSEB), and advisors [38].
- Section L instructions mirror Section M factors, include proposal structure, late proposal policy, and submission portal guidance per the FAR [31].
- <https://www.SAM.gov> synopsis posted (15 days before closing for most supplies/services; 30 days for R&D unless justified) in accordance with the FAR [31].
- Amendment template/change log prepared; routing matrix established for expedited approvals per NAVSEA guidance [39].

3.2.4 Navy Overlays and Tailoring

NMCARS overlays. Follow [36, § 5205.303/5205.301] for synopsis exceptions and NAVSEA contracting notices for Contract Requirement Package checklists and release approvals [36, 39].

Technical data. Coordinate with TA on [35, § 252.227] clauses, distribution statements, and CDRL tailoring to protect Navy equities.

Shipbuilding tailoring. For availabilities, modernization, or planning efforts, ensure Section C and the WBS matches the avail sequence and integrates warfare-center technical direction and installation responsibilities [41].

3.2.5 Industry Engagement & Communications

Pre-release. RFIs, sources-sought notices, draft RFPs, and industry days are vetted through the KO; responses shared with all potential offerors to avoid unequal access [31].

Post-release. Clarifications and Q&A must be issued as amendments for all offerors; exchanges stay within FAR [31, § 15.201] boundaries until competitive range is established [39].

Small business focus. Coordinate with the Small Business Professional and Competition Advocate on set-aside decisions, subcontracting plan reviews, and mitigation of potential limitations on subcontracting per the FAR [31].

3.2.6 Documentation & Approvals to Capture

- Signed Source Selection Plan and SSA/SSAC/SSEB appointment letters with confidentiality and OCI certifications [38].
- Competition Advocate memoranda where narrowed sources or synopsis deviations are requested under the FAR/NMCARS rules [31, 36].

- Legal sufficiency memorandum for the conformed solicitation (Sections C, H, L, and M) retained in the contract file per NAVSEA contracting guidance [39].
- KO release memorandum noting solicitation number, issue date, amendment log, and proposal due date/time [41].

3.2.7 Competitive Method Selection: Sealed Bidding v.s. Negotiated Procurement

Sealed bidding (FAR Part 14). Preferred when requirements are well-defined, award will be based on price alone, discussions are not needed, and the Government expects to receive more than one responsive bid. It uses public opening, responsiveness determinations, and fixed-price awards [31, Part 14].

Negotiated procurement (FAR Part 15). Used when the Government needs to evaluate technical approach or past performance, intends to conduct discussions, or when requirements or pricing demand flexibility. Negotiated acquisitions support best-value trade-offs and cost-reimbursement vehicles [31, Part 15].

When to choose. If the team can describe performance in terms of precise specifications with minimal risk and expects no need for exchanges, sealed bidding streamlines award. When innovation, risk, or integration complexity requires subjective evaluation, or when schedule/budget changes are likely, negotiated procurement is mandatory under the FAR [31, 38].

3.2.8 Contract Type Landscape and Risk Allocation

Fixed-price family. Contractor accepts the cost risk; Government locks price once requirements are stable. Includes:

- **Firm-Fixed-Price (FFP):** Used for mature, low-risk requirements, often in Production and Deployment [31, 32].
- **Fixed-Price Incentive (Firm Target) (FPIF):** Shares cost variance through a negotiated share ratio; encourages cost control while keeping final price bounded by a ceiling. Used during EMD or LRIP when cost risk remains but production discipline is needed [31, 32].
- **Fixed-Price with Economic Price Adjustment (FPEPA):** Adds indexed adjustments for volatile commodities or long-lead items [31].
- **Fixed-Price Redetermination (FPR):** Re-prices after defined milestones when initial uncertainty is expected to retire [31].

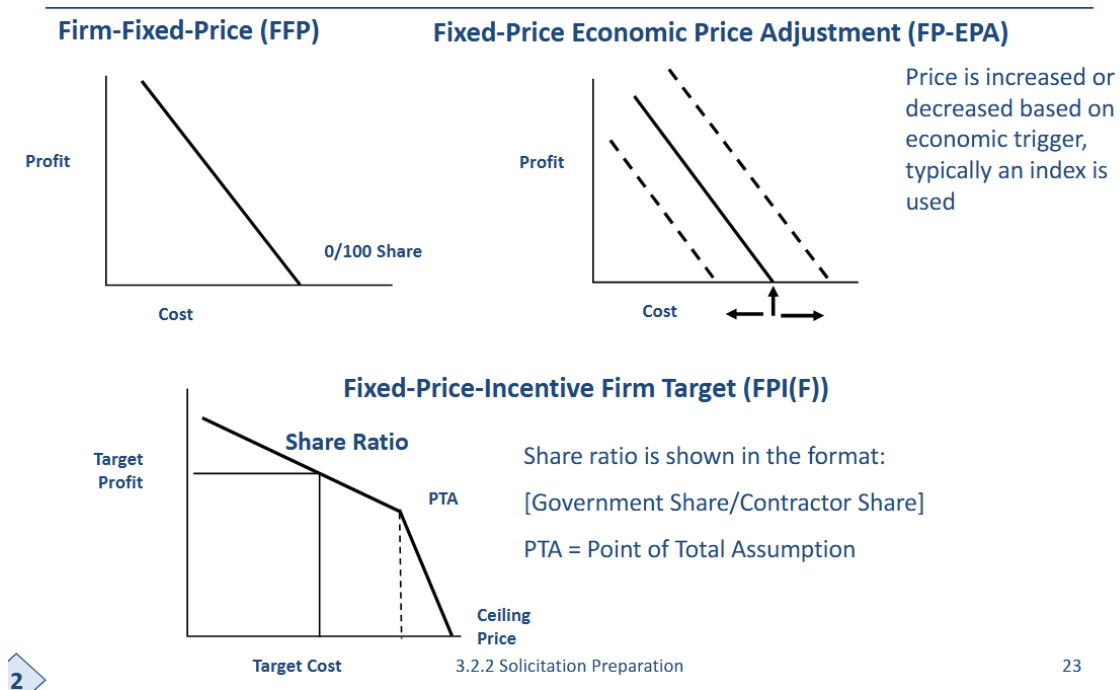


Figure 3.1. Fixed-price incentive geometry (cost, profit, PTA).

Cost-reimbursement family. Government carries more cost risk; contractor must deliver the best effort. Includes:

- **Cost-Plus-Fixed-Fee (CPFF):** Used in early R&D or prototyping (MSA/TMRR) when effort is exploratory [31, 32].
- **Cost-Plus-Incentive-Fee (CPIF):** Shares cost variance similar to FPIF but without a price ceiling; encourages efficiency while acknowledging uncertain cost baseline [31].
- **Cost-Plus-Award-Fee (CPAF):** Adds subjective award-fee evaluation for mission effectiveness or management performance that cannot be captured in objective metrics [31].

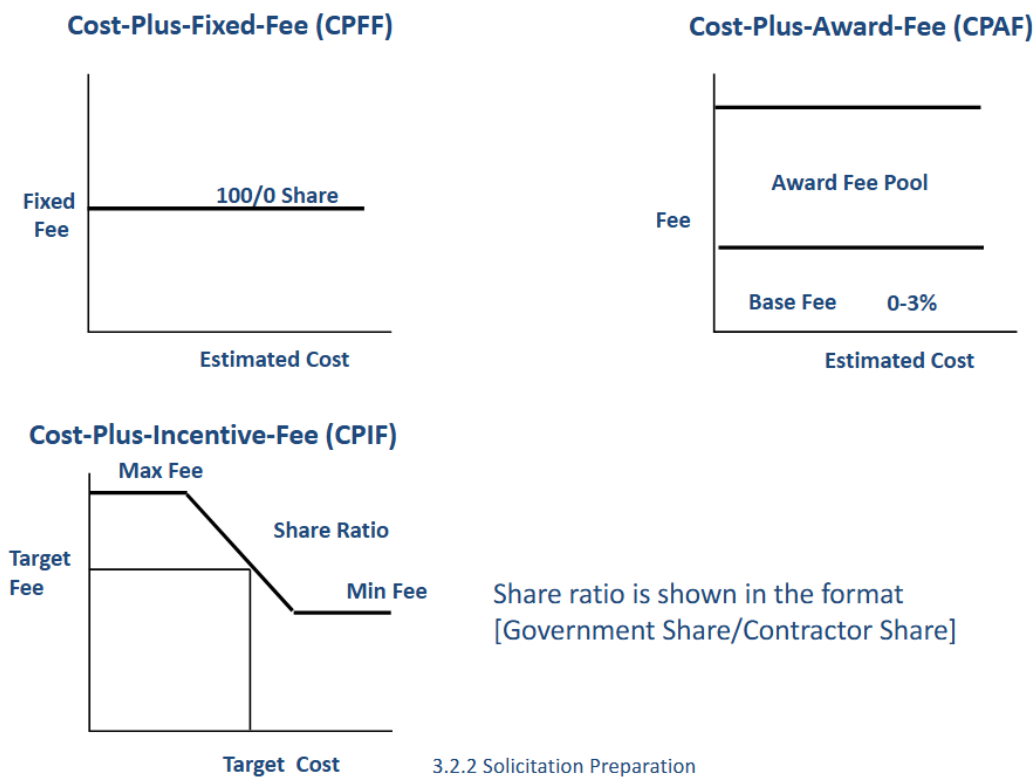


Figure 3.2. Cost-reimbursement contract risk/fee characteristics. Source: 3.2.2. Solicitation Preparation, 2025 [41].

Award v.s. incentive fees. Incentive fees (cost or performance) are calculated from predetermined formulas tied to measurable outcomes (cost, schedule, technical) under the FAR; award fees are earned through periodic board evaluations against tailored factors and may be unearned if performance is merely satisfactory [31, 39].

Award-fee governance. The Award Fee Determining Official (often the PM or SSA designee) chairs an Award Fee Board, uses the approved plan, and issues a determination memo. Fees are paid after each period and must be commensurate with value delivered; scores below the threshold earn zero dollars for that segment [31, 38].

MCA phase alignment. MSA and TMRR generally use CPFF or CPAF because of design uncertainty; EMD can transition to CPIF or FPIF as risk retires; Production and Deployment favors FFP/FPIF; Operations and Support relies on FFP or FFP Level of Effort (LOE) and can incorporate sustainment IDIQs for depot work [32].

MSA. Analytical trades and early prototypes benefit from CPFF flexibility as requirement scope evolves.

TMRR. CPIF/CPAF maintain incentives while the Government still absorbs major technical risk during competitive prototyping.

EMD. FPIF or CPIF balance contractor motivation and cost control once the design stabilizes, with ceilings to protect the Government.

PD. FFP/FPIF dominate because production baselines and learning curves are known; contractor should own execution risk.

OS. Sustainment leverages FFP, FFP LOE, or IDIQs task orders to manage repeatable workloads and retain competition for modernization.

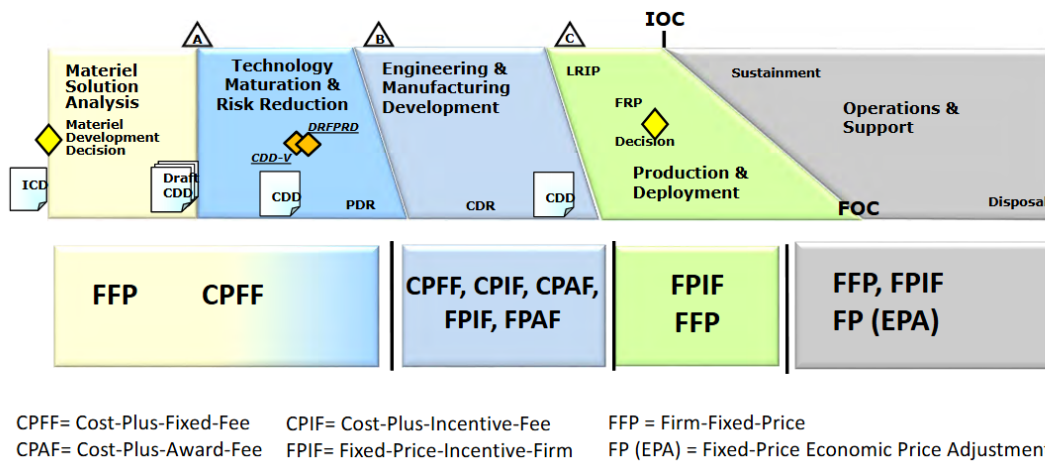


Figure 3.3. Contract-type emphasis across the Major Capability Acquisition phases.

3.2.9 Fixed-Price Incentive (Firm Target) Mechanics

Share ratio. Defines how cost overruns or underruns are split between Government and contractor (e.g., 70/30 means Government absorbs 70% of variance, contractor 30%) per the FAR [31].

Point of Total Assumption (PTA). The cost point beyond which every additional dollar of overrun is borne entirely by the contractor:

$$PTA = \frac{\text{Ceiling Price} - \text{Target Cost}}{\text{Government Share Ratio}} + \text{Target Cost}.$$

Example. Target cost \$100M, target profit \$10M (target price \$110M), ceiling price \$120M, share ratio 70/30. If actual cost is \$112M:

$$\text{Contract price} = \$110\text{M} + 0.3 \times (\$100\text{M} - \$112\text{M}) = \$106.4\text{M}.$$

The PTA occurs at \$113.3M; beyond that, profit erodes dollar-for-dollar until exhausted [31, 39].

3.2.10 Indefinite & Flexible Contract Forms

IDIQ. Indefinite-Delivery, Indefinite-Quantity contracts provide flexibility in quantity and timing with task/delivery orders; suited for recurring installations, sustainment, or services [31].

Time-and-Materials (T&M)/Labor-Hour (LH). Hybrid vehicles when it is not possible to estimate effort precisely; require ceiling price, surveillance, and justification because Government assumes cost risk on labor hours [31].

Unfixed Contract Actions (UCAs). Letter contracts or other actions authorized before all terms are settled (e.g., urgent ship repair). Must be definitized within the regulatory timeline, with strict management for obligation limits and profit [31, 35, 36].

3.2.11 Debriefings and Protests (Forums and Timelines)

Debriefings. Post-award (and certain pre-award) debriefings are conducted per the FAR [31, § 15.506]. For DoD enhanced debriefings, the question period extends the debrief conclusion for protest timeliness purposes; verify current Class Deviation guidance.

Protest forums. (i) Agency-level to the contracting activity (KO/Head of Contracting Activity) [31, § 33.103]; (ii) GAO bid protests [31, § 33.104]; (iii) U.S. Court of Federal Claims (Court of Federal Claims (COFC)) under 28 U.S.C. § 1491(b).

Who handles them. Agency: procuring activity and counsel; GAO: GAO decides, agency produces the record and implements any stay; COFC: Department of Justice litigates, COFC judge adjudicates and may grant injunctive relief.

Timelines (common rules). Pre-award challenges to solicitation terms must be filed before the proposal due date. For post-award protests at GAO: file within 10 calendar days of when the basis is known, or within 5 business days after debriefing conclusion to obtain an automatic stay of performance under Competition in Contracting Act (CICA). Agency-level protest timeliness follows similar 10-day knowledge rules in [31, § 33.103]. COFC has no fixed day-limit but relief is equitable; prompt filing is expected.

3.2.12 Why Cost-Plus After Low-Rate Initial Production? (PD and O&S Exceptions)

High residual technical risk. Post-LRIP, major engineering change proposals, tech refresh, or obsolescence-driven redesign can reintroduce uncertainty better handled under CPIF/CPAF than FFP.

Unstable scope or emergent defects. Corrective actions discovered during Initial Operational Test and Evaluation (IOT&E)/early fielding may require iterative investigation and rework where a best-effort *cost-reimbursable* vehicle protects schedule and access to talent.

Sustainment stand-up. Initial depot activation, organic transition, or complex PBL standing up new measurements and data flows can warrant short-duration CPFF/CPAF with incentives while processes stabilize.

Data rights/technical baseline gaps. Where the Government lacks detailed technical data to price FFP confidently, a temporary cost-plus arrangement with aggressive CDRL deliverables can bridge to FFP.

Governance. Treat as exceptions with a plan to transition to FFP/FPIF once uncertainty retires; document rationale in the Acquisition Strategy and obtain approvals consistent with [31, 35].

3.2.13 Simplified Acquisition and Small Business Programs

Simplified Acquisition Threshold (SAT). Generally \$250,000 (higher in contingency or overseas circumstances). Benefits include streamlined competition, quotation-based awards, and mandatory small-business reservations above the micro-purchase threshold [31].

Small Business Administration (SBA) programs. FAR Part 19 [31, Part 19] implements 8(a), Historically Underutilized Business Zone (HUBZone), Service-Disabled Veteran-Owned Small Business (SDVOSB), and Women-Owned Small Business (WOSB)

programs. The contracting officer must consider set-asides, sole-source thresholds, and subcontracting plans to meet DON goals.

3.2.14 Alternative Acquisition Authorities

Other Transaction (OT) agreements. Authorized by 10 U.S.C. § 4021 (prototype) and § 4022 (follow-on production) to rapidly engage nontraditional performers, provided cost share or significant participation criteria are met [42, 43].

Small Business Innovation Research (SBIR). 15 U.S.C. § 638 establishes Phase I (feasibility), Phase II (prototype), and Phase III (commercialization/production). SBIR awards leverage R&D funds to deliver Navy-unique technology with data rights protections for five years [44].

3.2.15 Technical Data Rights Overview

Unlimited rights. Government may use, disclose, and authorize others without restriction (typically for data developed exclusively at Government expense) per the DFARS [35].

Government purpose rights. Government may use or authorize contractors for Government purposes for five years, after which rights convert to unlimited. Applies to mixed-funding developments under the DFARS [35].

Limited/Restricted rights. Contractor retains control; Government use is constrained to internal purposes (limited for noncommercial technical data, restricted for commercial computer software). Negotiated licenses or specifically negotiated licenses can expand access per the DFARS [35].

3.2.16 Uniform Contract Format Essentials

Sections A through H (contract). Cover Standard Form 33, supplies/services, packaging, inspection, delivery, special contract requirements, clauses, and attachments per the FAR UCF [31].

Sections I through K (legal/representations). Incorporate FAR/DFARS clauses, representations, certifications, and instructions for completion [31, 35].

Sections L and M (solicitation/Evaluation). Instructions to offerors, proposal structure, page limits, and evaluation criteria that must remain synchronized to any change to factors or relative order requires an amendment under the FAR [31, 38].

3.3 Cost and Price Evaluation

[31] “Federal Acquisition Regulation (FAR)” “Federal Acquisition Regulation (FAR),” Regulation, 2025. Accessed: 2025-09-29. [Online]. Available: <https://www.acquisition.gov/far>

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- [42] 10 U.S.C. § 4021: Prototype projects: other transaction authority 10 U.S.C. § 4021, Prototype projects: other transaction authority, 2023. Accessed: 2025-10-12. [Online]. Available: <https://uscode.house.gov/>
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- [47] “3.6.2. EVM Data Analysis” Engineering Duty Officer School, “3.6.2. EVM Data Analysis,” Engineering Duty Officer (EDO) School, Coursebook Module 3.6.2, version 5.2, Mar. 26, 2025, CIN A-4N-0021/-0034.

3.3.1 Summary

KO accountability. The KO must determine that every negotiated price is fair and reasonable, integrating price, cost, technical, field-pricing, and risk analysis inputs before award per FAR 15.404-1(a)(1)–(5) [31, § 15.404-1(a)].

Cost v.s. price lens. Use price analysis when certified cost or pricing data are not required, and escalate to cost analysis when element-by-element scrutiny is needed to support the prenegotiation objective under the FAR [31, §§ 15.404-1(b), 15.404-1(c)].

Rate discipline. Forward pricing rates and structured profit analysis keep the Government prenegotiation objective current; the objective is always proposed cost plus profit or fee per FAR policy [31, §§ 15.401, 15.404–4, 15.407–3].

Financial health cue. Profitability ratios (Return on Sales (ROS), Return on Assets (ROA)) and cash-flow checks expose responsibility risk and focus Defense Contract Audit Agency (DCAA)/ DCMA field-pricing support requested by the KO [31, § 9.104-1] [45].

3.3.2 Practitioner Steps

1. Baseline the offeror's proposal: confirm certified cost or pricing data requirements, request data other than certified cost or pricing data when warranted, and map contractual requirement traces per the FAR [31, §§ 15.403-1, 15.404-1(a)].
2. Execute price analysis first—compare competition results, historical buys, catalog/commercial data, and independent estimates; pivot to cost analysis when price analysis alone cannot demonstrate reasonableness per the FAR [31, §§ 15.404-1(b), 15.404-1(c)].
3. Decompose direct labor, material, and Other Direct Costs (ODCs); verify indirect pools and allocation bases; and apply the five allowability tests before accepting proposed cost elements [31, §§ 31.201-2, 31.202, 31.203].
4. Establish the profit/fee objective using the agency's structured approach and update forward pricing rates or agreements as necessary to avoid stale factors in negotiations under the FAR [31, §§ 15.404-4, 15.407-3].
5. Pull in field-pricing support early: blend DCAA audit results, DCMA production assessments, and program team technical evaluations into the prenegotiation briefing and contract file [31, § 15.404-1(a)(5)] [31, § 42.101(b)] [35, § 242.302(a)(13)(A)].

3.3.3 Proposed Price, Cost, Profit, and Fee

The breakdown of the proposed price is showing in Figure 3.4 and described below.

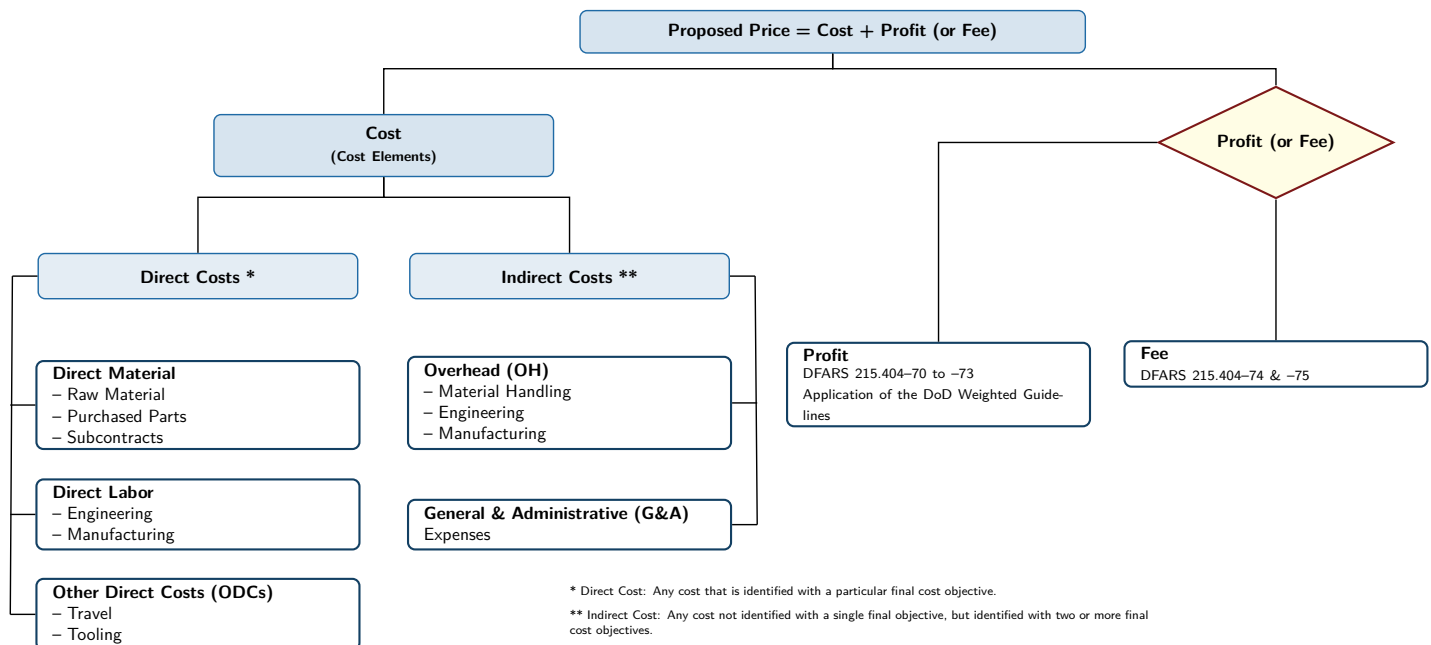


Figure 3.4. Mapping direct v.s. indirect contractor cost elements. Source: 3.2.4. Cost and Price Evaluation, 2024 [45]

Proposed price. The offeror's total price equals estimated cost plus any profit or fee applicable to the contract type per the FAR [31, § 15.401].

Cost. Sum of allowable direct and indirect costs required to deliver the contract—all subject to reasonableness, allocability, and cost-principle limits under the FAR [31, §§ 31.201-2, 31.202, 31.203].

Profit. Negotiated incentive element above allowable cost for fixed-price or incentive contracts, set through structured analysis per the FAR [31, § 15.404-4(a)].

Fee. Fixed remuneration on cost-reimbursement vehicles that does not vary with actual cost (e.g., CPFF fee) [31, § 16.306(a)].

3.3.4 Direct and Indirect Cost Structure

Direct cost pools. Direct labor, materials, and ODCs traceable to a single final cost objective; charged through project-specific accounts backed by bills of material, timecards, travel authorizations, or subcontract quotes [31, § 31.202] [45].

Indirect cost pools. Overhead (manufacturing, engineering) and G&A accrued across multiple objectives, then allocated using consistent bases that reflect benefits received [31, § 31.203].

3.3.5 Other Direct Costs and Cost Allowability

ODCs. Travel, subcontract services, specialized tooling, and similar charges that can be singled out for the contract even though they are not labor or bulk material categories [31, § 31.202] [45].

Cost allowability. A cost is billable only if it is reasonable, allocable, consistent with applicable Cost Accounting Standards (CAS) / Generally Accepted Accounting Principles (GAAP), compliant with contract terms, and within any Subpart 31.2 limitations [31, § 31.201-2].

Note

Slides may refer to “cash allowability”; FAR terminology is *cost* allowability.

3.3.6 Cost and Price Analysis

Price analysis. Examination of the total proposed price without breaking out cost elements using competition, historical buys, or market indicators per the FAR [31, § 15.404-1(b)].

Cost analysis. Evaluation of individual cost elements (direct, indirect, and profit) when price analysis alone cannot demonstrate reasonableness under the FAR [31, § 15.404-1(c)].

Integrated view. Technical analysis, field-pricing support, and risk assessments feed both techniques to substantiate the final fair and reasonable determination per the FAR [31, § 15.404-1(a)(5)].

3.3.7 Forward Pricing Rates and Fully Burdened Labor

Forward Pricing Rate Proposal (FPRP). Contractor-submitted forward pricing rate proposal that lays out projected indirect rates and factors for the pricing period [31, § 15.407-3(a)].

Forward Pricing Rate Agreement (FPRA). Negotiated agreement—often executed by DCMA—that locks indirect rates for one to three years, reducing repetitive audits and speeding negotiations [31, § 15.407-3(c)] [45].

Fully burdened rate. Applies direct labor, indirect burdens, and any negotiated profit/fee to a common labor-hour baseline so skill mixes can be compared on an apples-to-apples basis [45].

3.3.8 Indirect Rate Math and Profitability Checks

Apply the standard relationships from the cost principles and the coursebook when validating proposals; they are board favorites for quick-turn computations covering indirect pools, Total Cost Input (TCI), G&A, and profitability metrics such as ROS and ROA [31, §§ 31.201-2, 31.202, 31.203] [45].

$$\text{Indirect Cost Rate} = \frac{\text{Indirect Cost Pool}}{\text{Allocation Base}} \quad (3.1)$$

$$\text{TCI} = \text{Direct Cost} + \text{Overhead Cost} \quad (3.2)$$

$$\text{G\&A Cost} = \text{G\&A Rate} \times \text{TCI} \quad (3.3)$$

$$\text{Fully Burdened Labor Rate} = \frac{\text{Direct Labor Cost} + \text{Indirect Costs} + \text{Profit/Fee}}{\text{Direct Labor Hours}} \quad (3.4)$$

$$\text{ROS} = \frac{\text{Operating Profit}}{\text{Net Sales}} \quad (3.5)$$

$$\text{ROA} = \frac{\text{Net Income}}{\text{Total Assets}} \quad (3.6)$$

Track ROS and ROA against industry benchmarks, cash-on-hand, and debt loads to detect responsibility concerns or unsustainable buy-ins before negotiations [45].

3.3.9 Oversight and Field Pricing Support

DCMA production insight. Delegated contract administration offices deliver manufacturing surveillance, schedule risk analysis, and forward-pricing coordination; DFARS highlights that payment authority stays with the buying command even when DCMA supports rates [35, § 242.302(a)(13)(A)] [45].

DCAA audit coverage. DCAA is the responsible Government audit agency for most contractors, providing proposal adequacy reviews, incurred cost audits, and financial capability analyses [31, § 42.101(b)].

Integrated negotiation record. Document how DCMA, DCAA, technical, and program inputs influenced the prenegotiation objective and the final price reasonableness determination [31, § 15.404-1(a)(5)] [45].

3.3.10 Checklist

- File both price and cost analysis results (including technical inputs) showing how the fair-and-reasonable price was determined per the FAR [31, § 15.404-1].
- Verify each significant cost element passes the five allowability tests and that allocation bases match current practice under the FAR [31, §§ 31.201-2, 31.203].

- Confirm the current FPRA/FPRP is in the file or document why legacy rates remain valid [31, § 15.407-3].
- Capture DCAA audit opinions, DCMA rate guidance, and profitability ratio trends in the prenegotiation briefing to evidence responsibility diligence [31, §§ 9.104-1, 42.101(b)] [45].

3.4 Earned Value Management (EVM)

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3.4.1 What EVM is

Earned Value Management (EVM) integrates scope, schedule, and cost to quantify performance by comparing planned work to completed work and associated costs; it is widely used in government and defense programs to keep delivery on time and within budget.

3.4.2 Key Components of EVM

Core cost and schedule metrics establish the earned value performance baseline [46, 47].

Planned Value (PV) / Budgeted Cost of Work Scheduled (BCWS): The budgeted cost for the work scheduled to be completed by a specific date.

Earned Value (EV) / Budgeted Cost of Work Performed (BCWP): The budgeted cost for the work actually completed by a specific date.

Actual Cost (AC) / Actual Cost of Work Performed (ACWP): The actual cost incurred for the work completed by a specific date.

Budget at Completion (BAC): The total budget allocated for the project.

Estimate at Completion (EAC): The forecasted total cost of the project based on current performance.

Cost Performance Index (CPI): A measure of cost efficiency, calculated as $CPI = EV/AC$. A CPI less than 1 indicates a cost overrun.

Schedule Performance Index (SPI): A measure of schedule efficiency, calculated as $SPI = EV/PV$. A SPI less than 1 indicates a schedule delay.

Schedule Variance (SV): The difference between the earned value and the planned value, calculated as $SV = EV - PV$. A negative SV indicates a schedule delay.

Cost Variance (CV): The difference between the earned value and the actual cost, calculated as $CV = EV - AC$. A negative CV indicates a cost overrun.

Info

The first three are equivalent terms; this guide uses the PV, EV, AC forms.

3.4.3 When to Use EVM

EVM is mandated on cost or incentive contracts (and applicable subcontracts) when the program meets the policy thresholds and the effort is discretely measurable [35, 46]. Key decision points for the PM include:

\$100 million and above: Implement the full American National Standards Institute (ANSI)/Electronic Industries Alliance (EIA)-748 standard and ensure the contractor's system is formally validated by DCMA.

\$20 million to \$99.9 million: Implement ANSI/EIA-748 guidelines, with validation required when performance indicates risk.

Additional policy considerations follow [32, 46].

- Applicability determinations must confirm the work scope can be discretely measured before mandating EVM.
- The Milestone Decision Authority may approve EVM on FFP, time-and-materials, or LOE contracts, but such use is discouraged absent clear benefit.
- For efforts below \$20 million, the PM conducts a risk-based cost-benefit analysis to decide whether EVM adds value.
- Contracts that require EVM must also deliver Integrated Program Manager's Data Analysis Report (IPMDAR) data and complete an IBR within six months of award.

3.4.4 EVM Compliance

EVM clauses are flowed in the solicitation and award package (DFARS 252.234-7001 and 252.234-7002) alongside the Contractor Business Systems clause (DFARS 252.242-7005) to anchor validation and surveillance requirements [35]. Compliance expectations follow [32, 46].

- The program office integrates EVM planning into the WBS, maintains the Performance Measurement Baseline, and ensures timely receipt and analysis of IPMDAR submissions.
- DCMA leads EVM system acceptance, validation, and ongoing surveillance against ANSI/EIA-748 and the DoW EVM Implementation Guide; DCAA supports with accounting system audits.
- Service focal points such as Acquisition Data and Analytics and the DON Center for EVM adjudicate applicability determinations and coordinate policy updates.
- SUPSHIP can execute validation and surveillance responsibilities for shipbuilding programs on behalf of DCMA (*Appendix B.1: RDML Jonathan J. Jett-Parmer leads SUPSHIP; RDML Robert J. Dodson (USNR) is the reserve deputy*).

3.4.5 EVM Principal Players

Programmatic, contracting, and oversight roles synchronize to sustain disciplined performance management [46].

PMO: Implements EVM on the contract, ensures solicitations and awards contain the required clauses, and works with the contracting activity to tailor reporting and Integrated Master Schedule (IMS) requirements.

DCMA: Serves as the DoW EVM Executive Agent by validating, accepting, and surveilling contractor systems, maintaining the official acceptance roster, and analyzing IPMDAR and schedule submissions (including the 14-point IMS review) to focus government attention on emerging issues.

DCAA: Conducts accounting system audits, corroborates cost data, and supports DCMA during surveillance events and reviews.

Acquisition Data and Analytics (ADA): Acts as the department-wide focal point for EVM policy, guidance, and competency management; issues interpretations to maintain consistent application across programs.

DON Center for EVM: Coordinates applicability determinations with Deputy Assistant Secretary of the Navy (Acquisition Policy and Budget) (DASN (APB)) and OSD, and serves as the Navy's central point of contact for EVM matters.

SUPSHIP: Performs many DCMA/DCAA surveillance roles for shipbuilding contracts when delegated by the cognizant program office. (*Appendix B.1: See Appendix B.1 for current SUPSHIP flag billet holders*)

3.4.6 PMB

Figure 3.5 shows the characteristic S-curve depicting cumulative PV/BCWS, while Figure 3.6 summarizes the progression from defining the SOW to final Performance Measurement Baseline (PMB) adjustments. PMBs are:

- Scoped, scheduled, and budgeted plans for the authorized work
- Time-phased budgets that align to the master schedule
- The basis for cost and schedule performance management
- Effectively the PV for the entire project

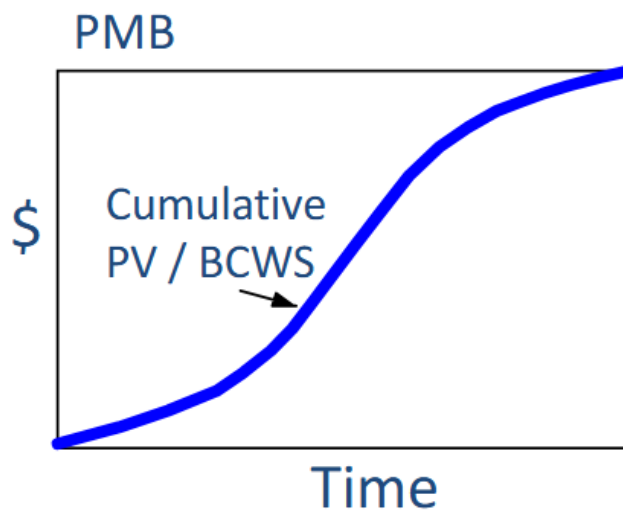


Figure 3.5. PMB Cost v.s. Time chart showing the “S”-curve. Source: 3.6.1. Introduction to EVM, 2025 [46].



Figure 3.6. The development flow for PMB. Source: 3.6.1. Introduction to EVM, 2025 [46].

PMB DEVELOPMENT

- Step 1: *Define all work*. Decompose the effort using the WBS and align it with the organizational structure. Control accounts are the natural management points where Control Account Managers (CAMs) are assigned, and each control account contains work packages and (if needed) planning packages.
- Step 2: *Schedule the work*. Develop the integrated schedule (often visualized with Gantt charts) that sequences the WBS elements, includes key milestones, and forms the backbone of the time-phased budget.
- Step 3: *Budget the work*. Assign time-phased budgets to each work package, establish management reserve for in-scope known unknowns, and confirm that the sum of the control accounts equals the contract budget base.

Changes to the PMB must be formally controlled and documented. Reasons for changes include:

- Contract changes
- Internal replanning
- Formal reprogramming

3.4.7 EVM Reviews and Reports

Formal reviews validate the integrity of EVM data products before and after system acceptance [46].

POST-ACCEPTANCE REVIEW

- Objective: ensure performance data remain accurate after system acceptance and identify any corrective actions required to reaffirm compliance.
- Timing: conducted as needed following system acceptance and prior to the next IBR.
- Led by the Review Director (typically DCMA) with membership tailored to the purpose of the review.
- Culminates in a formal report prepared by the Review Director.

INITIAL COMPLIANCE EVALUATION

- Objective: validate that the Contractor's EVM system description matches actual practice and satisfies the Earned Value Management System (EVMS) criteria.
- Timing: executed prior to the initial IBR (and subsequently only if needed).
- Led by the DCMA Review Director with cross-functional support as required.
- Results documented in a report signed by the Review Director.

INTEGRATED BASELINE REVIEW

- Participants: the contractor, PM and deputy, DCMA/DCAA/SUPSHIP personnel, and relevant technical staff.
- Purpose: joint assessment to verify the realism and accuracy of the PMB, confirm the entire technical scope is captured, and ensure resources and schedules are achievable.
- Timing: conducted within six months of contract award (or major replanning event) in accordance with the DoW EVM Implementation Guide and service policy [46].

INTEGRATED PROGRAM MANAGEMENT DATA ANALYSIS REPORT

This contractually required report delivers cost, schedule, and technical status derived from the Contractor's EVM system, enabling the PMO to identify performance problem areas and emerging risks. It is required on every contract that mandates EVM and must be delivered at least monthly. The data set is tailored for each contract based on risk, size, and integration considerations, but the canonical formats are:

1. WBS (most common)
2. Organizational categories
3. Program Management Baseline
4. Staffing
5. Explanation and Problem Analysis
6. IMS

A Contract Funds Status Report (CSFR) complements the IPMDAR by providing funding (price) information that reconciles to the cost-focused earned value data [46].

PROBLEMS

When using EVM, the following are indications that a problem exists [47].

- Use of management reserves
- Significant revisions to the PMB
- Zero variance
- Sudden change in trends
- Unreasonable Total Cost Performance Index (TCPI)

3.4.8 EVM Data Analysis

Figure 3.7 shows the S-curve representation of the EVM variables; the gaps between the curves are the cost and schedule variances that drive performance assessments [47].

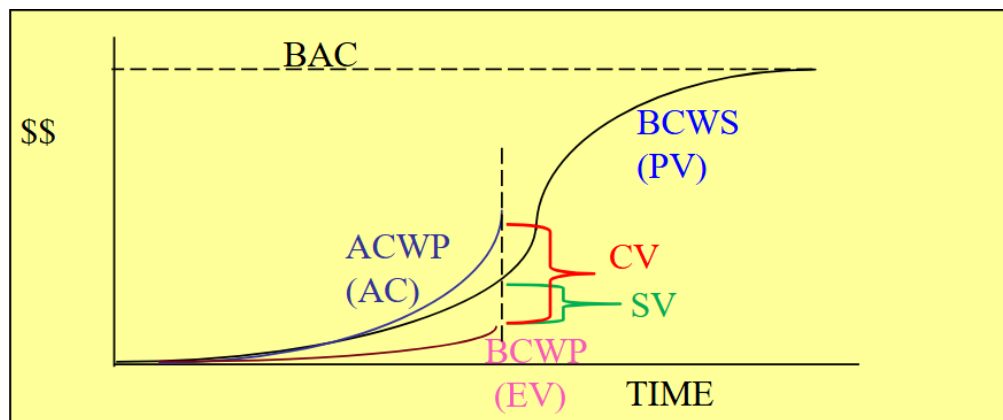


Figure 3.7. EVM chart showing the independent variables and variances. Source: 3.6.2. EVM Data Analysis, 2025 [47]

DATA ANALYSIS STEPS

1. **Get current status:** capture the latest cost and schedule performance using CPI, SPI, and narrative explanations.
2. **Identify trends:** analyze indices and variance trajectories over time to highlight emerging risks and opportunities.
3. **Predict completion:** develop independent EAC and Variance at Complete (VAC) assessments to forecast final outcomes.
4. **Determine management actions:** decide where to apply resources, contract changes, or risk mitigations to recover performance [47].

PERFORMANCE MEASUREMENT TECHNIQUES

Percent Complete: Applies to long-duration tasks lacking interim milestones; progress is reported as the earned value fraction of total budget.

Weighted Milestones: Uses milestone weights for long-duration tasks with clear waypoints, crediting earned value as milestones are completed.

Percent Start/Percent Finish: Suitable for short-duration tasks (less than two reporting periods) with credit apportioned at start and finish.

0/100 Method: Provides earned value only when the task is complete, offering a conservative measure for discrete, short tasks [47].

EQUATIONS

Standard cost and schedule performance calculations include [47].

$$CV = EV - AC$$

$$SV = EV - PV$$

$$CPI = \frac{EV}{AC}$$

$$SPI = \frac{EV}{PV}$$

$$CV\% = \left(\frac{CV}{EV} \right) \times 100$$

$$SV\% = \left(\frac{SV}{PV} \right) \times 100$$

$$\% \text{ Complete} = \left(\frac{EV}{BAC} \right) \times 100$$

$$\% \text{ Spent} = \left(\frac{AC}{BAC} \right) \times 100$$

$$EAC = \frac{BAC}{CPI}$$

$$EAC = AC + \frac{BAC - EV}{\left(\frac{EV}{AC} \right)_{3 \text{ months}}}$$

$$EAC = AC + \frac{BAC - EV}{CPI \times SPI}$$

$$EAC = AC + \frac{BAC - EV}{0.8 CPI + 0.2 SPI}$$

$$VAC = BAC - EAC$$

$$TCPI = \frac{\text{Work remaining}}{\text{Budget remaining}}$$

$$TCPI_{EAC} = \frac{BAC - EV}{EAC - AC}$$

$$TCPI_{BAC} = \frac{BAC - EV}{BAC - AC}$$

4 Defense Acquisitions

4.1 Acquisition Policy

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4.1.1 Summary

Statute drives authority. Title 10 charges the Military Departments to equip the force, while each annual NDAA refreshes acquisition authorities and reporting duties—know the latest delegation trail before advising a board [48].

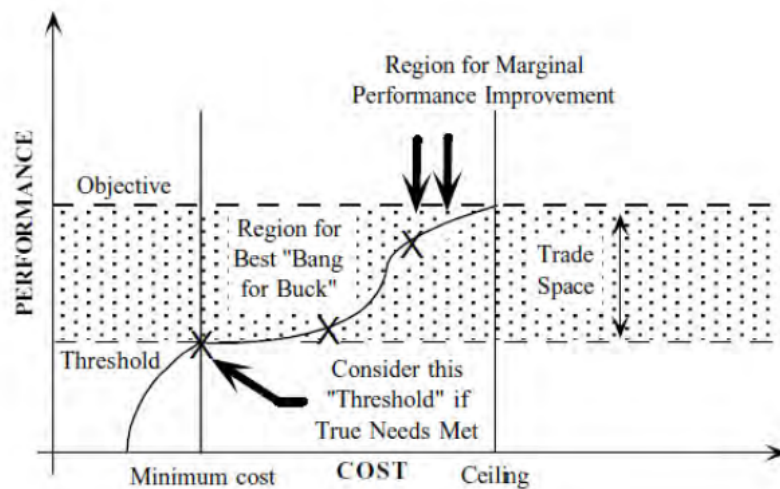
Policy = DoW 5000 series. DoDD 5000.01 establishes acquisition principles and senior roles; DoDI 5000.02 operationalizes the Adaptive Acquisition Framework (AAF) pathways; DoDI 5000.85 prescribes MCA execution details [32, 49, 50].

Regulation stack. FAR > DFARS > NMCARS translates statute and policy into enforceable contracting rules; Secretary of the Navy Instruction (SECNAVINST) 5000.2G tailors the AAF for DON programs and makes ASN(RD&A) the SAE [3, 31, 35].

The main drivers that a PM must balance is cost, schedule, and performance as shown in Figure 4.1. The trade space for performance versus cost is shown in Figure 4.2, which shows the relation of performance and cost with the threshold and objective requirements.



Figure 4.1. The cost, schedule, and performance triangle that must be balanced and traded. Source: 3.3.1. Acquisition Policy and Players, 2025 [48].



Graphic from Selected Topics in Assurance Related Technologies (START)
Volume 5, Number 2

*The “knee of the curve” helps answer this question:
“Is the additional performance worth the additional investment?”*

3.3.1 Acquisition Policy and Players

Figure 4.2. Trade space of performance v.s. cost showing the range of viable options with the region of best “value.” Source: 3.3.1. Acquisition Policy and Players, 2025 [48].

4.1.2 Practitioner Steps (Board Prep Focus)

1. Confirm statutory authority and delegation: identify the Defense Acquisition Executive (DAE) (Under Secretary of War for Acquisition and Sustainment (USW(A&S))) or delegated MDA, the SAE, ASN(RD&A), and the resource sponsor accountable for the requirement [3, 48, 49].
2. Select the correct AAF pathway (or hybrid) and align entry/exit criteria, statutory reports, and decision forums (MS A/B/C, MDD, Business Case Analyses (BCAs), Congressional notices) [32, 50].
3. Map contracting rules to the pathway: FAR/DFARS clauses, competition requirements, and Tailored Acquisition Strategy approvals [31, 35, 48].

4. Crosswalk Navy overlays: SECNAVINST 5000.2G, SECNAVINST 5400.15D organization responsibilities, and PEO/DRPM charters for technical authority touchpoints [2, 3].
5. Prepare decision documentation: update the Acquisition Strategy /Systems Engineering Plan (SEP) /Test and Evaluation Master Plan (TEMP), ensure statutory certifications (Clinger-Cohen, PPBE affordability caps) are current, and pre-brief the chain (Program Office → PEO → ASN(RD&A)) [32, 48].

4.1.3 Policy Stack and Authorities

Title 10, U.S.C. Establishes Service responsibilities (Subtitle C for DON organization, Subtitle A Part V for acquisition management) and empowers annual NDAAs to adjust acquisition thresholds or pilot authorities [48].

DoW Directives/Instructions. DoDD 5000.01 sets acquisition principles, governance forums, and senior leader responsibilities; DoDI 5000.02 implements the AAF with six scalable pathways; DoDI 5000.85 gives MCA-specific statutory requirements (Joint Requirements Oversight Council (JROC), cost reporting, baseline control) [32, 49, 50].

Regulations. FAR and DFARS governs all federal contracting; DFARS adds DoW-specific clauses (e.g., earned value, data rights, cybersecurity); NMCARS adds DON policy and ASN(RD&A) approval levels [31, 35, 36].

Service overlays. SECNAVINST 5000.2G tailors milestone documentation, Naval SYSCOM oversight, and Naval Accelerated Acquisition; SECNAVINST 5400.15D assigns PEO and SYSCOM responsibilities for acquisition program execution [2, 3].

Advisory guidance. The Defense Acquisition Guidebook (DAG) captures best practices—it is not directive authority, but boards expect you to know how it informs planning reviews and tailoring memoranda [51].

Info

Boards expect you to quote the controlling document *and* state who owns the decision. Memorize the policy ladder: Statute → Directive → Instruction → Regulation → Service overlay.

4.1.4 Acquisition Players and Decision Forums

DAE. USW(A&S) chairs the Defense Acquisition Board (DAB), is MDA for ACAT ID/Major Automated Information System (legacy ACAT I equivalent) (IAM) programs unless delegated, and approves key acquisition policies [48, 49].

SAE. ASN(RD&A) serves as the DON SAE; appoints PEOs, assigns MDA for ACAT II and below, and ensures Navy acquisition compliance with DoW policy [2, 3].

Chief of Naval Operations/Resource Sponsor. Validates requirements and PPBE resourcing, provides integrated warfare/community priorities to ASN(RD&A) and the PEOs [48].

PEO/DRPM. Executes programs within delegated authorities; maintains acquisition baseline control, briefs ASN(RD&A) and DAB-level forums, and ensures SYSCOM technical authority integration [2, 32].

PM. Accountable for cost, schedule, performance; leads the IPT, maintains statutory certifications, and readies milestone documentation [32, 48].

Governance forums. Milestone Decision Reviews, the DAB, Configuration Steering Boards, Overarching IPTs, and Navy Program Executive Council reviews provide structured oversight and risk adjudication [32, 48, 49].

Note

Edge case: Rapid acquisition authorities (e.g., UCA) compress governance. Ensure delegation letters document any waived statutory certifications before you recommend skipping a DAB or ASN(RD&A) review [50].

4.1.5 Adaptive Acquisition Framework Pathways

The AAF can be visualized in Figure 4.3. The figure shows all six pathways with their purposes and general flow of the acquisition for each.

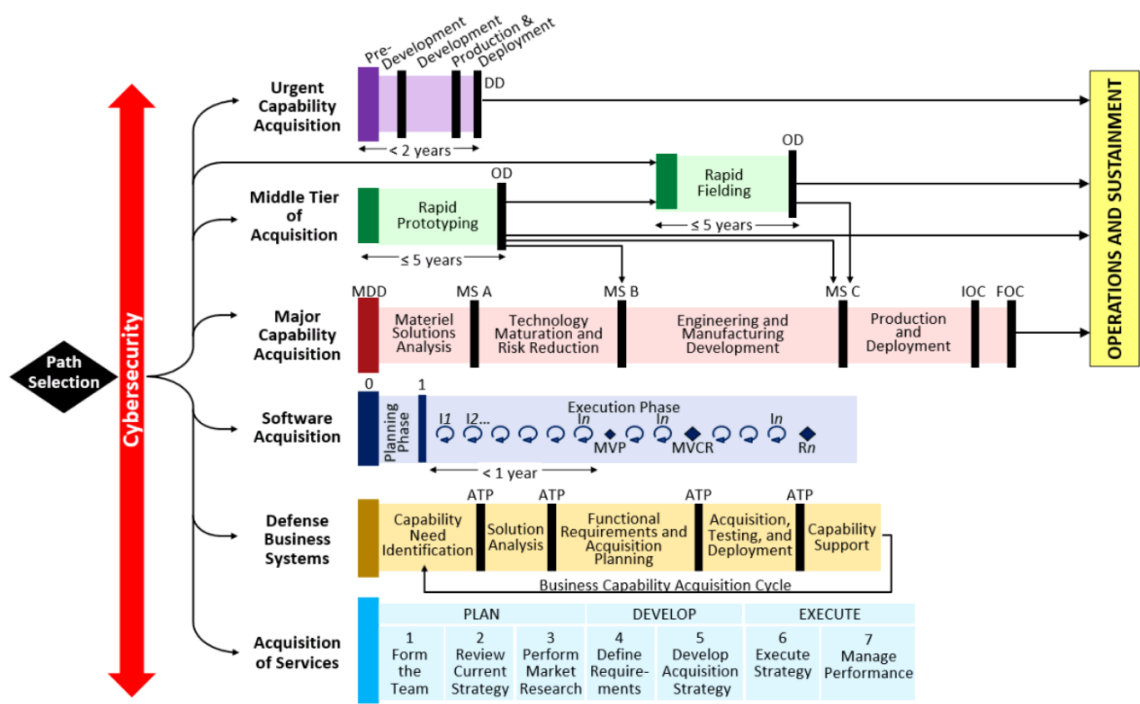


Figure 4.3. Adaptive Acquisition Pathways. Source: Adaptive Acquisition Framework, 2025 [34].

The MCA is the main acquisition pathway that EDOs will operate in. As shown in Figure 4.4, it can be broken down into five phases, four major decision points and three milestones.

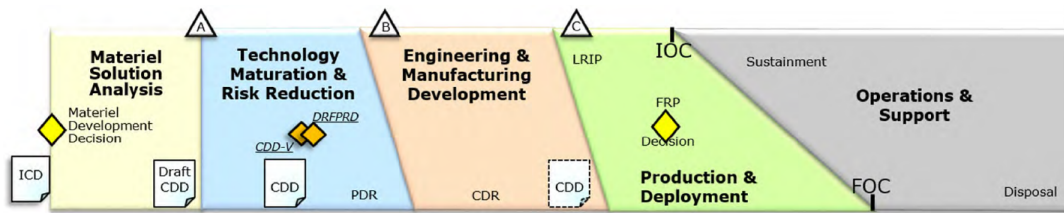


Figure 4.4. Major Capability Acquisition with phases, milestones, and decisions points. Source: 3.3.1. Acquisition Policy and Players, 2025 [48]

PATHWAY PLAYBOOKS

Major Capability Acquisition.

Entry. Default when a validated ICD, PPBE affordability certification, and Title 10 Major Defense Acquisition Program (MDAP) thresholds are triggered; the DAE (USW(A&S)) or delegated MDA authorizes the Material Development Decision [32, 50, 52]. *Major events.* Material Solution Analysis, TMRR, EMD, Production & Deployment, and Operations & Support framed by MS A/B/C, Production Readiness Reviews, Configuration Steering Boards, and Full-Rate Production Decisions [32]. *Outputs.* Acquisition Strategy, SEP, Life-Cycle Sustainment Plan (LCSP), TEMP, SAR/DAES, and when applicable Live Fire and survivability certifications [32, 53, 54]. *What happens.* The Program Manager leads integrated product teams, controls the Acquisition Program Baseline, and coordinates statutory certifications before each milestone package is staffed through the Navy Program Executive Council and the DAB [32]. *Pros.* Maximum statutory confidence, predictable decision forums, and the ability to tailor documentation while retaining congressional transparency [32]. *Cons.* Heavier documentation, longer lead times, and concurrency risks if requirements or funding discipline erodes between milestones [32].

Middle Tier of Acquisition.

Entry. Warfighter sponsor certifies the capability can be prototyped or fielded within 5 years, and the MDA approves a rapid prototyping or rapid fielding strategy in lieu of Joint Capabilities Integration and Development System (JCIDS) milestone paperwork [55]. *Major events.* Competitive prototyping sprints, operational demonstrations, and limited fielding increments documented in the Middle Tier of Acquisition (MTA) Strategy and semiannual execution reports to Under Secretary of War for Research and Engineering (USW(R&E))/USW(A&S) [55]. *Outputs.* MTA Acquisition Strategy, Test Strategy (tailored TEMP), transition plan to either MCA or sustainment funding, and congressional notifications when cost or schedule deviates [55]. *What happens.* Programs lean on OTs, digital engineering, and combined DT/OT events; logistics and cybersecurity artifacts are right-sized but must be ready before transition to production [55]. *Pros.* Compresses documentation, accelerates learning, and protects prototyping budgets from the full milestone stack. *Cons.* Sustainment planning and PPBE wedges still have to be established before transition; poor exit planning strands prototypes without a production home [55].

Software Acquisition Pathway.

Entry. Capability owner commits to iterative Development, Security, and Operations (DevSecOps) delivery, establishes a software factory or enterprise pipeline, and designates a product manager vice a traditional milestone PM [56]. *Major events.* Plan, Execute, and Deliver phases with quarterly (or faster) capability drops, authority-to-operate reciprocity reviews, and portfolio synchronization reviews in lieu of MS A/B/C [56]. *Outputs.* Software Acquisition Strategy, Product Roadmap, backlog burn-down metrics, continuous test/accreditation evidence, and sustainment metrics for cloud hosting costs [56]. *What happens.* The government product owner jointly manages the pipeline with contractors, employs automated testing,

and updates requirements via configuration-controlled roadmaps instead of static CDDs [56]. *Pros.* Continuous user feedback, cyber resiliency baked into each sprint, and the ability to budget in capability releases instead of monolithic blocks [56]. *Cons.* Requires mature pipelines, disciplined product management, and close PPBE coordination to realign color-of-money profiles from hardware-centric norms [56].

Urgent Capability Acquisition.

Entry. A validated Joint Urgent Operational Need (JUON)/Joint Emergent Operational Need (JEON)/Urgent Operational Need (UON) identifies a combatant-commander risk that cannot wait two years; ASN(RD&A) or USW(A&S) approves entry with an affordability determination and sustainment exit criteria [57, 58]. *Major events.* Rapid assessment within days, solution selection inside 30 days, fielding no later than 24 months, and close-out/transition reviews that assign residual sustainment ownership [58]. *Outputs.* Urgent Capability Acquisition Strategy, tailored test/safety releases, congressional notification packages, and a signed disposition plan for residual hardware [58]. *What happens.* The Rapid Acquisition Cell coordinates funding reprogramming, waivers, and senior integration group decisions while the program office executes minimal-but-sufficient contracting and test [58]. *Pros.* Weeks-to-months fielding, ability to leverage existing production lines or Commercial Off-the-Shelf (COTS) solutions. *Cons.* Limited longevity (normally five years or less), intense oversight on waivers, and explicit requirement to transition to another pathway or divest [58].

Defense Business Systems.

Entry. A portfolio exceeds the statutory Defense Business System (DBS) cost thresholds or is designated critical; requires Business Process Reengineering certification, a Functional Sponsor, and alignment to the Business Enterprise Architecture before funds may be released [59]. *Major events.* Capability Needs Statement approval, Functional Requirements Board adjudication, incremental capability releases, and annual Investment Review Board certification to Congress [59]. *Outputs.* Business Case, Functional Strategies, Capability Support Plan, data standards, and interoperability determinations to satisfy Net-Ready Key Performance Parameter (KPP) expectations [59]. *What happens.* Functional leads co-chair the IPT with the Program Manager, enforcing process improvements before automating legacy workflows [59]. *Pros.* Forces portfolio-level prioritization, ties funding to measurable process change. *Cons.* Extra governance (Investment Review Board (IRB), Defense Business Council) and Clinger-Cohen certifications can slow schedules if stakeholders are not aligned early [59].

Defense Acquisition of Services.

Entry. Non-IT services exceeding simplified acquisition thresholds, or any effort designated special interest, must use DoDI 5000.74 governance and appoint a Services Category Manager before solicitation [60]. *Major events.* Service Requirement Review Board, Acquisition Strategy Panel, independent cost estimate for Category I/II services, and annual review of contractor performance metrics [60]. *Outputs.* Services Acquisition Plan, Quality Assurance Surveillance Plan, Performance Work Statement, and documented affordability/cost realism analyses [60]. *What happens.* Multifunctional teams (requirements owner, contracting, legal, comptroller) collaborate early, determine best-fit contract type, and tailor competitive ranges for recurring availabilities or modernization services [60]. *Pros.* Aligns recurring services with category management, encourages outcome-based metrics. *Cons.* Additional reviews for Category I/II services can add lead time if the requirement was scoped as a product effort originally [60].

Info

Pathways can be combined (e.g., prototype under Middle-Tier, transition to MCA, then adopt the Software Pathway for continuous upgrades). Boards expect you to articulate which statutory obligations remain even when tailoring [50].

4.1.6 Warfighting Acquisition System Imperatives (2025)

- Purpose.**
SECWAR redesignated the Defense Acquisition System (DAS) as the Warfighting Acquisition System (WAS) on 7 Nov 2025 to treat acquisition as a warfighting function, make speed-to-field the primary metric, and push decisions to the accountable portfolio leaders [61].
- Portfolio accountability.**
Portfolio Acquisition Executives (PAEs) chair Capability Trade Councils, own integrated portfolio scorecards, and are tasked to deliver minimum viable combat capability in months—not years—while continuously iterating increments; they are expected to waive non-statutory documents, reallocate within portfolio controls, and document trades in decision memoranda [61, 62].
- Mission engineering and experimentation.**
The Mission Engineering and Integration Activity (MEIA) replaces lengthy requirements staffing with rapid kill-chain analysis, cross-Service experiments, and industry demos, so requirement adjustments are anchored in real data before the DAE or PAE commits resources [61, 63].
- Requirements–budget coupling.**
Key Operational Problems (KOPs) and the Requirements and Resourcing Alignment Board (RRAB) ensure every joint problem statement is paired with funding guidance (including Joint Acceleration Reserve (JAR) taps) inside the same PPBE cycle, eliminating the historical gap between validated requirements and appropriated dollars [63].
- Implications for EDOs.**
Expect Capability Trade Council taskers in lieu of Configuration Steering Boards, portfolio scorecards that track cycle time from need to Initial Operational Capability, and requests for mission-thread data packages to feed MEIA campaigns—ships, systems, and industrial-base briefs must now quantify time-based performance alongside cost/schedule [61, 62].

Info

Board cue: explain how you would tailor milestone documentation under the WAS memo while preserving statutory certifications (e.g., Clinger-Cohen, cybersecurity) and how you will capture trade rationales now that Capability Trade Councils replace legacy Configuration Steering Boards [61].

4.1.7 Urgent Operational Needs Governance

JUON.

Combatant commanders submit JUONs to the Joint Rapid Acquisition Cell when an unforeseen operational gap threatens lives or mission success within the current fight; the Joint Staff validates within days, assigns a lead Service, and requires fielding in ≤ 24 months with a plan to transition or terminate after the contingency [57].

JEON.

A JEON covers emerging threats with a slightly longer horizon (2–5 years) but still demands joint prioritization, Functional Capabilities Board staffing, and explicit affordability bounds before the Joint Requirements Oversight Council renders a decision [64].

UON.

Fleet, Type, and numbered Force Commanders elevate UONs per Office of the Chief of Naval Operations Instruction (OPNAVINST) 5000.42E; requests must include the operational vignette, Doctrine, Organization, Training, Materiel, Leadership and Education, Personnel, Facilities, and Policy (DOTMLPF-P) analysis, recommended sponsor, and incremental funding estimate before Gate 1 of the two-pass/seven-gate process [65].

Request content.

Every urgent package documents the risk of not acting, desired operational effect, timeframe to theater, sustainment/retrograde plans, cybersecurity implications, and classification handling; incomplete data delays Joint Staff or OPNAV acceptance [57, 65].

Acquisition handoff.

Validated packages flow to the UON Management Board and then to ASN(RD&A) for alignment with the Urgent Capability Acquisition (UCA) pathway, rapid contracting, and source selection; the MDA records any tailored documentation and must still close on lifecycle ownership before the temporary solution sunsets [58].

4.1.8 Security Cooperation and Capital Alignment (2025 Reforms)

Arms-transfer realignment.

SECWAR shifted the Defense Security Cooperation Agency (DSCA) and the Defense Technology Security Administration (DTSA) under USW(A&S) to unify Foreign Military Sales, Direct Commercial Sales, technology protection, and industrial-base production decisions under one accountable enterprise that can plan exportability features early and keep allied demand synced with U.S. production lines [66].

National Security Capital Forum.

Section 1092 of the FY 2025 NDAA and subsequent SECWAR direction established the National Security Capital Forum (NSCF), chaired by the Office of Strategic Capital, to convene federal and non-federal financiers, monitor capital flows that affect defense production, and flag transactions that threaten supply-chain resilience [67].

EDO relevance.

Program briefs now need to show how exportability, co-production, and investment incentives are built into acquisition strategies because DSCA/DTSA decisions and NSCF findings will increasingly shape schedule risk (supplier solvency, capital availability) and board questions on allied burden-sharing [66, 67].

4.1.9 Acquisition Categories and Delegated Decision Authority

USW(A&S) serves as the DAE; SECNAV designates ASN(RD&A) as both the Navy SAE and CAE, while SECNAVINST 5000.2G and the Warfighting Acquisition Strategy empower PAEs to manage portfolios and request tailored delegation [3, 49, 62].

ACAT I (MDAP).

Programs that meet the MDAP thresholds ($\geq$ \$525M RDT&E or $\geq$ \$3.6B procurement in FY 2020 constant dollars) default to DAE oversight; ACAT ID programs (e.g., *Columbia*-class SSBN) keep USW(A&S) as MDA, whereas ACAT IC efforts (e.g., DDG(X) combat systems refresh) are normally delegated to ASN(RD&A) with reporting to Congress via SAR/DAES [32, 48, 52].

ACAT IA.

Major Automated Information Systems exceeding \$360 M life-cycle cost or designated by Under Secretary of War (Comptroller) (USW(C))/CAE; the Navy SAE is the default MDA and may assign PEO Digital or the relevant PAE to execute under the Software or Business Systems pathways (examples: Navy ERP, Maritime Maintenance IT portfolio) [50, 59].

ACAT II.

Programs below MDAP thresholds but above \$200 M RDT&E or \$920 M procurement (FY 2020 constant dollars) are typically delegated from ASN(RD&A) to the cognizant PAE/PEO; examples include shipboard radars, launch-and-recovery systems, or complex weapon upgrades where the PEO retains the MDA [32, 48].

ACAT III.

Efforts below ACAT II thresholds (but still acquisition programs) normally have the PAE, SYSCOM commander, or PM acting as MDA; documentation is tailored, yet APBs, TEMPs, and configuration control still apply [3, 32].

ACAT IV (DON unique).

SECNAVINST 5000.2G divides ACAT IV into IVT (test) and IVM (monitor) for sustainment, modernization, and In-Service Engineering efforts; NAVSEA/NAVAIR/NAVWAR commanders often serve as MDA, delegating narrowly scoped efforts to PMs while ensuring technical authority concurrence [3].

Delegation chain. The DAE retains ACAT ID decisions unless a signed memo delegates to ASN(RD&A); the Navy SAE typically keeps ACAT IC/IA programs and can further delegate ACAT II/III/IV to PAEs. SECNAVINST 5000.2G requires each delegation letter to specify statutory certifications retained by ASN(RD&A), while the Warfighting Acquisition Strategy directs PAEs to chair Capability Trade Councils and reassign funds across their portfolios when trades stay within cost/schedule thresholds [3, 62].

4.1.10 Project Overmatch Status

Project Overmatch is the Navy's operational architecture initiative to link platforms, weapons, sensors, and data fabrics so that Fleet Commanders can execute CJADC2 and Joint Fires at decision speed [13, 14]. VADM Seiko Okano (EDO) serves as the direct reporting program manager (DRPM) and steering committee chair while serving as PMILDEP; she sponsors Five Eyes data-sharing arrangements, Maven Smart Systems overlays, and Open DAGIR integrations that move capability from concept to fielded software in weeks [13, 14].

The DRPM title matters: a DRPM reports directly to ASN(RD&A), can tailor milestone documentation, and retains signature authority for integration decisions that cut across multiple PEOs. Under the Oct 2024 Five Eyes Project Arrangement, Capt. Remil Capili (PM), Shayna Bond (international deputy), and David Flowers (international lead) are embedding Cooperative Project Personnel from Australia, Canada, New Zealand, and the United Kingdom in San Diego, building a coalition lab, and preparing RIMPAC 26 demonstrations following Trident Warrior 24 and Talisman Sabre 25 events [13]. Capili, as PM, drives day-to-day execution and reports to Okano; Okano retains DRPM/steering-chair authority even after moving from NAVWAR to PMILDEP, preserving direct reporting to ASN(RD&A) and cross-PEO tailoring authority [13].

As PMILDEP, Okano now chairs Program Decision Meetings when delegated, synchronizes PAEs across portfolios, and uses Open DAGIR/Maven to inform requirements trades and PPBE issue papers; her FY26–27 focus is expanding the coalition network, institutionalizing rapid AI/ML onboarding, and ensuring Overmatch digital threads stay visible in milestone packages so new ship, submarine, and aircraft programs inherit the data standards by default [3, 13, 14].

4.1.11 Navy Overlays and Touchpoints

- SECNAVINST 5000.2G.** Implements ASN(RD&A) Program Decision Meetings, requires risk-based tailoring plans, and codifies DON-level documentation (Acquisition Plan, LCSP, Plan of Action and Milestones (POA&M)) [3].
- SECNAVINST 5400.15D.** Aligns PEOs and SYSCOMs, clarifies technical authority (NAVSEA 05, NAVWAR 5.0) integration, and designates Deputy ASN(RD&A) portfolio responsibilities [2].
- Resource Sponsor coordination.** OPNAV N9, N4, or equivalent sponsor must endorse requirements and PPBE positions before ASN(RD&A) decisions [48].

Note

When SECNAV policy conflicts with older DoW memoranda, defer to the most current directive from the higher authority unless ASN(RD&A) grants written tailoring. Document the rationale in the Acquisition Strategy.

4.2 Milestones and Phases

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4.2.1 Defense Acquisition System Overview

The DAS is the governance framework that aligns requirements, resourcing, and acquisition execution to deliver warfighter capability. It is policy-driven by DoDD 5000.01 and process-enabled through DoDI 5000.02, using the AAF to tailor pathways while preserving statutory oversight and Navy-unique controls [49, 50].

4.2.2 Adaptive Acquisition Framework Pathways (Board Snapshot)

MCA. Default for complex weapon systems; uses phased milestones with mandatory statutory certifications; emphasizes affordability trades, test rigor, and Navy two-pass / seven-gate touchpoints [32].

MTA. Rapid prototyping or rapid fielding to deliver an operational capability within five years; minimizes documentation, but demands robust transition planning to MCA or sustainment funding [55].

UCA. Accelerated path to field urgent capability within 2 years or provide an interim solution in less than 5; leverages tailored test, streamlined acquisition decision memoranda, and dedicated senior oversight [58].

Software Acquisition Pathway. Iterative, DevSecOps-focused delivery of software increments with continuous integration, enterprise cloud alignment, and recurring user feedback cadences [56].

Acquisition of Services Pathway. Structures service requirements, performance metrics, and contracting strategies to assure mission outcomes and oversight thresholds (e.g., Multi-Functional IPT requirements review) [60].

Defense Business Systems Pathway. Rationalizes business processes, data standards, and Business Capability Acquisition Cycle (BCAC) to unlock enterprise interoperability before investing in solutions [59].

Note

Pathways are not exclusive—an EDO may begin in MTA for prototyping, then transition to MCA for production once requirements and sustainment plans stabilize. Programs can also leverage UCA for an initial bridge while planning a transition to MCA.

4.2.3 Baseline Discipline: Acquisition Program Baselines, Deviations, and Nunn–McCurdy

APB. The cost-schedule-performance contract between the MDA, resource sponsor, and PEO; defines objective and threshold values that bound trade space [32].

Program Deviations. Triggered when performance, schedule, or cost thresholds breach; the Program Manager must notify the MDA within 30 days and submit a corrective action plan within 60 [32].

Nunn–McCurdy Act. Statutory reporting when Program Acquisition Unit Cost (PAUC) or Average Procurement Unit Cost (APUC) breaches significant ($\geq 15\%$ current / 30% original) or critical ($\geq 25\%$ current / 50% original) thresholds; critical breaches require certification to Congress within 60 days or program termination [68].

4.2.4 JCIDS Flow into the Defense Acquisition System

ICD. Supports the MDD with validated capability needs and initial attributes [64].

CDD. Provides KPPs and Key System Attributes (KSAs) to anchor Milestone B and the Technical Requirements Baseline [64].

Capability Production Document (CPD). Updates KPP/KSA values for production decisions and sustainment planning at Milestone C [64].

Note

Maintain requirements traceability matrices linking KPP thresholds to test plans and APB values for board discussions.

4.2.5 KPPs v.s. KSAs (Know the Difference)

KPP (Key Performance Parameter). A performance attribute of a system critical to mission success. Each KPP has a threshold (minimally acceptable) and objective (desired) value. Failure to meet a *threshold* normally triggers program reassessment or revalidation of the requirement. *Why it matters:* Drives design trades, tempers requirement creep, and is reportable in baselines and decision reviews [64].

KSA (Key System Attribute). Important performance attribute not critical enough to be a KPP. KSAs are managed and traded to balance cost/schedule/performance without jeopardizing mission effectiveness. *Why it matters:* Provides disciplined trade-space for the Program Manager and requirements sponsor [64].

Difference. KPPs are mission-critical and tightly controlled; KSAs are important but offer more flexibility. Both appear in the CDD/CPD with threshold/objective values, but KPPs carry higher visibility and risk if breached.

4.2.6 Acquisition Strategy Essentials

Purpose. Documents tailored acquisition approach, contract strategy, intellectual property posture, and affordability constraints for MDA approval [32].

Key Contents. Pathway selection, phasing plan, incentive structure, test and evaluation strategy, risk burn-down, sustainment concept, cybersecurity, international cooperation, and business case for software/data management [32, 49].

Approval Event. MDA concurrence at Milestone A (initial) with updates prior to Milestone B and significant re-baselines; Navy 2-pass/7-gate integrates resource validation before signature [65].

4.2.7 International Acquisition Objectives

Foreign Military Sales (Foreign Military Sales (FMS)). Strengthen partner capability while leveraging U.S. economies of scale; requires disclosure review and Letter of Offer and Acceptance alignment [32].

Direct Commercial Sales (Direct Commercial Sales (DCS)). Contractor-led exports; Program Office ensures configuration control and technology-security compliance [32].

- Cooperative Development.** Shared investment with allies (e.g., Memoranda of Understanding) to reduce life-cycle cost and enhance interoperability [32].
- Armaments Cooperation / International Programs Security.** Integrates disclosure, export licensing, and technology protection plans into the Acquisition Strategy [32].

4.2.8 Navy Two-Pass / Seven-Gate Review

- Pass 1 – Capability.** Led by OPNAV N9 to mature the requirement: Gate 1 validates the capability gap, Gate 2 refines solutions and mission analysis, Gate 3 assesses affordability and alignment with Navy enterprises, and Gate 4 authorizes program initiation (supports MDD) [65].
- Pass 2 – Resourcing.** Led by OPNAV N8 to align budgeting: Gate 5 verifies contracting/test readiness before Milestone B (often synchronized with the Development RFP Release Decision), Gate 6 confirms production and sustainment resources prior to Milestone C, and Gate 7 evaluates sustainment performance and modernization plans [65].

Note

For board prep, highlight where SECNAV Gate Reviews or Defense Acquisition Board touchpoints overlay Navy gates to demonstrate cross-tier synchronization.

4.2.9 Technology Readiness Levels

Table X summarizes Technology Readiness Level (TRL) definitions with representative naval examples.

TABLE XI: TECHNOLOGY READINESS LEVELS WITH REPRESENTATIVE NAVAL EXAMPLES.

TRL	Definition	Example (EDO context)
TRL 1	Basic principles observed	Novel corrosion-resistant alloy proven in laboratory
TRL 2	Technology concept formulated	Hypothesis for integrating alloy into hull structures
TRL 3	Analytical and experimental proof-of-concept	Coupon tests validating stress performance in relevant environment
TRL 4	Component validation in laboratory	Scale panel fabrication and testing under controlled loads
TRL 5	Component validation in relevant environment	Module integrated into land-based test site with salt-fog exposure
TRL 6	System/subsystem prototype in relevant environment	Full-scale hull section installed on test article or surrogate vessel
TRL 7	System prototype in operational environment	Prototype hull segment installed on operational ship during limited sea trials
TRL 8	Actual system completed and qualified	Final production drawings released, inspections passed for lead ship
TRL 9	Actual system proven in mission operations	Sustained operational deployment meeting performance metrics

Source: *Technology Readiness Assessment Guide*, 2020 [69]

Hint: Match TRLs to TMRR exit criteria: critical technologies should be ≥ TRL 6 before Milestone B.

Note

Operational environment testing typically begins at **TRL 7** (system prototype in an operational environment). **TRL 8** means the actual system is completed and qualified (e.g., qualification/acceptance testing, certifications). **TRL 9** means the actual system is *proven in mission operations* (e.g., successful fleet employment meeting mission metrics).

4.2.10 T&E Organizations and Roles

OSD Director, Operational Test and Evaluation (DOT&E). Independent oversight of operational test adequacy, approves/concurs in TEMP for covered programs, and reports to Congress on operational suitability and effectiveness.

Service Operational Test Agencies. Navy *Commander, Operational Test and Evaluation Force (COMOPTEVFOR) (COTF)*, Air Force *Operational Test and Evaluation Center (AFOTEC)*, U.S. Army *Test and Evaluation Command (ATEC)*, Marine Corps *Operational Test and Evaluation Activity (MCOTEA)*. Plan/execute IOT&E and report independent findings to the Service and DOT&E.

Developmental Test (DT). Program Offices with SYSCOM Warfare Centers (e.g., Naval Air Warfare Center Aircraft Division (NAWCAD), Naval Air Warfare Center Weapons Division (NAWCWD), NSWC, NUWC) plan and run DT to verify requirements, qualify designs, and feed Operational Test Readiness Review (OTRR).

Interoperability/Joint. *Joint Interoperability Test Command (JITC)* assesses joint interoperability/Net-Ready KPP compliance where applicable.

Ranges/Land-based sites. Service ranges and land-based engineering/test sites provide representative and operational environments to support DT/Operational Test (OT).

Artifacts. TEMP aligns DT/OT scope and phases; OTRR certifies readiness for IOT&E; SAR / DAES capture results for governance [32].

4.2.11 Major Capability Acquisition Lifecycle Overview

MSA. Shape the trade space and select a preferred materiel solution; conclude with an AoA recommendation and technology maturation plan [32].

TMRR. Mature technologies, refine requirements, and reduce integration risk to enter EMD; manage competitive prototyping where feasible [32].

EMD. Complete detailed design, verify system performance, and prepare for production and deployment decisions [32].

P&D. Produce capability, conduct IOT&E, field initial units, and decide on Full-Rate Production (FRP) [32].

O&S. Sustain readiness, execute modernization, and plan for disposal while feeding affordability back into PPBE cycles [32].

Figure 4.5 Shows the major phases, review and decision points within MCAs. The following are the reviews that occur during the phases:

Alternative Systems Review (ASR). Validates the preferred materiel solution and technology maturation approach emerging from the AoA; inputs include threat assessments, cost-schedule trades, draft system concepts, and updated Technology Development Strategy / SEP outlines, and the primary output is an ASR report that authorizes drafting authoritative system-level requirements for the upcoming System Requirements Review (SRR) [32, 51].

SRR. Confirms that system-level performance, interface, cyber, and logistics requirements trace to the validated ICD / CDD and are testable; requires a draft System Specification, mission threads, interface control documentation, and technical performance measures, and produces a baselined system requirements set and updated verification strategy feeding System Functional Review (SFR) planning [32, 51].

SFR. Demonstrates that the functional architecture and allocated requirements can satisfy the mission needs within cost/schedule constraints; leverages functional allocation matrices, preliminary hazard analyses, and draft reliability growth plans, and yields an approved functional baseline plus a risk/opportunity burn-down plan ahead of Preliminary Design Review (PDR) [32, 51].

PDR. Assesses whether the allocated baseline and preliminary design (drawings, models, manufacturability assessments, support concepts) are mature enough to enter detailed design; the review outputs include an approved allocated baseline, configuration-controlled action list, and direction for completing qualification hardware prior to Critical Design Review (CDR) [32, 51].

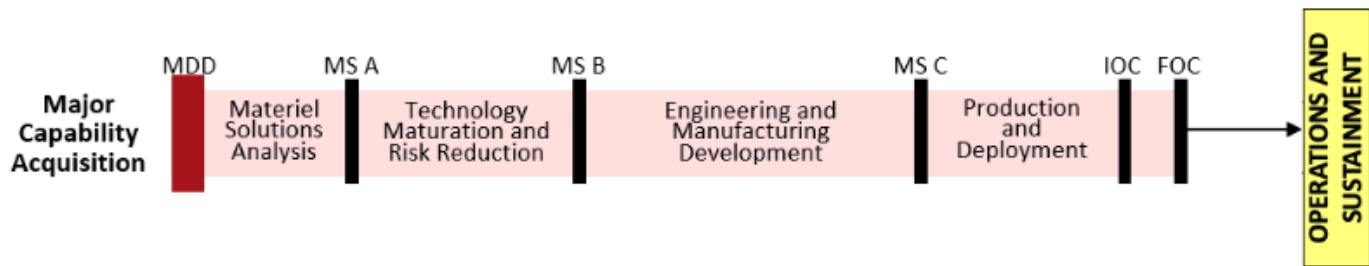
CDR. Confirms the detailed design meets all requirements with acceptable technical risk; inputs include product baseline technical data packages, qualification/analysis results, producibility assessments, and cybersecurity artifacts, and the decision produces an approved product baseline plus direction to release build-to packages and procure long-lead items [32, 51].

Test Readiness Review (TRR). Evaluates readiness to execute a specific test event by reviewing test articles, instrumentation, facilities, procedures, safety releases, and deficiency dispositions; a successful TRR outputs an authorization to start test with documented entrance/exit criteria and risk mitigations [32, 51].

System Verification Review (SVR). Verifies the integrated system meets the product baseline and is ready for production decision reviews; it uses completed verification cross-reference matrices, configuration audits, and qualification / DT results to produce a verified system baseline, deficiency adjudication list, and recommendation for Full-Rate Production Decision Review (FRPDR) preparation [32, 51].

OTRR. Certifies that production-representative assets, training, logistics products, safety artifacts, and test ranges are ready to support initial operational test; the review outputs the operational test readiness certification and any restrictions or corrective actions before DOT&E concurrence [32, 51].

Production Readiness Review (PRR). Determines whether manufacturing processes, supply chain, quality systems, and sustainment enablers can support LRIP / FRP quantities; requires control plans, process capability data, staffing/budget burn rates, and yields a decision to proceed into production along with corrective actions for any shortfalls [32, 51].



Lifecycle View of Major Capability Acquisition

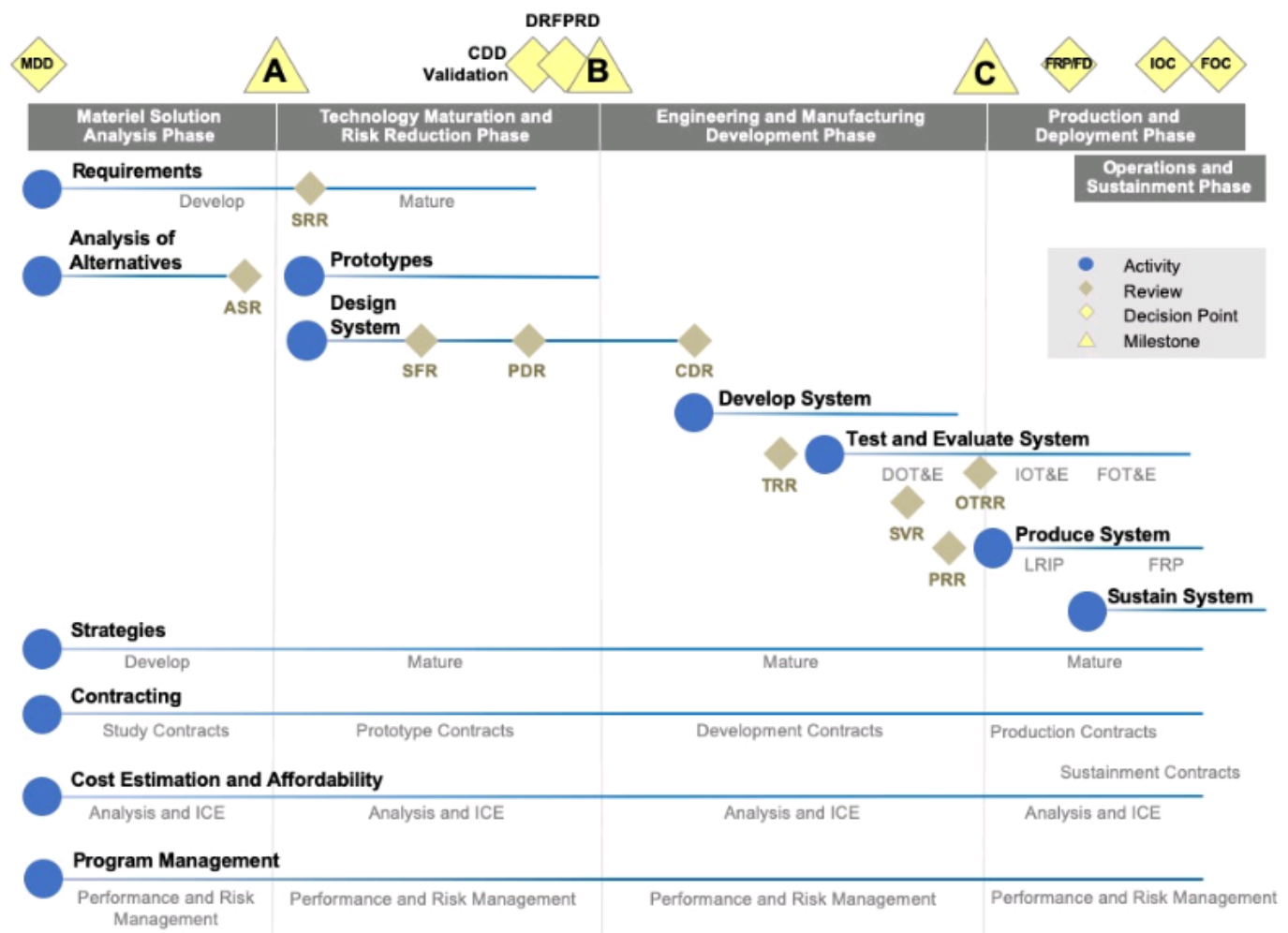


Figure 4.5. Lifecycle view for a MCA showing all functions and activities for each phase. Source: DoDI 5000.85, 2020 [32]

4.2.12 Materiel Development Decision

Inputs. Validated ICD, initial affordability analysis, preliminary acquisition strategy, and resource sponsor endorsement [50].

Decision. MDA authorizes entry into MSA (or an alternate pathway) and designates the lead PEO/PM [32].

Outputs. Acquisition Decision Memorandum (ADM), competitive AoA guidance, and direction to form the IPT structure [32].

4.2.13 Materiel Solution Analysis Phase Essentials

Purpose. Analyze alternatives, define the preferred materiel solution, and scope technology maturity and risk reduction tasks [32].

Key Activities. Conduct AoA, establish initial system performance metrics, develop early test strategies, and plan technology maturation for critical subsystems [32].

Entrance Criteria. Approved MDD, resourced AoA study guidance, and draft acquisition strategy [50].

Exit Criteria. Approved TMRR acquisition strategy update, technology risk assessments (i.e., TRL $\geq$ 3–4), and draft CDD inputs for Joint staffing [32].

4.2.14 Milestone A Focus Areas

Purpose. Authorize entry into TMRR and approve the initial Acquisition Strategy and APB [32].

Board Topics. AoA results, technology maturation plans, affordability caps, cost estimates, and early test strategies [32].

Outputs. ADM, approved initial APB, direction for competitive prototyping, and signed Technology Development Strategy if required [32].

4.2.15 Technology Maturation and Risk Reduction Phase Essentials

Purpose. Reduce integration risk, mature the Technical Baseline, and prepare for Engineering & Manufacturing Development source selection [32].

Key Activities. Competitive prototyping (if directed), systems engineering trade studies, reliability growth, cyber-resilience planning, and Capabilities Development Document Validation (CDD-V) [32].

Entrance Criteria. Approved Milestone A ADM, funded risk-reduction plan, and established IMS aligned to Navy Gate 5 timeline [32, 65].

Exit Criteria. Technology Readiness Assessments showing TRL $\geq$ 6, draft Milestone B documentation (e.g., SEP, TEMP), and configured product baseline ready for the Development Request for Proposal Release Decision (DRFPRD) [32].

4.2.16 Milestone B Focus Areas

Purpose. Authorize entry into EMD and establish the program as a major defense acquisition program with formal reporting [32].

Board Topics. Affordability caps, independent cost estimates, TEMP approval, LRIP planning, and production-representative prototype evidence [32].

Outputs. Updated APB, approved Acquisition Strategy revision, and contractual authority to release EMD RFPs [32].

4.2.17 Engineering and Manufacturing Development Phase Essentials

Purpose. Complete detailed design, build and test engineering development models, and validate manufacturing processes [32].

Key Activities. CDR, system verification, developmental test events, cybersecurity accreditations, and sustainment baseline development [32].

Entrance Criteria. Approved Milestone B ADM with resourced contracts, finalized technical baseline, and product-support planning aligned to Navy Gate 6 [32].

Exit Criteria. Successful IOT&E readiness, manufacturing readiness $\geq$ TRL 8, and validated sustainment strategies supporting Milestone C [32].

4.2.18 Milestone C Focus Areas

Purpose. Authorize production and deployment following demonstration of operational effectiveness and suitability [32].

Board Topics. IOT&E results, sustainment and supply-chain readiness, affordability assessments, and updated manpower/training plans [32].

Outputs. FRPDR plan, updated APB, and authority to obligate procurement funding for production lots [32].

4.2.19 Production and Deployment Phase Essentials

Purpose. Produce and deliver the system while validating operational performance and support concepts [32].

Key Activities. LRIP execution, IOT&E, Configuration Steering Board (CSB) governance, and ramp to Full-Rate Production [32].

Exit Criteria. FRP approval, attainment of Initial Operational Capability (IOC), and sustainment plans ready for fleet release [32].

4.2.20 Operations and Support Phase Essentials

Purpose. Sustain mission effectiveness, execute modernization, and manage total ownership cost until disposal [32].

Key Activities. Performance-Based Logistics execution, Depot maintenance planning, Post-Deployment reviews, and CSBs for major updates [32].

Milestones. Full Operational Capability (FOC), disposal decision documentation, and updates to capability requirements feeding future planning cycles [32].

5 Requirements

5.1 Joint Capabilities Integration & Development System

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5.1.1 Requirements Demand Versus Acquisition Supply

JCIDS. Establishes the joint demand signal by translating strategic guidance into validated capability requirements, risk statements, and DOTMLPF-P change recommendations that scope non-materiel and materiel options [70, 71].

DAS. Converts validated requirements into executable acquisition strategies, cost/schedule baselines, and lifecycle sustainment plans under DoDD 5000.01 and DoDI 5000.85 [32, 49].

Note

Board cue: be ready to explain where a program sits in both systems; Gate 2 packages are dead on arrival if the JCIDS paperwork is stale even when the DAS strategy is current [71].

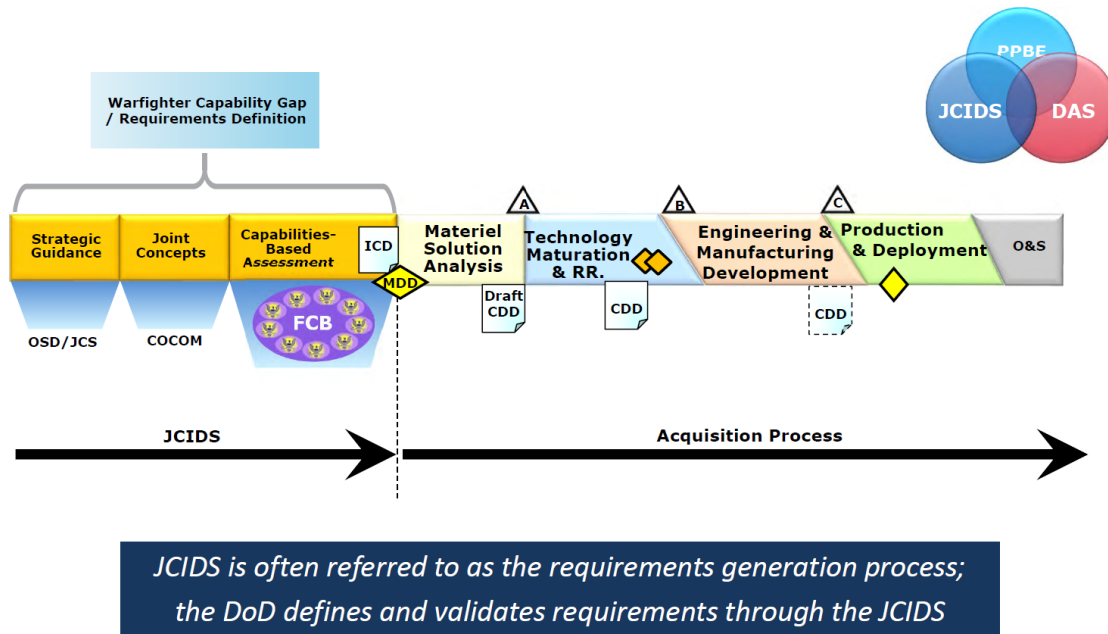


Figure 5.1. Relationship between the Joint Capabilities Integration and Development System and the Defense Acquisition System. Source: 3.3.3. Joint Capabilities Integration and Development System, 2025 [71]

5.1.2 Capability Requirements and the Capability-Based Assessment Cadence

Capability requirements are mission-task statements (task, condition, standard) that are solution-agnostic yet measurable enough to drive trades, and Capability-Based Assessments (CBAs) provide the disciplined analysis to justify them [70]. Each CBA follows repeatable phases:

Study initiation. The sponsoring component issues the study notice, nominates a lead Functional Capabilities Board (FCB), scopes timelines, and aligns analytical support and governance expectations [70, Annex B].

Operational context & mission threads. Analysts decompose national / Combatant Command (CCMD) guidance into joint concepts, mission threads, and integrated architectures to keep the study “concept-based and threat-informed” [70, Encl. A].

Gap and risk assessment. Existing and programmed capabilities are mapped against required mission effects to quantify gaps, assess risk, and prioritize by warfighting impact and threat timelines [70].

Solution analysis. DOTMLPF-P are explored first; only when risk remains unacceptable are materiel approaches carried forward, together with preliminary affordability bounds and schedule realism [70].

Documentation & validation path. The resulting evidence package (mission context, gaps, DOTMLPF-P recommendations, risk, and proposed documents) feeds the Gatekeeper, FCB, and validation authority decision on which JCIDS artifact (ICD, DOTMLPF-P Change Recommendation (DCR), update) to staff [70, Encl. A].

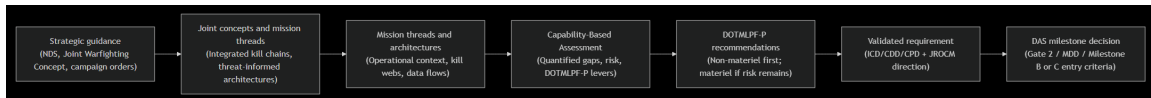


Figure 5.2. Concept-to-requirement flow that must stay synchronized with acquisition execution. Source: 3.3.3. Joint Capabilities Integration and Development System, 2025 [71].

5.1.3 Strategy-to-Requirement Synchronization

Joint demand starts with strategic direction (National Defense Strategy, Joint Warfighting Concept, campaign orders) and is progressively tailored until a validated requirement enters the DAS [70, 71]. Key touchpoints:

Strategic guidance. SECWAR and CJCS translate national objectives into force design problems and assign lead sponsors; the Joint Staff J-7 houses concept development and readiness assessments that set the analytical agenda [70].

Joint concepts & mission threads. Mission engineers decompose guidance into mission areas, integrated kill chains, and threat-informed architectures so that requirements stay “concept-based and threat-informed” before solutioneering [70, Encl. A].

CBAs. Sponsors conduct the structured CBA described above to document measurable capability requirements, prioritize risks, and recommend DOTMLPF-P levers with quantitative evidence [70].

Gatekeeping & staffing. The Joint Staff Gatekeeper assigns a Joint Staffing Designator (JSD), routes packages through Knowledge Management and Decision Support (KM/DS), and tasks the appropriate FCB for certification prep; timelines and suspense assignments live inside KM/DS so Navy leads can monitor in real time [70].

Validation. Depending on the JSD, the JROC, Joint Capabilities Board (JCB), or delegated Service/CCMD authority validates requirements, issues memoranda Joint Requirements Oversight Council Memorandums (JROCMs), and records mandatory actions (e.g., certifications, follow-on DCRs) that acquisition teams must honor [70, 72].

Acquisition handoff. Validated artifacts, with their JROCM direction and Capability Trade Council (CTC) or CSB trade expectations, become entry criteria for Gate 2, the MDD, and milestone reviews, ensuring the PAE and PEO communities inherit coherent demand signals [71].

5.1.4 Major Capability Acquisition Requirement Set

ICD. Captures the operational context, quantified gaps, and risk that justify materiel and non-materiel approaches; it is required to support the Materiel Development Decision and is deliberately solution-agnostic [70, Appendix B-A].

CDD. Establishes system-level performance attributes (KPPs, KSAs, Additional Performance Attributes (APAs)) that anchor TMRR planning and must be validated before the Development RFP release and Milestone B; it becomes the authoritative source for the Technical Requirements Baseline [32, 70].

CDD updates. Refine thresholds/objectives when an EMD phase spans multiple lots or when KPPs trades are needed before Milestone C; updates keep production-representative values authoritative without re-opening earlier portions of the document [70].

CPD. Translates the validated CDD into production-ready parameters, sustainment metrics, and deployment considerations for Milestone C and Full-Rate Production/Fielding decisions [32, 70].

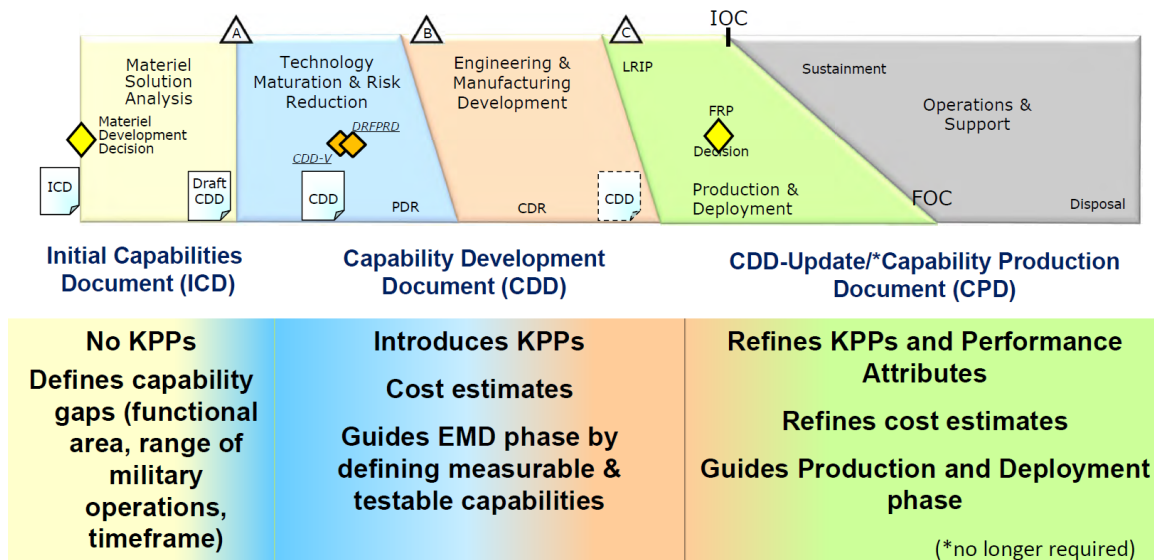


Figure 5.3. Placement of Joint Capabilities Integration and Development System artifacts inside the phased Major Capability Acquisition pathway. Source: 3.3.3. Joint Capabilities Integration and Development System, 2025 [71].

5.1.5 How KPPs Stay Integrated with DAS

Validated KPPs define the trade space that Milestone Decision Authorities use to approve baselines, certifications, and contracts. DoDI 5000.85 requires KPP ownership to be traceable from the CDD into the System Requirements Review, Developmental Test plans, and Acquisition Program Baseline; breaches trigger Gatekeeper tripwire reviews and Configuration Steering Boards [32, 70]. Navy PMs must therefore align CDD/CPD updates with SEA 05 warrant packages and ASN(RD&A) Gate 6 readiness to keep statutory certifications (Net-Ready (NR)-KPP, cybersecurity, safety) synchronized [65, 71].

5.1.6 Prioritizing Materiel and Non-Materiel Solutions

JCIDS policy directs sponsors to work through every DOTMLPF-P lever before defaulting to expensive materiel approaches, both to reduce risk and to speed delivery [70]. Practical applications include:

Doctrine. Update joint/combined warfighting concepts or tactics to mitigate gaps quickly (e.g., change Anti-Submarine Warfare (ASW) Concept of Operations (CONOPS) before buying sensors).

Organization. Re-align Fleet/NAVWAR command relationships or mission tailoring to close gaps without new hardware.

Training. Modify Fleet Readiness Training Plan events or simulator syllabi so crews can exploit existing systems better.

Materiel. Only after non-materiel levers fail should new materiel increments be scoped, with clear linkage to DAS milestones.

Leadership. Insert leadership development or acquisition workforce upskilling when governance bottlenecks (e.g., JROC literacy) create the gap.

Personnel. Adjust billets, Navy Enlisted Classifications, or civilian skill codes to align manpower with capability employment.

Facilities. Invest in pier-side infrastructure, cyber ranges, or test venues necessary to realize capability performance.

Policy. Seek changes to SECNAV/OPNAV directives or coalition agreements when authorities constrain capability delivery.

5.1.7 Governance Roles and Validation Chain

JROC. Executes Title 10 § 181 duties—prioritizing capability gaps, validating KPPs, delegating authority, and issuing JROCM guidance that binds both requirements and acquisition communities [70, 72].

Joint Staff Gatekeeper & JCB. The J-8 Gatekeeper screens every submission, assigns Joint Staff suspense, and recommends whether the JROC or JCB (chaired by the VCJCS) is the right validation venue; the JCB focuses on issues that are joint but lower risk, accelerating queue time for most Navy programs [70].

FCBs. Functional Capabilities Boards (e.g., C4/Cyber, Protection, Logistics) chair working groups, broker certifications (Net-Ready, Force Protection, System Survivability), monitor implementation of DCRs tasks, and elevate tripwires to the JCB/JROC [70].

Service validation authorities. OPNAV N9/N9I act as the Navy Gatekeeper, ensuring Gate 0/1 packages satisfy OPNAVINST 5000.42E expectations before forwarding to the Joint Staff, and they remain responsible for Service-specific “Joint Integration” decisions even when joint validation is delegated [65].

Requirements are therefore validated either by the JROC, the JCB (if delegated), or by the cognizant Service/CCMD authority when the Joint Staff assigns a Joint Staffing Designator of “Joint Integration/Information” [70].

5.1.8 IT, C4I, and Coalition Interoperability Implications

Net-Ready. Every Information System (IS) ICD, IS CDD, and Capability Drop version of the CDD must secure Net-Ready certification from the C4/Cyber FCB chair, demonstrating compliance with enterprise architecture, data standards, and mission thread-level interoperability [70].

Architecture and reuse. Sponsors must deliver Department of War Architecture Framework (DoDAF) views (All Viewpoint (AV), Operational Viewpoint (OV), Systems Viewpoint (SV), Capability Viewpoint (CV)) that trace requirements to reusable services, Application Programming Interfaces (APIs), and data models, enabling software pathways and rapid iteration [50, 70].

C4I interoperability. DoDD 5000.01 compels all acquisition programs to plan for cross-Service and allied interoperability; JCIDS enforces this by requiring coalition releasability considerations and Combined/Joint Mission Threads inside every requirement [49].

Allied impacts. Capability sponsors must document how U.S. and allied systems exchange data, share bandwidth, and remain cyber survivable before validation; failure to do so can halt Gate 2 or Net-Ready KPPs endorsements [70, 71].

5.1.9 Warfighting Acquisition System (JCIDS Replacement)

SECWAR's 7–10 Nov 2025 memoranda and Acquisition Transformation Strategy direct the Department to replace legacy JCIDS paperwork with the Warfighting Acquisition System (WAS), an enterprise approach built on Capability Portfolio Management (Capability Portfolio Management (CPM)) and Capability Trade Councils that keep requirements, resourcing, and acquisition synchronized [61, 62].

Key Operational Problems.

The JROC now publishes a short list of KOPs—problem statements tied to the National Defense Strategy—and stops validating Service-level CDDs except where statute forces it, returning day-to-day authority to the Services [63].

Mission Engineering & Integration Activity.

USW(R&E) and USW(A&S) created the MEIA to run rapid experiments, collect live data, and produce mission-thread analyses that replace lengthy JCIDS studies before resource decisions are made [61, 63].

Requirements & Resourcing Alignment Board.

The new RRAB, co-chaired by the DepSecWar and VCJCS, aligns each prioritized KOP with PPBE guidance, reallocations, and Joint Acceleration Reserve decisions inside the same budget cycle so validated needs show up in the FYDP immediately [63].

Joint Acceleration Reserve.

The JAR is a centrally managed pool that bridges the “valley of death” by funding promising prototypes or industrial-base surges between POM builds; RRAB recommendations trigger its use when a KOP solution proves viable [63].

Inputs.

Mission-thread data, kill-chain metrics, threat assessments, PPBE affordability guidance, experimentation results, and digital engineering artifacts flow into each CPM cell so that operational problems are quantified before trades begin [62].

Core events.

The DepSecWar and Vice Chairman convene quarterly CPM reviews; PAEs call Capability Trade Councils whenever trades exceed delegated cost/schedule/performance thresholds; unresolved items elevate to the Deputies Management Action Group [62].

Outputs.

Portfolio scorecards, capability roadmaps, trade memoranda, and data packages that substitute for the ICD/CDD/CPD sequence while still feeding Program Decision Memoranda, issue papers, and Acquisition Program Baselines [62].

Key players.

DepSecWar, VCJCS, USW(A&S), USW(R&E), USW(C), PAEs, and functional analytics teams replace the traditional Gatekeeper/FCB/JROC staffing chain; Capability Trade Councils provide the audit trail for requirement changes [62].

Integration with PPBE and the DAS.

The strategy instructs USW(C) and USW(A&S) to budget and execute at the portfolio level, letting PAEs move resources inside those controls while simultaneously updating APBs and milestone documentation when trades occur [62].

Differences from JCIDS.

WAS relies on live data and executive accountability instead of protracted document staffing, collapses requirements/resourcing/acquisition touchpoints, and uses Capability Trade Councils in lieu of formal Configuration Steering Boards [62].

Legislative status.

The strategy is policy guidance pending FY 2026 NDAA language; during FY 2025 the Department is executing pilot portfolios (munitions industrial base, Project Overmatch) under DepSecWar memoranda while still producing any JCIDS artifacts explicitly mandated by statute [62].

5.1.10 What EDOs Must Master Under WAS**Time-based metrics.**

Portfolio scorecards now track median days from KOP identification to MDD/IOC; EDOs must baseline their programs against those targets and be ready to discuss what trades are required to shrink the timeline [61].

Capability Trade Councils.

Expect recurring requests to convene or support these councils with mission-thread data, digital models, and industrial-base options so that requirement, schedule, and cost trades are justified without defaulting to a JROC package [62].

Experiment-to-budget pipeline.

Because MEIA experimentation outputs feed the RRAB, EDOs must deliver credible prototype results (with cost/sustainment implications) fast enough to influence the same PPBE build; missing the window means losing JAR opportunities [63].

Security cooperation touchpoints.

DSCA/DTSA realignment under USW(A&S) means exportability, tech-release, and allied production plans are now discussed alongside Capability Trade Councils—EDO briefs should cover how security cooperation choices affect industrial-base capacity and delivery risk [66].

5.1.11 Current Events: SECWAR Acquisition Transformation Strategy

SECWAR's 10 Nov 2025 Acquisition Transformation Strategy directs the Warfighting Acquisition System to eliminate legacy JCIDS bureaucracy, convert Configuration Steering Boards into CTCs, and empower PAEs to make requirements trades inside cost/schedule thresholds [62]. The memo explicitly calls for JCIDS elimination and streamlining of DAS reviews, meaning Navy teams must plan for near-term transition periods where PAEs convene CTCs to validate trades in lieu of traditional Joint Staff staffing, while DAS documentation is simultaneously pushed to lower approval levels [62]. Expect joint governance updates (Gatekeeper rules, FCB charters) and plan to capture trade rationales in Decision Memoranda so that future auditors can trace how requirements changes affected APBs.

5.2 Test & Evaluation

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5.2.1 Board Prep Summary

Test and Evaluation (T&E) provides the objective evidence that a system can achieve the warfighter outcomes promised in requirements documents before obligating the fleet to fielding, sustainment, and congressional reporting commitments [32, 73]. A test is the controlled activity (lab, range, digital thread, fleet event) that produces data, while an evaluation is the analytic judgment that compares that data to thresholds, suitability expectations, and risk acceptance criteria [73]. Effective T&E integrates developmental, operational, and live-fire evidence so that each MCA milestone is backed by verified design data and validated mission performance insights rather than optimistic modeling alone [53]. Programs that synchronize T&E with systems engineering and program control expose performance gaps early, recycle fixes through the technical baselines, and enter OTRR with a defensible TEMP, resource plan, and deficiency scrub.

5.2.2 Definition and Relationship to Verification and Validation

Developmental Test and Evaluation (DT&E) verifies “did we build the system right” by showing the technical baseline meets allocated requirements, standards, and interface controls, while Operational Test and Evaluation (OT&E) validates “did we build the right capability” by proving Sailors and Marines can accomplish missions under representative threat, environment, and sustainment conditions [50]. Verification relies on repeatable measurements (e.g., propulsion plant efficiency in a land-based engineering site) and feeds configuration control; validation uses operational scenarios, tactics, and logistics supportability to demonstrate mission effectiveness and suitability. DoDI 5000.88 further directs programs to execute frequent, iterative end-user

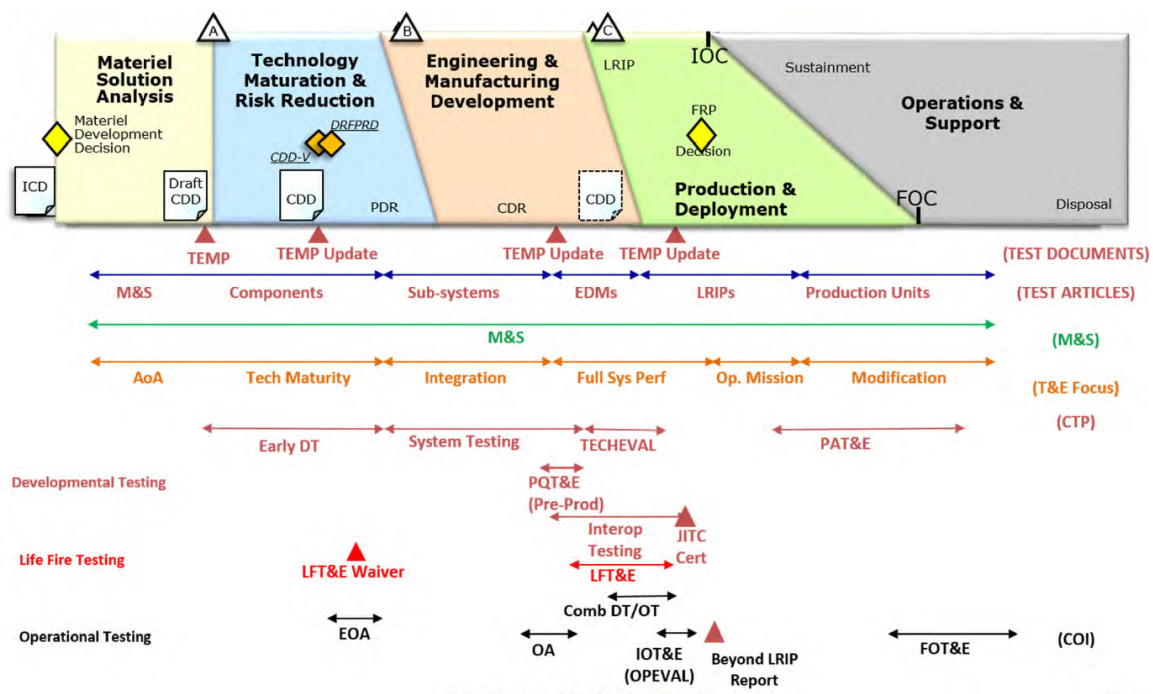


Figure 5.4. Key MCA test and evaluation touchpoints that frame DT, OT, and certification decisions. Source: 3.5.5. Test and Evaluation Part I, 2025 [73].

validation of features and usability so mission thread owners can accept the delivered capability before major decisions [74]. Verification and Validation (V&V) and Verification, Validation, and Accreditation (VV&A) activities must be planned in the TEMP and SEP so that models, simulations, hardware-in-the-loop facilities, and live events share common data assumptions and confidence levels before milestone reviews [53]. *Verification* therefore proves compliance with the documented specification (analysis, inspection, test, demonstration), whereas *validation* proves the warfighter accepts the delivered capability and can close the mission threads; both sides of the V&V matrix must be satisfied before the MDA can certify readiness for production or fleet release [53].

5.2.3 Statutory Requirements and Key Roles

10 U.S.C. § 139. Establishes the DOT&E as an independent advisor to SECWAR and Congress; DOT&E approves operational test plans, monitors execution, and must issue a public report for covered programs [75].

10 U.S.C. § 2399. Prohibits a program from proceeding beyond LRIP or requesting Full-Rate Production approval until DOT&E and the lead Operational Test Agency submit their independent reports to Congress [76].

10 U.S.C. § 2366. Requires Live Fire Test and Evaluation (LFT&E) or an approved waiver for covered systems (personnel, platform, and munitions lethality/survivability) before beyond-LRIP decisions; it also directs reporting of vulnerability/lethality findings to defense committees [54].

Deputy Assistant Secretary of War for Test and Evaluation (DASW(T&E)). Serves as the OSD focal point for test resources, designates Chief Developmental Testers (CDTs) for ACAT I programs, charters Lead Developmental Test and Evaluation Organizations (LDTOs), and co-chairs Defense-level T&E Working Integrated Product Teams [53].

CDT and LDTO. The CDT orchestrates the integrated test schedule, data requirements, and deficiency resolution cadence, while the LDTO (often a warfare center or system command test directorate) furnishes instrumentation, modeling, and range execution authority for DT&E [53].

Operational Test Agencies. COMOPTEVFOR, AFOTEC, ATEC, and MCOTEA plan and execute OT&E Missile Defense Agency (MDA) record [77].

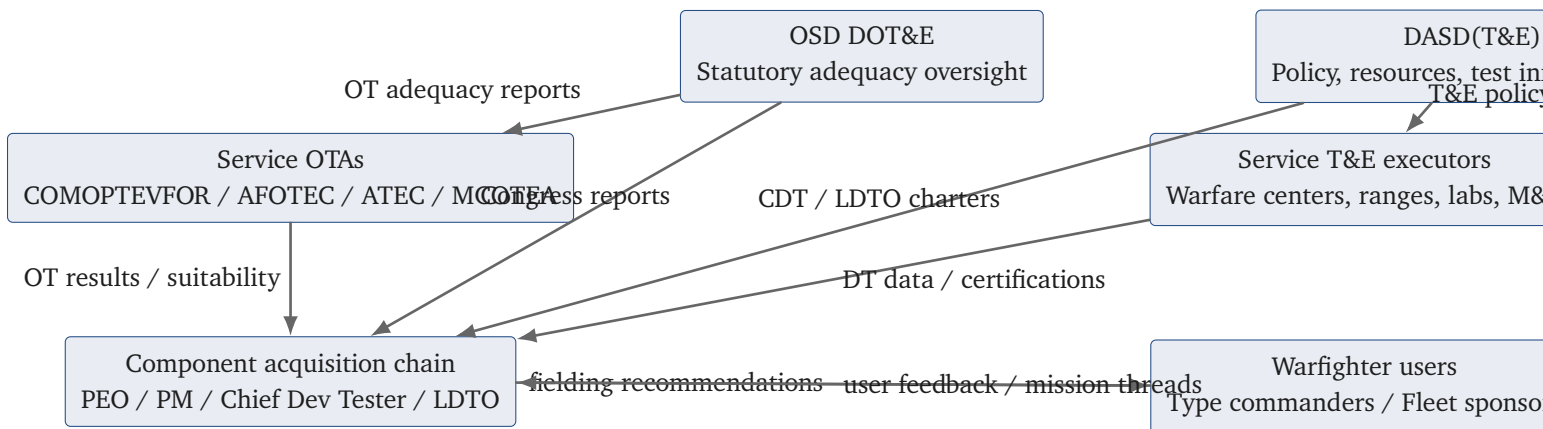


Figure 5.5. DoW test and evaluation governance chain linking OSD oversight, Service OTAs, warfare centers/ranges, and program-level test leadership.

5.2.4 Developmental Test Activities

Component and qualification testing. Bench, lab, and land-based engineering site events that verify materials, software builds, cyber hardening, interoperability, and safety factors before integration; data feeds configuration audits and reliability growth curves [73].

Subsystem and integration testing. Hardware-in-the-loop, systems integration labs, and shore-based platform integration tests that expose interface defects and mission thread timing issues before at-sea trials [73].

System-level DT&E. Full-up prototypes or first-of-class hulls execute combined developmental/operational scenarios, cybersecurity assessments, and logistics demonstrations to validate readiness for TRR and Technical Evaluation (TECHEVAL) [32].

Reliability growth and maintainability demonstrations. Planned stress events and failure-mode tracking prove that Mean Time Between Operational Mission Failure (MTBOMF) and corrective maintenance timelines align with sustainment strategies before Operational Test Readiness [53].

Certification testing. Airworthiness, safety, information assurance, weapon certification, and electromagnetic environmental effects tests that deliver statutory certifications needed for fielding [73].

Note

Edge case: Software pathway efforts may collapse TECHEVAL/Operational Evaluation (OPEVAL) into continuous DevSecOps assessments, but the program must still show how combined testing and user evaluations satisfy Title 10 adequacy requirements before fielding [50].

5.2.5 Live Fire Test and Evaluation

When required.

Title 10 demands LFT&E for every new or significantly upgraded crew-occupied platform, combat vehicle, major munitions program, or missile defense component that will face threat lethality; full-up system-level shots must be realistically stressed prior to requesting Full-Rate Production approval [53, 54].

When waived.

Only the SECWAR can waive full-up LFT&E when the system is unmanned, when destructive testing would be unreasonably expensive or impractical, or when alternative testing (component shots, validated modeling, allied data) provides equivalent insights; Congress must be notified of the rationale and the alternative plan [53, 54].

Execution and use.

Programs plan vulnerability and lethality events in the TEMP, employ full-up articles plus critical subsystems and surrogates, and collect battle damage, crew survivability, and cascading-effect data to inform design hardening, tactics, and survivability equipment upgrades; the results become an annex to the DOT&E report before the FRPDR [32, 53].

5.2.6 Operational Test Phases

Early Operational Assessment (EOA). A qualitative review, often using simulations or limited field events, that informs CDD refinement and acquisition strategy decisions before Milestone B [77].

Operational Assessment (OA). Focused operational events (sometimes embedded in Fleet Battle Problems or Fleet Exercises) that characterize mission effectiveness, training load, and logistics demands prior to LRIP or dedicated IOT&E [77].

Initial Operational Test and Readiness Evaluation (IOT&RE). Navy programs often combine IOT&E and Readiness Evaluation to prove both mission performance and Fleet introduction readiness; Title 10 requires that the IOT&E portion be adequate, realistic, and reported to Congress before FRP [76].

Follow-on Operational Test and Evaluation (FOT&E). Conducted after FRP or major configuration changes to verify corrective actions, CPD thresholds, cybersecurity resilience, and software drops prior to wide deployment [53].

5.2.7 Fundamental Test and Evaluation Terms

Critical Operational Issue (COI). The mission-outcome questions (“Can the ship defend itself against a raid?”) that structure OT&E plans and reports [77].

Measure of Effectiveness (MOE) & Measure of Suitability (MOS). Measures of Effectiveness and Suitability quantify warfighting value (probability of raid annihilation) and fleet readiness attributes (reliability, maintainability, training burden) tied to COIs [77].

Measure of Performance (MOP). Measures of Performance (e.g., radar detection range) show lower-level technical behaviors that roll up to MOEs/MOS [73].

Deficiency categories. Category I deficiencies are mission-critical failures that could cause death, severe injury, or major mission loss; Category II items are significant but less critical issues requiring tracking and correction [77].

Combined testing. Planned events that collect developmental and operational data simultaneously to save schedule and align datasets for evaluation teams [50].

5.2.8 Critical Technical Parameters and Technical Performance Measures

Critical Technical Parameters (CTPs) are the small subset of design attributes that must be controlled to assure a KPP or KSA remains achievable; they are derived from the ICD/CDD and baselined inside the APB [32]. Each CTP maps to one or more Technical Performance Measures (TPMs) that engineers track continuously using lab data, digital models, or DT&E results; TPMs trends highlight when technical debt threatens TEMP and SEP assumptions so leadership can adjust scope, schedule, or funding. Programs are expected to show a clean trace from COI → KPP/KSA → CTP → TPM → planned test points in the TEMP, ensuring every test objective underwrites an operational outcome and every requirement has an observable verification method [53, 73].

5.2.9 Test and Evaluation Master Plan

The TEMP captures the integrated T&E strategy, schedule, resources, and data management approach across the lifecycle; it harmonizes the SEP, LCSP, and APB so that every milestone package shows how risk will be retired [51]. Part I summarizes program context and key performance requirements, Part II defines the integrated test program (DT, OT, certification, modeling, and analysis), Part III outlines resource needs (ranges, threat surrogates, workforce, instrumentation), and annexes address data rights, stats, and cyber accreditation [53]. ACAT I(D) TEMPs require DOT&E approval; other TEMPs are approved by the Component Acquisition Executive. Updates are mandatory when requirements change, critical deficiencies emerge, or significant schedule/funding revisions alter test sequencing [32].

5.2.10 Modeling and Simulation

Modeling and simulation provide the only practical way to explore the mission design space before hardware exists, rehearse high-risk scenarios (e.g., hypersonic raid defense), and extend sparse live-test data across the operational envelope [73]. Programs must document model purpose, pedigree, uncertainty, and VV&A status in the TEMP; accredited models can bound live-test needs, size instrumentation, and support decision-quality evaluations when safety, cost, or treaties limit destructive testing [53]. Mission engineering digital threads (system-of-systems simulations, campaign models) allow combined DT/OT teams to choreograph events, align data management plans, and feed statistical confidence calculations before and after live events.

5.2.11 Navy Overlays and Integration

OPNAVINST 3960-series policy (captured in the coursebook) requires Navy programs to plan TECHEVAL (led by the engineering agent) before OPEVAL, to obtain OPNAV N94/N9 concurrence on test objectives that bear on Fleet introduction risk, and to route OTRR packages through COMOPTEVFOR for certification that logistics, training, and safety enablers are in place [77]. First-of-class ships also execute Builder's and Acceptance Trials, Board of Inspection and Survey (INSURV) events, and dedicated combat system Ship Qualification Trials that integrate with TECHEVAL/OPEVAL data packages. Navy Commander, Operational Test Force (COTF) retains authority to recommend restricted Fleet release if Cat I deficiencies remain; those actions feed the Key Roles appendix update for DASW(T&E), CDT, and COTF responsibilities.

5.2.12 Fiscal and PPBE Touchpoints

RDT&E appropriations fund most DT&E events, test articles, range time, and contractor support, while procurement colors such as SCN or Weapons Procurement, Navy buy dedicated OT assets when production-representative hardware is required for statutory adequacy [18, 73]. Operations and Maintenance funds cover Fleet participation, temporary duty, and sustainment of government test ranges. Because OT adequacy is a precondition for FRP and sustainment decisions, Program Objective Memorandum issue papers must protect test range modernization, threat-representative targets, and Modeling & Simulation accreditation costs or risk a PPBE disconnect when COMOPTEVFOR identifies unfunded test objectives.

5.2.13 Assumptions and Uncertainties

- Coursebook Modules 3.5.5 and 3.5.6 (March 2025 editions) remain the latest DON-specific T&E references.
 - COMOPTEVFOR manual updates post-2024 were not available offline; Navy-specific terminology (e.g., OPEVAL taxonomy) is based on coursebook guidance.
 - Upcoming NDAA changes to Title 10 T&E statutes have not been incorporated pending official publication.
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6 Maintenance

6.1 Fleet Maintenance Assets

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6.1.1 Echeloned Maintenance Levels

Organizational-level. Ship’s force keeps the hull, mechanical, and electrical plants safe and combat-ready by executing preventative checks, battle damage control, and light repairs with organic tools, allowing the platform to remain as self-sufficient as possible while deployed [78, 79].

Intermediate-level. Intermediate Maintenance Activities (IMAs) and tenders augment the crew with certified artisans (e.g., welders, riggers, calibration labs) who can disassemble components, correct corrosion, and return equipment to specification without docking the ship; Navy Afloat Maintenance Training Strategy (NAMTS) builds those Navy Enlisted Classifications (NECs), and so strike groups can recover faster when forward [78, 79].

Depot-level. Naval Shipyards (NSYs), private yards, and SUPSHIPs conduct heavy industrial work such as reactor refueling, shaft and underwater body repairs, and major modernizations that exceed shipboard capability; every task is tied to the Class Maintenance Plan (CMP) and engineered work instructions [79, 80, 81].

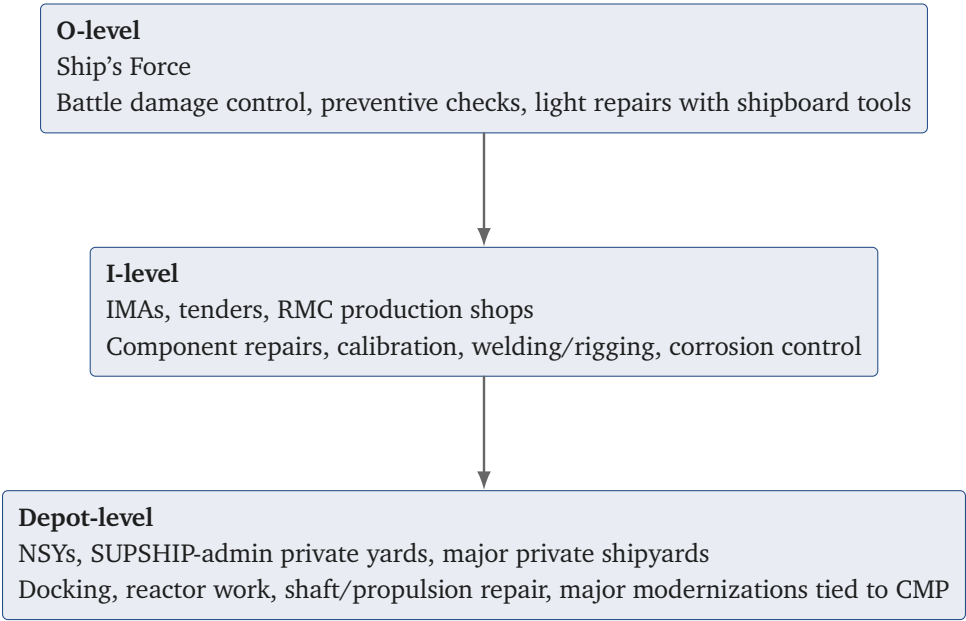


Figure 6.1. Maintenance echelon coverage and typical executing organizations from O-level through depot-level work.

Note

Board cue: reference at least one example task in each echelon (e.g., IMA balancing pump rotors vs. depot replacement of a main reduction gear) to show command of maintenance depth.

6.1.2 Shore Maintenance Organizations and Roles

NSYs. Public yards (e.g., Norfolk, Pearl Harbor, Puget, Portsmouth) report to NAVSEA 04 and integrate nuclear workforces, planning codes, and lifting/handling departments that mirror Naval Sea Systems Command Instruction (NAVSEAINST) 5450.14D standard organization [80].

Regional Maintenance Centers (RMCs). Regional Maintenance Centers deliver voyage repairs, execute CMP work, and act as the local TA for availabilities; they synchronize contracting, production shops, and Total Ship Readiness Assessment (TSRA) under Commander, Navy Regional Maintenance Center (CNRMC) [79, 81].

SUPSHIP. Supervisors of Shipbuilding administer new construction and conversion/modernization contracts as ACOs, coordinate budget execution, and provide technical oversight for nuclear refueling overhauls (e.g., CVN RCOH) (Appendix B.1:

RDML Jonathan J. Jett-Parmer serves as NAVSEA Deputy Commander, Supervisor of Shipbuilding; RDML Robert J. Dodson (USNR) is the reserve deputy; Commander, Naval Sea Systems Command (COMNAVSEA) is ADM William J. Houston (Submarine officer, non-EDO))[9, 79, 81].

Private shipyards. Huntington Ingalls, Bath Iron Works, General Dynamics NASSCO/Electric Boat, Fincantieri Marinette Marine, and Austal execute modernization packages, submarine maintenance, and surge depot work as part of the industrial base portfolio directed by OPNAV and PEO contracts [79].

6.1.3 Regional Maintenance Command Structure

Flag oversight (EDO). CNRMC is an EDO flag billet (RDML Dan L. Lannamann) reporting to COMNAVSEA; CNRMC sets standards and performance goals for all RMCs and aligns with TYCOM production priorities (*Appendix B.1: COMNAVSEA is VADM James P. Downey; CNRMC is RDML Dan L. Lannamann*)[81, 82].

RMC reporting. RMCs (e.g., Mid-Atlantic, Southeast, Southwest, Japan, and Puget Sound Naval Shipyard and Intermediate Maintenance Facility (IMF) detachments) report administratively to CNRMC and operationally to the supported TYCOM, with competency-aligned codes (100 fleet support, 200 availability management, 300 production shops, 400 quality assurance/technical authority) to standardize execution[81].

Work control touchpoints. Regional Maintenance and Modernization Coordination Office (RMMCO), availability work package managers, and contracting officers integrate with ship's force, Port Engineers, and TYCOM maintenance officers; EDO Port Engineers and RMC production officers bridge technical authority to the TYCOM readiness owner[79, 81].

6.1.4 Command, Funding, and Technical Authority Chains

Operational chain. CNO delegates readiness to USFF and PACFLT, who assign Type Commanders to own production schedules and risk acceptance for their forces [78].

Technical chain. NAVSEA 21/04 issue modernization and maintenance standards, while warfare centers and TAs publish CMP-derived work packages; disputes escalate through the Technical Authority chain independent of the operational chain [78, 81].

Funding chain. Fleet maintenance (Operations and Maintenance, Fleet (OC&MF)) flows from OPNAV N82 to BSOs and on to execution activities, while modernization is forecast via Fleet Modernization Design (engineering) funding (FUNDS-D)/Fleet Modernization Alteration funding (FUNDS-ALT)/Fleet Modernization Procurement funding (FUNDS-K) and, in limited pilots, OPN funds for selected private-yard availabilities to accelerate contract awards [18, 79].

Maintaining clarity between supported/supporting chains prevents the common failure mode where operational priorities outpace technical warrant approvals or funding authority, triggering rework and availability churn.

6.1.5 Workforce Readiness and Assessments

Readiness assessments. TSRA events and Condition Found Reports feed the Current Ship's Maintenance Project (Current Ship's Maintenance Project (CSMP)) so planners can refresh the CMP and future availability work packages [79].

Training pipelines. Programs such as NAMTS, SurgeMain reservists, and civilian apprentice pipelines grow targeted skills (e.g., non-skid application, inside machinist) to close gaps identified in Fleet readiness reviews [79].

Digital enablers. Class maintenance databases, Navy Data Environment (NDE) instances, and tool control systems give RMCs and NSYs shared configuration visibility that underpins quality assurance and auditability [81].

Timely feedback from these mechanisms is what allows SURFMEPP/SUBMEPP to rebalance work between the O-, I-, and depot echelons before a ship misses a CNO availability window.

6.2 Navy Modernization Program

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6.2.1 Modernization vs. Maintenance

Modernization. Improves capability or readiness above the baseline by installing approved Ship Alterations (SHIPALTs), Modernization Work Orders, and Fleet modernization program items that change the ship's configuration, safety posture, or mission systems [78, 83].

Maintenance. Restores the system to its original performance and reliability; it cannot change capability without an approved modernization directive per OPNAVINST 4700.7M [83, 84].

Repair. Corrects degradations to return the item to the current baseline; repairs that change form, fit, or function require an approved modernization document to avoid unauthorized configuration changes under OPNAVINST 4700.7M [78].

EDOs must articulate which work package element they are defending, because it dictates the funding source (e.g., FUNDS-D/FUNDS-ALT for modernization versus OC&MF for maintenance) and who holds approval authority.

6.2.2 Governance, Roles, and Documentation

Modernization governance. The Navy Modernization Program (NMP) integrates OPNAV resource sponsors, PEOs, Participating Acquisition Resource Managers (PARMs), Warfare Centers, and Fleet forces through the NMP Executive Steering Group and Sub-Class Modernization Coordinating Groups; the process is codified in SL720-AA-MAN-030, TS9090.310, and the Joint Fleet Maintenance Manual (JFMM) [84, 85, 86].

Key documents. Modernization Change Documents capture the requirement, Justification and Cost Forms allocate funding, SHIPALT Records (Record of Change) provide lifecycle traceability, installation drawings direct work, and the Alteration Equivalent to Repair (Alteration Equivalent to Repair (AER)) process keeps urgent fixes legal [83, 86].

Execution agents. Alteration Installation Teams (AITs), RMMCOs, and SUPSHIPS integrate modernization tasks into availabilities while Naval Supervising Activities verify that work followed the approved package (*Appendix B.1: SUPSHIP is led by RDML Jonathan J. Jett-Parmer with RDML Robert J. Dodson (USNR) as reserve deputy (Appendix B.1))* [85, 86].

TABLE XII: MODERNIZATION DOCUMENT OWNERSHIP AND FUNDING TIE POINTS

Document	Primary Owner	When Used	Funding / Approval Tie
Change Initiation Notice	Fleet sponsor or PARM	Starts the change process; screens technical need and logistics impacts before engineering begins	Drives resource sponsor endorsement and PARM tasking; no funding commitment yet
Justification and Cost Form (JCF)	PARM and resource sponsor	Allocates modernization dollars and prioritizes hulls; accompanies Change Install planning	Aligns FUNDS-D (engineering) and FUNDS-K (procurement); documents budget line and phasing
ShipAlt Record of Change	Alteration Engineering Agent	Provides authoritative configuration record across hulls and life of the change	Required for configuration audits, warranty, and to show work stayed within authorized scope
Installation Drawing / LAR	Alteration Engineering Agent and installer	Directs production work and testing during the scheduled availability or pier-side install	Must match funded scope; links to MOA/RMMCO package and turn-over tests before closeout

6.2.3 Process Flow and Funding Integration

1. **Identify need.** Fleet feedback, reliability data, or new threats drive a Change Initiation Notice, which is screened for technical feasibility and logistics impacts [83].
2. **Engineer and budget.** The Alteration Engineering Agent develops Ship Check data, installation drawings, and metrics while the PARM aligns FUNDS-D (lifecycle engineering) and FUNDS-K (procurement) appropriations; pilot efforts now allow limited OPN use for private-yard execution when approved by OPNAV N9/N83 [83, 85].
3. **Select availability.** Modernization Work Package Integration (Work Package Integration (WPI)) events match each change to a specific CNO availability, Continuous Maintenance availability, or pier-side AIT window based on lead time material and platform readiness [86].
4. **Install and certify.** AITs execute under a Memorandum of Agreement (MOA) that spells out safety, quality, and turnover requirements; completion data is entered into the NDE to update the authoritative configuration record [85, 86].

Failure to follow this sequence is why unauthorized equipment has been discovered during Board of Inspection and Survey (INSURV) events—a career ender for the responsible EDO.

6.2.4 Execution Enablers and Lessons Learned

Configuration management. Every modernization line item in the Availability Work Package (AWP) must trace back to an approved change document, logistics support analysis, and allowance parts list to avoid sparing gaps [84].

MOA discipline. The MOA between the AIT, RMMCO, the Naval Supervising Activity (NSA), and Ship's Force defines boundary isolations, tag-out responsibilities, testing, and habitability; lessons learned from Consolidated Afloat Networks and Enterprise Services (CANES) and Aegis Weapon System (AWS) installs show that unsigned MOAs invite production and safety conflicts [83, 86].

AIT employment. Use AITs when the task requires repeat installations across multiple hulls, unique Original Equipment Manufacturer (OEM) skills, or when it is more cost-effective to mobilize the capability than to train an entire shipyard trade [85].

6.3 Fleet Maintenance Policy and Implementation

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6.3.1 Policy Baseline

OPNAVINST 4700.7M. Establishes the maintenance policy for United States Navy ships: maintain the fleet at the highest practical readiness, perform work at the lowest capable level, and apply Reliability-Centered Maintenance (RCM)/Condition-Based Maintenance (CBM) methods to manage Total Ownership Cost [78].

Office of the Chief of Naval Operations Notice (OPNAVNOTE) 4700. Publishes CNO availability intervals, durations, man-day targets, and Repair Days for each platform, providing the authoritative baseline for the Long-Range Maintenance Schedule (Long-Range Maintenance Schedule (LRMS)) [87].

JFMM and Condition-Based Maintenance Plus (CBM+). USFF/PACFLTINST 4790.3 (JFMM) assigns responsibilities to TYCOMs, RMMCOs, and NSAs, while OPNAVINST 4790.16 issues CBM+ policy so sensors and analytics can replace purely time-based maintenance when justified [84, 88].

TABLE XIII: MAINTENANCE POLICY MAP AT A GLANCE

Document	Owner	Decisions Touched	Compliance Talking Points
OPNAVINST 4700.7M	CNO (N9)	Sets fleet maintenance policy baseline (lowest capable level, RCM/CBM, color-of-money discipline)	Use as the anchor for deferral justifications, craft risk acceptance to its hierarchy, and cite when defending public/private yard mix and maintenance levels
OPNAVNOTE 4700	CNO (N9)	Publishes class-by-class availability intervals, durations, manday targets, and repair days that drive the Long-Range Maintenance Schedule	Treat as the authoritative schedule baseline; deviations require TYCOM/CNO alignment and supporting maintenance metrics
JFMM (COMFLTFORCOMINST 4790.3)	USFF/PACFLT	Defines execution standards for work control, assessments, certifications, and availability gate reviews	Align DFS/TSO closure, QA evidence, and RMMCO coordination to JFMM chapters; reference for watch team and AIT roles
CBMP (OPNAVINST 4790.16)	CNO (N4)	Governs condition-based maintenance approvals, data pedigree, and sensor analytics used to replace time-directed tasks	Document the analytics package (sensor health, thresholds, reliability data) and secure CBMP authority approval before removing directed tasks

The entire maintenance enterprise hinges on tying ship assessments to these references; citing a different standard during a board (e.g., OEM manual) without reconciling it to the JFMM and policy hierarchy is a red flag.

6.3.2 Strategy Frameworks

Optimized Fleet Response Plan (OFRP). OPNAVINST 3000.15 drives Fleet Response Plans that align readiness, training, and maintenance phases; maintenance windows in the OFRP billet plan are how TYCOMs defend availability lengths and sequencing [89].

Surface Force Readiness Manual (SFRM). Naval Surface Force Atlantic (SURFLANT)/Naval Surface Force Pacific (SURFPAC) 3502-series guidance aligns training milestones (Basic, Advanced, Integrated) with maintenance health metrics so a ship cannot progress if critical maintenance or material conditions remain open [90].

Naval Sustainment System—Shipyards (NSS-SY). NSS-SY injects commercial best practices (Theory of Constraints, value-stream management, digital dashboards) into the four public yards with the goal of 80% on-time Attack Submarine, Nuclears (SSNs) deliveries and no idle submarine backlog by FY26 [91].

6.3.3 Availability Approval and Execution Discipline

1. **Baseline work.** SURFMEPP/SUBMEPP compile CMP work, TYCOMs repairs, and modernization candidates into the Baseline Availability Work Package (BAWP) at A-36 months; work then flows through the CNO scheduling conference for final approval [87, 91].
2. **Directed vs. condition-based maintenance.** Directed Maintenance Tasks (Directed Maintenance Task (DMT)) must be executed regardless of condition (e.g., structural checks mandated by Submarine Safety Program (SUBSAFE)), while CBM tasks rely on evidence such as oil analysis, smart sensors, or TSRA discoveries to justify deferral [78, 88].
3. **Approval boards.** TYCOMs Availability Work Package Reviews validate that all Departure from Specifications (DFSs) and Temporary Standing Orders (TSOs) have closure plans before endorsing the package for CNO approval [84].
4. **Metrics.** Schedule adherence, manday performance, and quality escape metrics (e.g., repeat maintenance, Objective Quality Evidence (Objective Quality Evidence (OQE)) rejects) appear in USFF/PACFLT readiness briefs and feed the next LRMS update [91].

6.3.4 Trends, Risks, and Talking Points

- **Industrial base mix.** Navy policy seeks roughly a 50/50 public-private mix for surface depot work, with CVNs and SSNs/SSBNs remaining predominantly public-yard to retain nuclear competencies [78].
- **Manpower constraints.** NSS-SY reviews highlight persistent shortages in key trades (nuclear welders, radiological controls, rigging) that drive docking delays; referencing NSS-SY corrective actions shows awareness of current risk mitigations [91].
- **Digital thread.** The NDE modernization module is now the authoritative source for installed configuration, allowing planners to reconcile modernization with maintenance data without resorting to ad-hoc spreadsheets [86].
- **Fiscal discipline.** Maintenance execution must honor color-of-money boundaries (OC&MF vs. OPN/SCN); unauthorized obligation adjustments are Anti-Deficiency Act (ADA) violations [18, 31].

6.4 AIT Fundamentals and Lessons Learned

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6.4.1 Purpose and Advantages

- **Concentrated expertise.** An AIT is a trained unit (government, contractor, or hybrid) directed by an AIT manager to install specific modernization changes that require unique skills, tooling, or repetitive production-line execution [86, 92].
- **Speed and repeatability.** Using the same team across hulls captures lessons learned, reduces quality escapes, and accelerates delivery of high-priority combat system upgrades (e.g., CANES, AWS, GPS-based Positioning, Navigation, and Timing Service (GPNTS)) without overloading the shipyard workforce [85].
- **Safety and configuration control.** Dedicated teams enforce technical work documents, boundary isolations, and logistics support for each alteration, reducing the risk of unauthorized changes that could violate SUBSAFE or Level I controls [86].

6.4.2 Roles and Responsibilities

Sponsor / PARM. Assigns and funds the AIT, prioritizes hulls, and ensures logistics (allowance updates, training, tech manuals) are resourced before work begins [85].

AIT manager. Leads planning, safety, schedule integration, and quality execution; serves as the single point of contact for the Naval Supervising Activity and Ship's Force [86].

On-Site Installation Coordinator (OSIC). Coordinates day-to-day shipboard actions, manages tag-outs and system turnovers, and resolves emergent issues between the AIT, NSA, and Ship's Force [92].

RMMCO. Screens alteration packages, aligns them with availability windows, and enforces waterfront integration and reporting requirements [86].

NSA/Lead Maintenance Activity (LMA). Controls work authorization, boundary isolations, tests, and Objective Quality Evidence for work accomplished under the availability contract [84].

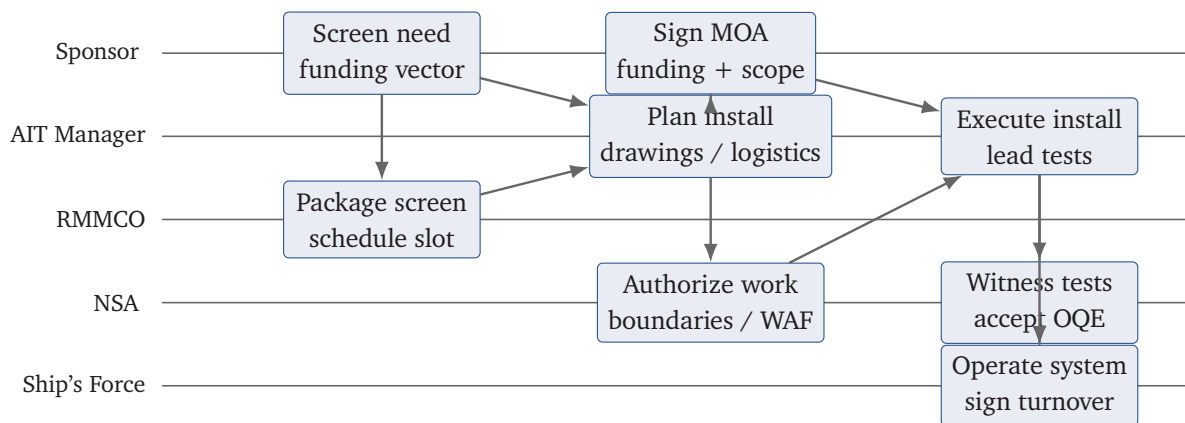


Figure 6.2. AIT swim-lane showing sponsor, AIT manager, RMMCO, NSA/LMA, and Ship's Force interactions from screening through certification.

6.4.3 Planning Controls and MOA Discipline

1. **MOA.** The MOA outlines scope, funding, schedule, safety, habitability, quality, testing, and turnover; it must be signed by the AIT sponsor, RMMCO, Ship's Commanding Officer, and NSA before work starts [86].
2. **Installation planning.** Each alteration requires a ship check, logistics certification, and completion of prerequisite changes; the NMP databases confirm that planned work matches the approved Change Document [85].
3. **Work Authorization Form (WAF)/tag-out alignment.** Integration with the Work Authorization Form process and Tag-out Users Manual (TUM) prevents conflicting isolations between the AIT and shipyard trades [93].
4. **Testing.** Entry/exit criteria, including pre-installation checkout, operational tests, and system certifications (e.g., Combat System Ship Qualification Test), are tailored per alteration and witnessed by NSA [86].

6.4.4 Lessons Learned for Board Prep

- **Late drawing changes drive rework.** Lock engineers early—AITs report that late drawing revisions, missing Interface Control Documents, or mismatched configuration data cause most delays [92].
- **Habitability and safety must be priced.** Many AITs arrive without berthing or industrial hygiene plans; ensuring the MOA captures services, hot work permits, and fall protection prevents emergent cost growth [85].
- **Document close-out.** Uploading OQE, system configuration updates, and post-install logistics changes to the NDE is mandatory—omitting them erodes Fleet supply support [86].

6.5 Waterfront Organizations, Work Controls, and Safety

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6.5.1 Naval Propulsion Program Oversight

Naval Nuclear Propulsion Program (Naval Reactors). Established by executive order and statute, Naval Reactors owns cradle-to-grave authority for nuclear propulsion design, maintenance, training, and operations; they enforce uncompromising safety rules regardless of warfighting urgency [94].

Naval Reactors Representative Office (NRRO). Resident Naval Reactors Representative Offices monitor each nuclear-capable waterfront, enforce program standards, and can halt work if safeguards are threatened; they are independent of yard leadership to prevent production pressure from eroding safety [94].

Reactor Plant Contractor's Office (RPCO). Reactor Plant Contractor Offices represent the prime contractors (e.g., Newport News Shipbuilding) and ensure contract compliance and technical rigor during construction, refueling, and depot maintenance [94].

Joint Test Groups. Department of Energy (DOE) and Navy personnel lead integrated test teams that certify plant reconstitution before turnover; they align with NRRO, SUPSHIP, and Ship's Force to execute rigorous testing sequences [94].

Board insight: referencing the Thresher, Guitarro, and Miami investigations underscores why NRRO authority and independent QA cannot be waived, even under combat timelines.

6.5.2 SUBSAFE and Submergence Systems Controls

- **SUBSAFE.** NAVSEA 0924-062-0010 mandates design, material, process, and test controls for the seawater boundary and critical systems on submarines; any work that touches a SUBSAFE boundary requires certified Level I material, approved procedures, and documented OQE [95].
- **Submergence Systems Certification.** Submergence Systems Certification manual series (SS800) manuals reinforce submarine rescue and diver support equipment requirements; the Submergence Systems Certification Board mirrors SUBSAFE rigor for non-platform systems [96].
- **Level I / critical material.** NAVSUP/NAVSEAINST 4440.16 controls procurement, receipt, and storage of Level I and SUBSAFE material to prevent counterfeit or degraded items from entering the supply chain [97].
- **Industrial safety.** S0570-AB-SAF-010 (fire protection) and S9002-AB-MAN-010 (submarine industrial safety) govern hot work, confined space, and damage control readiness during availabilities [98, 99].

6.5.3 Work Authorization and Tag-out Discipline

1. **WAF.** The NSA/LMA generates WAFs to define scope, isolations, and boundary conditions; all trades and AITs must align their work with the active WAF or request revisions through the Project Manager [84].
2. **Tag-out Users Manual.** S0400-AD-URM-010 prescribes red/amber tags, double barrier requirements, and control authority; the Ship's Force owns the tag-out log even when a contractor executes the work [93].

3. **Special evolutions.** Diving, radiological work, and ordnance moves require additional approvals (e.g., Naval Sea Systems Command Supervisor of Salvage and Diving Supervisor of Salvage and Diving (SUPSALV), Naval Ordnance Safety); EDOs must ensure these are embedded in the availability plan [94, 100].

6.5.4 Waterfront Organization Insights

- **Unified waterfront teams.** Naval Shipyards, Regional Maintenance Centers, RMMCOs, and Ship's Force form integrated production and test teams; weekly Project Management Reviews should cover safety, quality, schedule, and funding status [94].
- **Culture of questioning attitude.** The Naval Propulsion Program teaches that any worker can stop a job; referencing the Miami (SSN-755) fire (shipyard worker's vacuum) shows how minor lapses escalate without that culture [94].
- **Documentation.** Work packages, technical work documents, and checklists must be updated in the NDE to preserve traceability for SUBSAFE and other certifications [95].

6.6 Battle Damage Assessment and Repair

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6.6.1 Strategic Context

- **Global competition.** Renewed great-power competition with the People’s Republic of China (PRC) and Russia (plus persistent Democratic People’s Republic of Korea (DPRK) provocations) means distributed maritime operations must survive massed fires and cyber/electromagnetic attacks [101, 102].
- **Wartime Acquisition Response Plan (Wartime Acquisition Response Plan (WARP)).** SECNAV’s WARP memo directs every SYSCOM/PEO to maintain plans, contracts, and trained people to pivot from peacetime acquisition to wartime recovery of ships and systems [103].
- **GAO findings.** GAO-21-246 warns that current Battle Damage Assessment and Repair (BDAR) capabilities are insufficient for contested logistics, highlighting gaps in expeditionary repair, pre-positioned materiel, and command-and-control [104].

6.6.2 BDAR Continuum

- Damage Control.** Ship’s Force fights fires, floods, and casualties to stabilize the platform, preserve combat power, or keep it afloat until assistance arrives [102].
- Rescue and Assistance (R&A).** Military Sealift Command (Military Sealift Command (MSC)) tugs, Mobile Diving and Salvage Units (Mobile Diving and Salvage Unit (MDSU)), and SUPSALV assets conduct emergency towing, underwater repairs, object retrieval, and environmental mitigation [100].
- Damage Assessment.** Ship’s Force, salvage teams, and ashore Damage Assessment Teams from RMCs, NSYs, or SEA 21 analyze hull integrity, stability, and mission systems to recommend disposition [102].
- Expeditionary Repair.** Forward Deployed Ship Repair Team (FDSRT) and Expeditionary Maintenance and Repair Facility (EMRF) deploy with Shop-in-a-Box (SIB) sets to restore mobility or key mission systems without returning to a depot [102, 105].
- Depot Repair / Disposition.** NSYs, RMCs, and private yards perform major repairs, partial restorations, or decide to scrap; choices depend on threat, capability needs, cost, and industrial capacity [102].

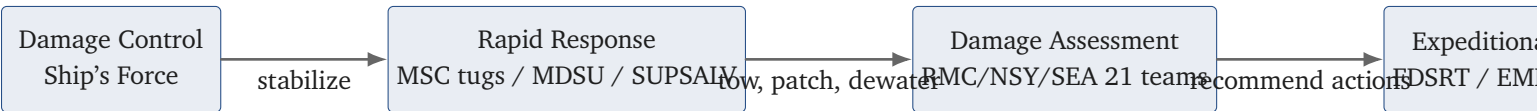


Figure 6.3. BDAR continuum from damage control through expeditionary repair to depot-level recovery, with primary responsible organizations at each step.

6.6.3 Command, Control, and Technical Authority

- **Operational control.** Fleet commanders designate BDAR task forces; Task Force logistics commanders orchestrate salvage, FDSRT, and EMRF employment while coordinating with theater Joint Logistics Enterprise nodes [106, 107].

- **Technical authority.** Wartime does not suspend Technical Authority—NAVSEA, Naval Reactors, and Warfare Centers still adjudicate waivers; however, WARP allows pre-planned delegated authorities and acceptance of graded risk when documented [103].
- **Reporting.** CASREP 4.0 timelines and Battle Damage Assessment reports flow through OPNAVINST 3040.2B to inform reconstitution priorities and Congressional notifications [108].

6.6.4 Logistics, Contracting, and Fiscal Readiness

Prepositioned materiel. Maritime Support Plan (MSP) force packages require pre-kitted SIBs, hull patches, and mobile power kits staged throughout the Indo-Pacific, United States European Command (USEUCOM), and United States Central Command (USCENTCOM) theaters [105].

Contracting. FAR Part 18 and DFARS Part 218 emergency authorities (e.g., oral solicitations, undefinitized contract actions, Defense Priorities and Allocations System ratings) enable rapid award, but commands must still document justification and maintain traceability for later definitization [31, 35].

Funding. BDAR relies on OMN for emergent repairs, SCN for major restorations, and Weapons Procurement, Navy (WPN) when combat system replacement constitutes procurement; Department of War (DoW) FMR Volume 3 prescribes reprogramming thresholds [18].

Supply chain resilience. The Resilient Maritime Logistics initiative (Project 33) aligns OPNAV N4, Fleet Logistics Centers, and Defense Logistics Agency to keep people, parts, and processes flowing under contested logistics [101, 102].

6.6.5 Maritime Support Plan Force Packages

1. **Afloat Augmentation.** 25–50 artisans augment the shipboard IMA to enable 24/7 casualty response within a strike group [105].
2. **Afloat Independent Maintenance Activity.** Tender-like capability distributed across vessels of opportunity that integrates EMRF, FDSRT, and logistics cells under Task Force Operational Control (OPCON) [105].
3. **Ashore Lean / Robust / Austere.** Scalable shore nodes that combine contractor oversight teams, NAVSEA contracting cells, and host-nation support to execute depot-equivalent work forward [105].

EDOs should be ready to explain which force package they would recommend for a given scenario, how long it takes to activate, and how technical authority, contracts, and funding are synchronized.

6.7 CNO Availability Planning

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6.7.1 Balancing Requirements, Resources, and Readiness

Customers. Fleet commanders and TYCOMs drive capability requirements and availability length based on the Optimized Fleet Response Plan and readiness gaps [89, 109].

Planners. Shipyards, RMCs, and planning activities must sequence work, tooling, infrastructure, and LLTM so the schedule is executable [109].

Resources. Budgets (OC&MF, OPN, FUNDS-D/FUNDS-K), workforce, and pier/dock capacity must align with CNO-directed Repair Days [87].

EDOs should articulate how they reconcile conflicting demands—for example, deferring a lower-criticality modernization so mandatory SUBSAFE work fits within the allotted man-days.

6.7.2 Availability Work Package Development

1. **Baseline (A-36 to A-24).** SURFMEPP/SUBMEPP derive the BAWP from the CMP, TYCOMs repairs, and modernization candidates; Ship Checks, Condition Found Reports, and TSRA results refine the package [84, 109].
2. **Initial Planning (A-21).** Planning Activities (Carrier Planning Activity, Submarine Planning Activity, RMCs) reconcile the BAWP with Class Plan priorities and issue the Initial Availability Work Package (AWP) [109].

3. **Final Planning (A-12 to A-3).** Results of pre-availability tests (Pre-Availability Test (PATs)/Pre-Overhaul Test (POTS)), DFSs, TSOs, and new modernization approvals are adjudicated; the final AWP is issued no later than three months before A-0 [87, 109].
4. **Contracting.** For surface private-yard availabilities, Multiple Award Contract, Multi-Order (MAC-MO) constructs increase competition and flexibility; public yards rely on Advanced Industrial Management (AIM) processes [109].

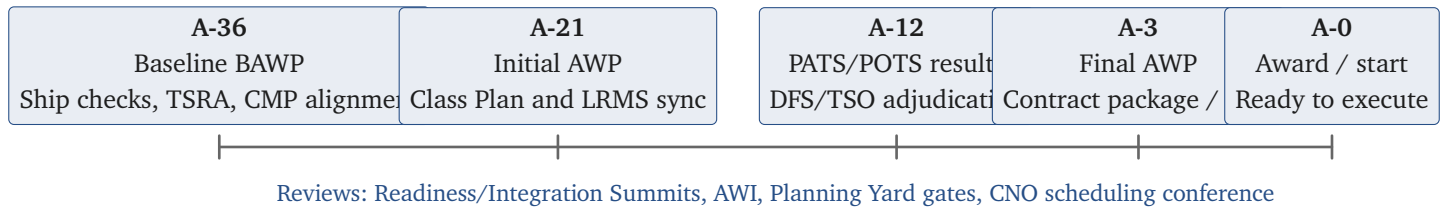


Figure 6.4. CNO availability planning timeline (A-36 through A-0) with key reviews and deliverables.

6.7.3 Planning Teams and Integration Forums

- **Carrier Team One.** Aligns NAVSEA 08/04/21, NAVAIR, Newport News Shipbuilding, and Fleet stakeholders to drive aircraft carrier availability excellence [109].
- **Surface Team One.** Unites SEA 21, SEA 05, SURFMEPP, TYCOMs, and PEOs to tackle systemic issues (e.g., late drawings, material delays) across destroyer/cruiser/amphibious availabilities [109].
- **Submarine Team One.** Led by PMS 392/SUBMEPP, integrates CNRMC, NAVSEA 08/21, and public yards to sustain SSNs/SSBNs availabilities and implement NSS-SY initiatives (*Appendix B.1: RDML Dan L. Lannamann holds the CNRMC billet*) [109].
- **Planning summits.** Readiness Summits, Availability Work Integration (Availability Work Integration (AWI)) events, and Integrated Production Planning (Integrated Production Planning (IPP)) meetings synchronize modernization, maintenance, logistics, and testing products [109].

6.7.4 Risk Controls and Board-Ready Talking Points

Critical path management. Identify the top three critical path evolutions (e.g., tank blasting/coating, propulsion plant maintenance, combat system rip-outs) and ensure early start requirements—barge services, LLTM, training—are executable before A-0 [109].

Material readiness. Demand 100% receipt of LLTM and government-furnished material before award; highlight mitigation plans for supply-chain risk (dual sources, cannibalization, additive manufacturing) [109].

Growth vs. new work. Explain how your team budgets anticipated growth work (historical allowances) versus truly new work (unique casualty), as well as the screening process for adding either to the contract [84].

Metrics. Use NSS-SY dashboards (Earned Value, manday burn, schedule adherence) and weekly risk reviews to show how you keep leadership informed and trigger help early [109].

6.8 CNO Availability Execution

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- [107] *Naval Warfare Publication 4-12: Navy Salvage Operations*Naval Warfare Publication 4-12: Navy Salvage Operations, NWP 4-12, Naval Warfare Publication, Navy Salvage Operations, Naval Warfare Development Command, Apr. 30, 2015. Accessed: 2025-11-14. [Online]. Available: <https://doctrine.navy.mil>
- [108] *OPNAVINST 3040.2B, Reporting of Significant Events and CASREPs*OPNAVINST 3040.2B, Reporting of Significant Events and CASREPs, OPNAVINST 3040.2B, Instruction, Casualty Reporting (CASREP) policy, Office of the Chief of Naval Operations, May 22, 2019. Accessed: 2025-11-14. [Online]. Available: <https://www.secnnav.navy.mil/doni/default.aspx>
- [109] “4.2.1. CNO Availability Planning”Engineering Duty Officer School, “4.2.1. CNO Availability Planning,” Engineering Duty Officer (EDO) School, Coursebook Module 4.2.1, version 5.1.26, Mar. 26, 2025, CIN A-4N-0021/-0034.
- [110] “4.2.2. CNO Availability Execution”Engineering Duty Officer School, “4.2.2. CNO Availability Execution,” Engineering Duty Officer (EDO) School, Coursebook Module 4.2.2, version 5.1.26, Mar. 26, 2025, CIN A-4N-0021/-0034.

6.8.1 Project Teams and Responsibilities

Shipyard team. The Project Superintendent (overall cost/schedule/safety owner), Project Manager (budget, earned value, planning integration), Assistant Project Superintendents (zone leads), Engineering & Planning (technical work documents, boundaries), and Business & Planning (funding, schedule coordination) form the core team [110].

RMC/private-yard team. The Port Engineer, Ship’s Maintenance Manager (Ship’s Maintenance Manager (SMM)), Ship’s Force Maintenance Officer, TYCOM representative, Contracting Officer, and contractor Project Manager manage planning yard availabilities [110].

Ship’s Force. Commanding Officer (CO)/Executive Officer (XO)/Chief Engineer (CHENG) maintain safety, tag-out, and configuration control; Combat Systems and Supply Officers coordinate tests, Shipboard Operational Verification Tests (SOVTs), and material receipt [84].

Specialists. Integrated Test Engineers, Project Support Engineers, Quality Assurance Specialists, Alteration Coordinators, and Administrative Contract Specialists provide focused oversight on testing, configuration, QA, and change management [110].

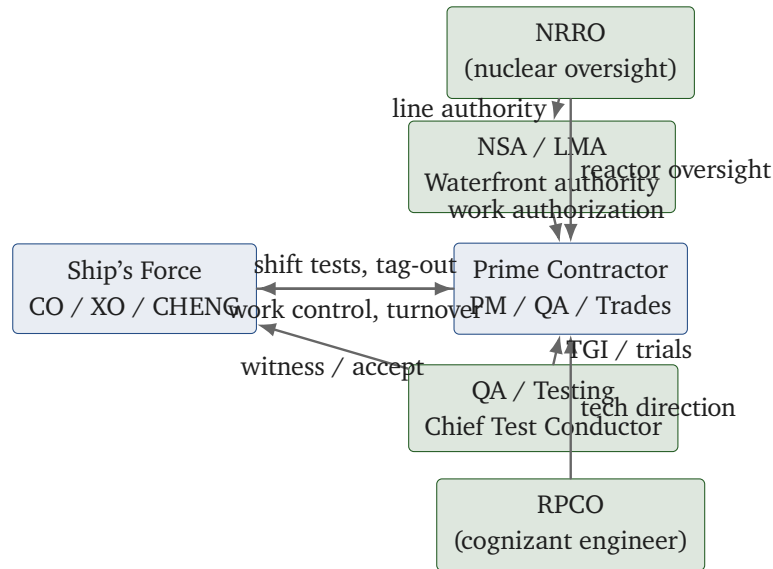


Figure 6.5. Nuclear waterfront project-team interactions across NRRO, RPCO, NSA/LMA, Ship's Force, and the prime contractor.

6.8.2 Execution Phases

1. **Nesting / Arrival.** Establish temporary services, fire watches, habitability, industrial hygiene controls, and turnover from Ship's Force to the Maintenance Activity; complete mandatory briefings (JFMM Chapter 2, Tag-out Users Manual) [110].
2. **Rip-out & Access.** Remove interferences, open systems, and conduct initial WAF reviews; late identification of hazardous material and access conflicts is a top schedule killer [84].
3. **Open & Inspect.** Execute inspections (tank, structural, piping) and document growth work; per NAVSEA Standard Item 009-01, inspections should be 100% complete by A+20% to preserve schedule margin [110].
4. **Repair & Production.** Complete component tasks, manage new vs. growth work approvals, and track critical path milestones using Task Group Instruction (TGI) [110].
5. **Testing, Trials, Certification.** Execute compartment turnover, Light-Off Assessment (LOA), SOVT, Combat System Ship Qualification Trial (CSSQT), Builder's/Acceptance Trials, and Sea Trials before Production Completion Date (PCD) [84].

TABLE XIV: CNO AVAILABILITY EXECUTION PHASES

Phase	Key Deliverables	Controlling Documents	Responsible Leads
Nesting / Arrival	Temporary services, fire watch plan, habitability, turnover briefs, tag-out baseline	JFMM Chapter 2; Tag-out Users Manual; local safety manuals	Project Superintendent, Safety Officer, Ship's Force Department Heads
Rip-out & Access	Interference removal, WAF approvals, confined space/hazardous material plans	WAF/Tag-out procedures; hot work and gas-free engineering instructions	Zone Assistant Project Managers, Production Shop Leads, Gas Free Engineer
Open & Inspect	Inspection reports, growth work candidates, material condition snapshots	Standard Item 009-01; QA/inspection instructions; class maintenance plans	Engineering & Planning, Quality Assurance, Class Maintenance Planner
Repair & Production	Completed work items, change control (JSNs/contract mods), schedule metrics, resource leveling	Contract/SOW; work package and change control processes; integrated master schedule standards	Project Manager, Port Engineer/KO, Business & Planning, Contractor PM
Testing, Trials, Certification	Test agendas, TGI alignment, LOA/SOVT/CSSQT completion, sea trial cards, turnover certificates	JFMM testing chapters; platform test manuals; INSURV/Builder's trial requirements	Chief Test Conductor, Ship's Force Test Coordinator, NSA/LMA witnesses

6.8.3 Governance, Quality, and Work Control

- **NSA / LMA.** Certifies that work is authorized, technically compliant, and properly tested before acceptance; reviews OQE prior to signing completion certificates [84].
- **Work control.** WAFs, TGI sequencing, and boundary tags must remain synchronized; growth work requires screening panels (Ship's Force, TYCOM, KO) to verify funding, schedule, and technical impacts [110].
- **New vs. growth work.** New work is undiscovered scope added post-award; growth work expands existing items. Both demand TYCOM approval, funding source identification, and contract mods (or Job Sequence Number (JSN) adjustments for public yards) [84].
- **Testing governance.** Chief Test Conductors manage test agendas; engineering agents and Ship's Force witness tests, and discrepancies must be adjudicated before certification events (e.g., LOA, CSSQT) [110].

6.8.4 Lessons Learned

- **Momentum matters.** Availability performance often collapses after undocking; maintain daily production meetings, emphasize housekeeping/safety, and enforce material staging to avoid second-half slumps [110].

- **Communication is a weapon system.** Cross-functional representatives (yard, Ship's Force, TYCOM, QA, Supply) must keep each other informed about configuration, testing, and material risks; unresolved disconnects manifest as trial failures [110].
 - **Document everything.** Capture change incorporation, OQE, and lessons in the NDE immediately—waiting until the end invites errors and delays closeout [84].
-

7 Civilian Personnel

7.1 Civilian Personnel Fundamentals

- [111] “6.1.1. Civilian Personnel Overview”Engineering Duty Officer School, “6.1.1. Civilian Personnel Overview,” Engineering Duty Officer (EDO) School, Coursebook Module 6.1.1, version 5.0, Dec. 18, 2024, CIN A-4N-0021/-0034.
- [112] “6.1.2. Workforce Shaping and Classification”Engineering Duty Officer School, “6.1.2. Workforce Shaping and Classification,” Engineering Duty Officer (EDO) School, Coursebook Module 6.1.2, version 5.2, May 29, 2025, CIN A-4N-0021/-0034.
- [113] “6.1.3. Equal Employment Opportunity”Engineering Duty Officer School, “6.1.3. Equal Employment Opportunity,” Engineering Duty Officer (EDO) School, Coursebook Module 6.1.3, version 5.1, Apr. 23, 2025, CIN A-4N-0021/-0034.
- [114] “6.1.4. Employee Relations”Engineering Duty Officer School, “6.1.4. Employee Relations,” Engineering Duty Officer (EDO) School, Coursebook Module 6.1.4, version 5.3, May 29, 2025, CIN A-4N-0021/-0034.
- [115] “6.1.5. Labor Relations”Engineering Duty Officer School, “6.1.5. Labor Relations,” Engineering Duty Officer (EDO) School, Coursebook Module 6.1.5, version 5.1, Mar. 21, 2025, CIN A-4N-0021/-0034.

7.1.1 Why CIVPERS management matters

Workforce weight. Civilian Personnel (CIVPERS) form the majority of the engineering and logistics talent at SYSCOMs and NWCs; how you engage and develop them directly sets Fleet readiness outcomes [111].

Human Resources Office (HRO) lanes. Core services are staffing, classification/compensation, employee relations, and labor relations; pull them in early for hiring plans, performance actions, and workplace changes [111].

Guidance stack. Primary references flow from Title 5, Office of Personnel Management (OPM) guidance, Department of the Navy policy, and local agreements; document assumptions and keep counsel/HRO informed to avoid missteps [111].

7.1.2 Workforce shaping and classification

Pay systems. Know the differences between General Schedule (GS), Wage Grade (WG), pay-band systems, and Senior Executive Service (SES) tiers; align the billet to the right system before recruiting [112].

Classification and Fair Labor Standards Act (FLSA). The Position Description (PD) drives pay plan, series, and grade/band; verify FLSA exempt/non-exempt status and ensure duties match the description to avoid backpay claims [112].

Filling vacancies. Compare reassignment/merit promotion, Direct Hire Authority (DHA), or competitive hiring; honor mandatory placement programs and capture recruiting timelines up front [112].

Downsizing levers. Sequence Reduction in Force (RIF), Separation Incentive Program (SIP), Voluntary Early Retirement Authority (VERA), and furlough options carefully; pair with workforce development to retain critical skills while meeting budget constraints [112].

7.1.3 Equal Employment Opportunity anchors

Bases and theories. The ten protected bases (race, color, religion, retaliation, sex, national origin, equal pay, age, disability, genetic information) and theories (disparate treatment, disparate impact, harassment, reprisal) frame every Equal Employment Opportunity (EEO) review [113].

Process discipline. Preserve timelines: timely contact with a EEO counselor, alternative dispute resolution where appropriate, investigation, and decision/hearing; supervisors must avoid reprisal and secure records [113].

Harassment prevention. Distinguish hostile environment from quid pro quo scenarios, act on any notice (verbal or written), and coordinate with HRO/Equal Employment Opportunity Commission (EEOC) liaisons for training and documentation [113].

7.1.4 Employee relations essentials

Work schedules and leave. Use standard, flexible, or Compressed Work Schedule (CWS) structures thoughtfully; apply annual/sick leave and leave without pay rules consistently and reinforce timekeeping discipline to avoid Absent Without Leave (AWOL) issues [114].

Performance cycle. Set clear standards early, conduct midpoint feedback, and close out on time; tailor awards (time-off, cash, quality step increases) to reinforce desired behaviors [114].

Conduct v.s. performance. Separate misconduct (e.g., AWOL, falsification) from poor performance; use counseling and a documented Performance Improvement Plan (PIP) before escalating to adverse action, and keep counsel/HRO informed [114].

7.1.5 Labor relations touchpoints

Bargaining unit scope. Confirm who is in the bargaining unit (non-supervisory, non-confidential employees) and what the collective bargaining agreement covers; track official time and steward access obligations [115].

Management rights v.s. bargaining. Management retains the right to assign work, direct employees, and determine budget; bargain procedures and impact when changing conditions of employment and document notice-to-union steps [115].

Meetings and disputes. Provide the union a chance to attend formal discussions and Weingarten meetings; log official time and resolve Unfair Labor Practices (ULPs) through grievance procedures or Federal Labor Relations Authority (FLRA) channels [115].

Appendices

A Platinum Card

The *Platinum Card* is a primary study tool that will have most of the information required for your boards. Sources: EDO Coursebook Modules 3.1.1, 3.1.3, 3.1.4, and 3.1.6. Be able to draw and talk through Figures A.1–A.9.

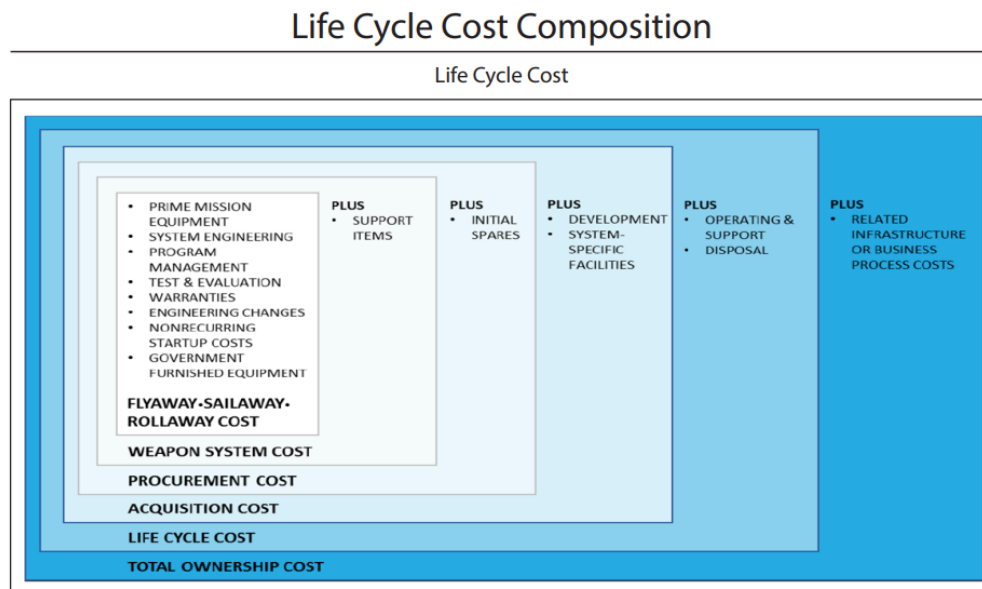


Figure A.1. Platinum Card: Life Cycle Cost Composition

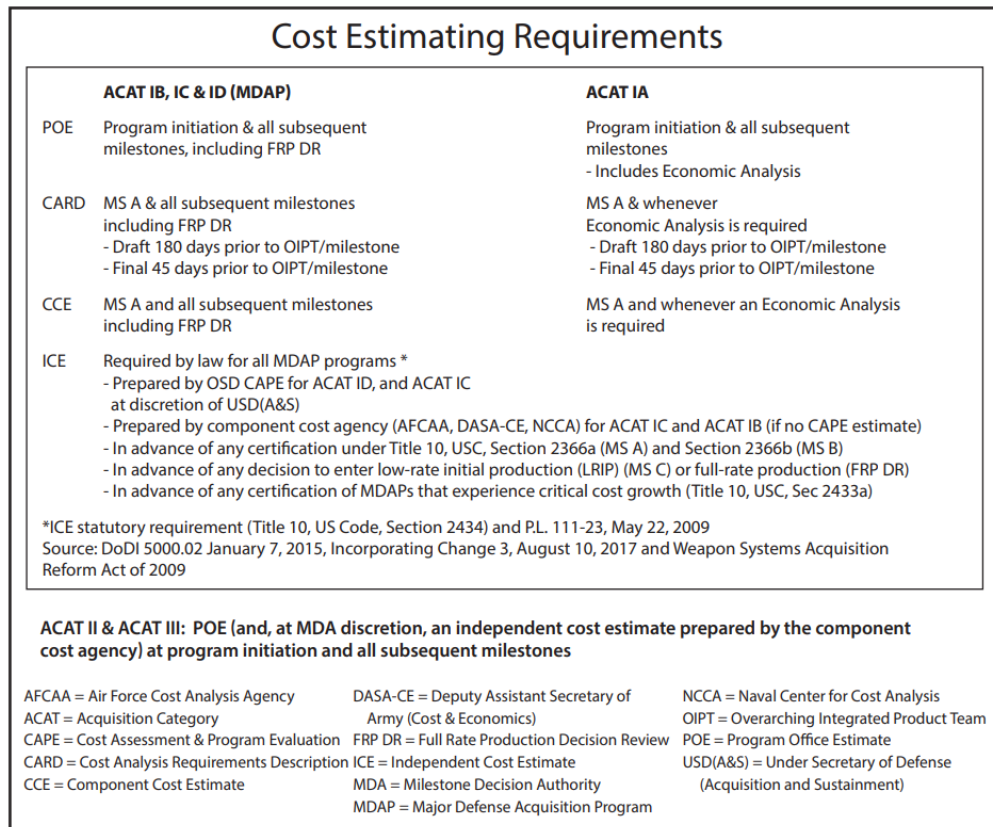


Figure A.2. Platinum Card: Cost Estimating Requirements

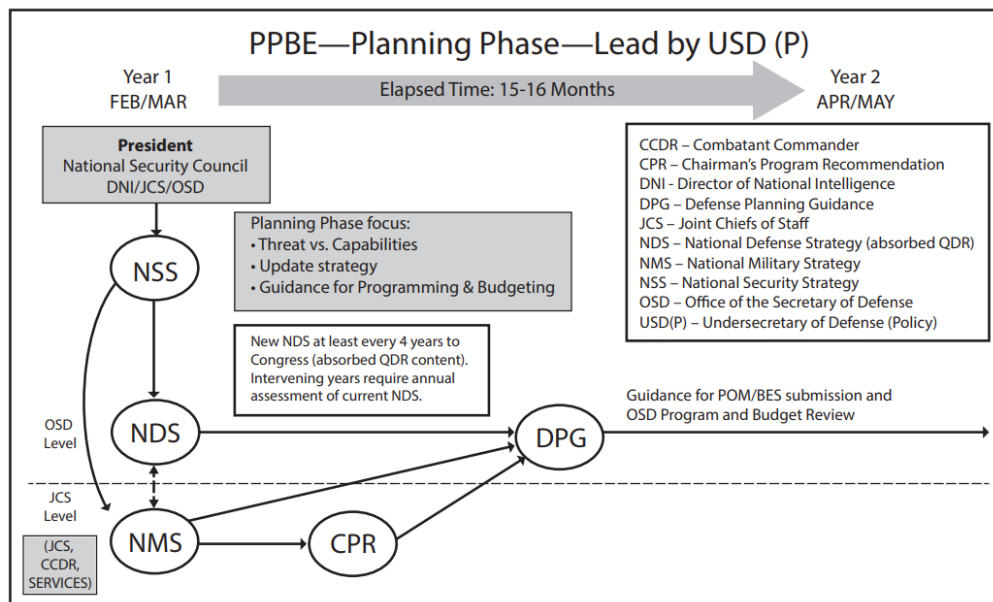


Figure A.3. Platinum Card: PPBE Planning Phase (led by Under Secretary of Defense for Policy (USD(P)))

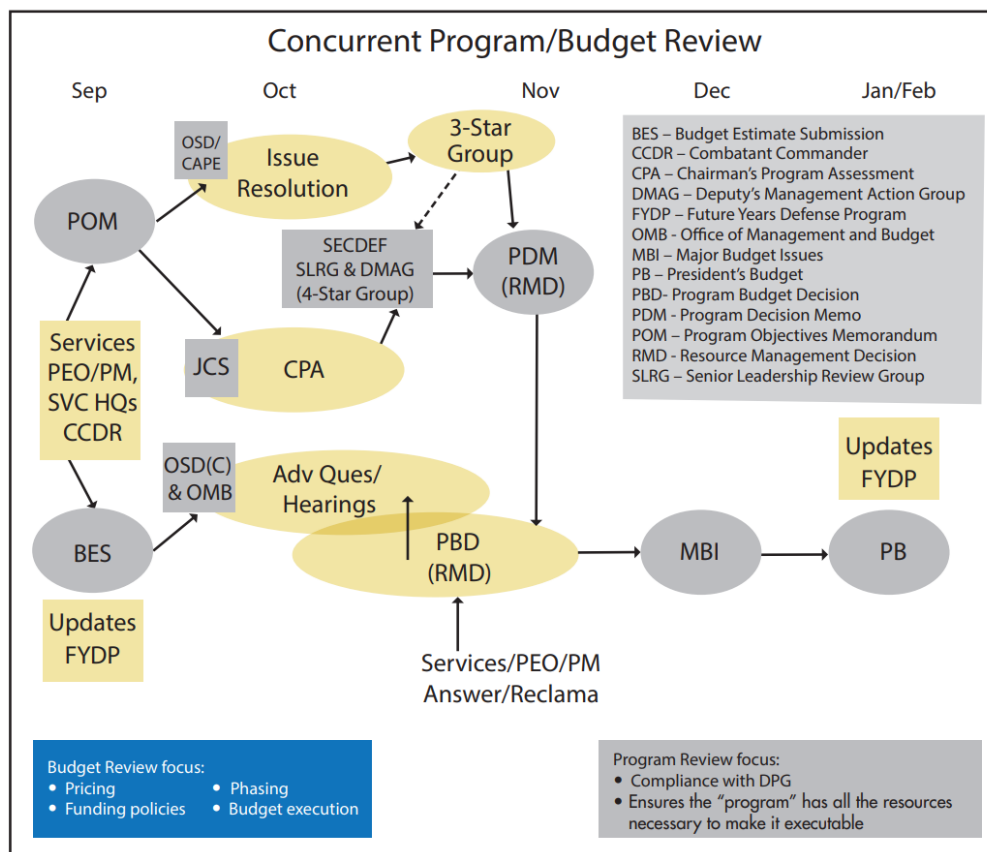


Figure A.4. Platinum Card: Concurrent Program/Budget Review

Below Threshold Reprogramming				
Amounts are cumulative over Entire Period of Obligation Availability				
APPRN	MAX INTO	MAX OUT	LEVEL OF CONTROL	OBL AVAIL
RDT&E	Lesser of +\$15M or + 20%	Lesser of - \$15M or - 20%	Program Element	2 Years
PROC	Lesser of + 15 M or + 20%	Lesser of - \$15M or - 20%	Line Item	3 years SCN: 5 Years
O&M	+ \$15M	- \$15M	Budget Activity (or Defense Agency) Some Sub-Activity Limitations on Decreases (see reference below)	1 Year
MILPERS	+ \$15M	- \$15M	Budget Activity	1 Year
MILCON	Lesser of +\$2M or + 25%	No Specific Congressional Restriction	Project	5 Years

Reference Sources: DoDFMR 7000.14-R, Volume 3, Chapter 6 (Sept 2015) and Chapter 7 (Mar 2011) Joint Explanatory Statement of Congress Making Appropriations for the Department of Defense for Fiscal Year 2024

Figure A.5. Platinum Card: Below Threshold Reprogramming

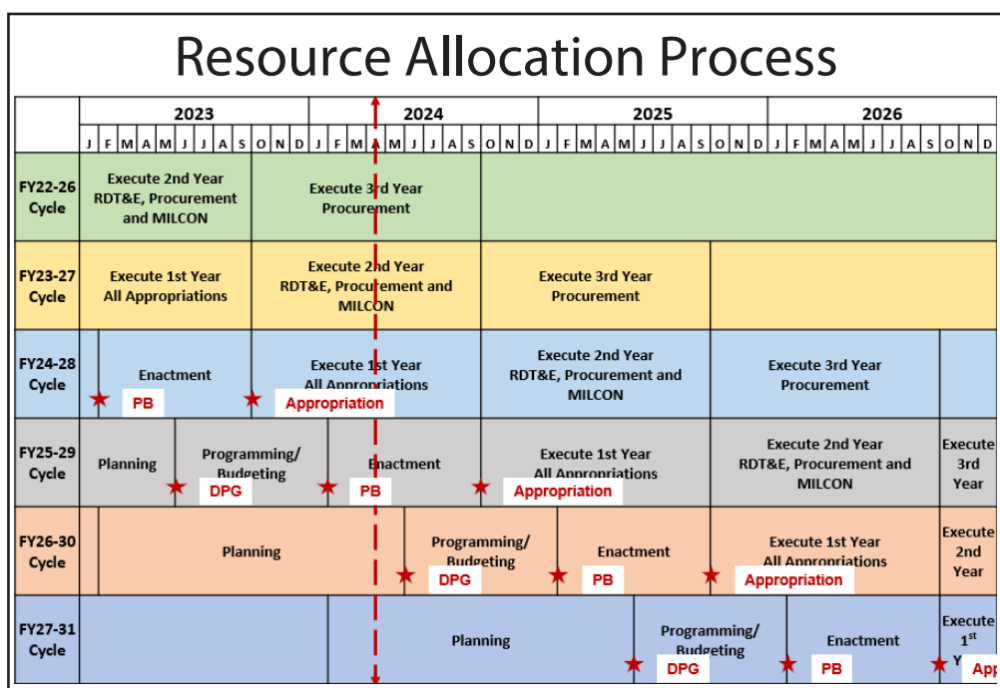


Figure A.6. Platinum Card: Resource Allocation Process

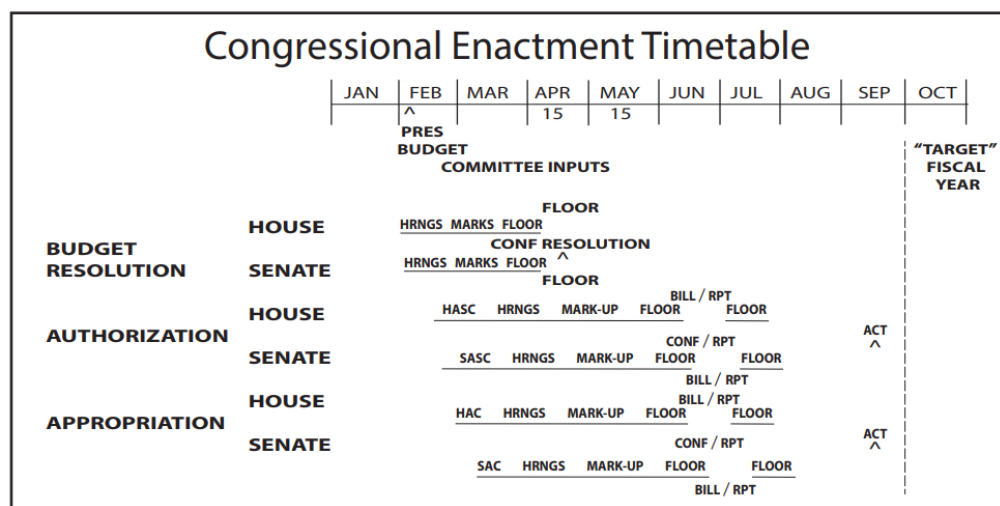


Figure A.7. Platinum Card: Congressional Enactment Timeline

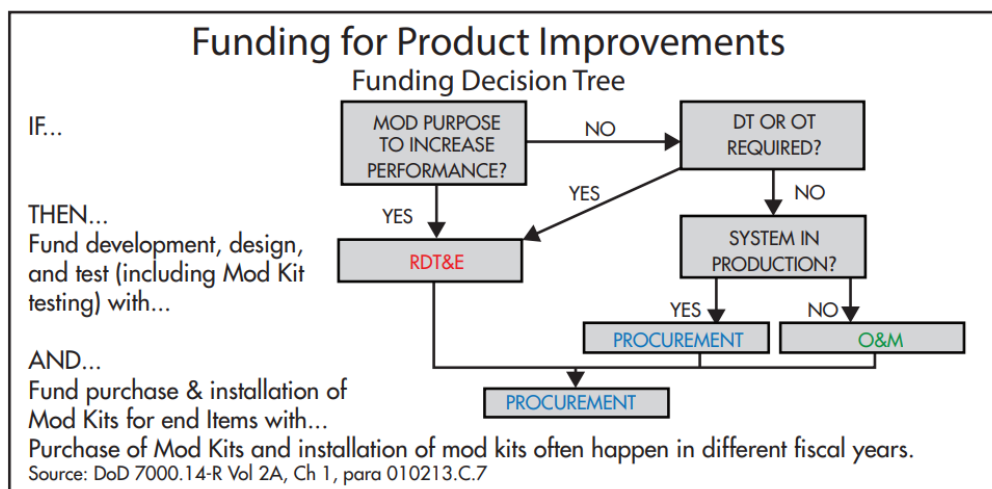


Figure A.8. Platinum Card: Funding for Product Improvements

RDT&E. Engineering changes requiring development/test.

Procurement. Production/installation of approved mods/end-items.

O&M. Minor mods/installation labor when authorized and not creating a new end-item.

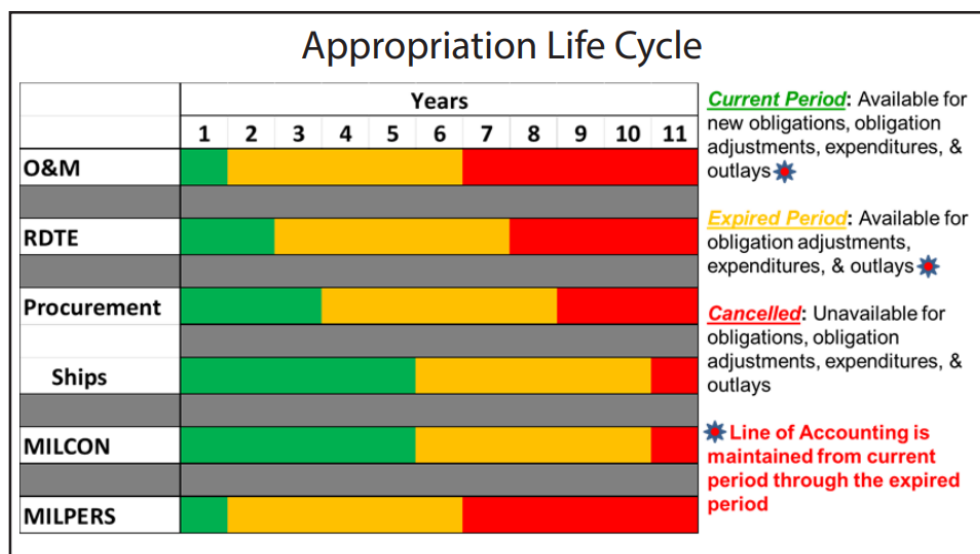


Figure A.9. Platinum Card: Appropriation Life Cycle

B Key Roles in Acquisition and Modernization

B.1 EDO Flag Officers (snapshot)

Verify names/billets the week of your board. Current as of: 05 Dec 2025 (sourced from Navy flag officer biographies).

NON-EDO BILLETS (FOR AWARENESS ONLY). The NAVSEA December 2025 org chart shows several key billets led by non-EDO officers: COMNAVSEA (ADM William J. Houston, Submarine officer), PEO IWS (RDML Thomas J. Dickinson, Surface Warfare AP), PEO MLB (RDML Kevin P. Biehn, Surface Warfare AP), NAVSEA 05 (RDML Jay A. Seif, Submarine AP), PEO UWS (RDML Douglas J. Adams), PEO SSBN/SEA 07 (RDML Todd Weeks), and PEO USC (currently SES; RDML Kevin R. Smith was relieved in May 2025)[9].

VICE ADMIRALS

VADM Seiko Okano.

PMILDEP to ASN(RD&A); former COMNAVWAR and PEO C4I / Project Overmatch lead[4].

REAR ADMIRALS (UPPER HALF)

RADM Brian A. Metcalf.

PEO Ships; delivers surface combatants, auxiliaries, and amphibious ships[6].

RADM Casey J. Moton.

PEO CVN; leads CVN new construction and in-service nuclear availability portfolios[7].

RADM Jonathan E. Rucker.

PEO SSN; oversees attack submarine acquisition and lifecycle support[8].

RADM J. Kurt Rothenhaus.

COMNAVWAR; commands NAVWAR and its PMWs delivering C4I/cyber capabilities[5].

RADM Douglas L. Williams.

Director for Test, MDA; synchronizes joint missile defense test campaigns[116].

REAR ADMIRALS (LOWER HALF)

RDML Peter D. Small.

NAVSEA Chief Engineer and Deputy Commander for Naval Systems Engineering; Commander, NSWC/NUWC[117].

RDML Dan L. Lannamann.

CNRMC; oversees regional maintenance centers and depot availabilities[82].

RDML Dianna Wolfson.

Director, Fleet Maintenance, USFF; integrates TYCOMs readiness and ship maintenance execution[118].

RDML Daniel W. Ettlich.

Director, Fleet Maintenance (N43), PACFLT; leads west-coast and forward-deployed maintenance readiness[119].

RDML Scott M. Brown.

Deputy Commander, Industrial Operations, NAVSEA; oversees shipyard and industrial base sustainment[120].

RDML Huan Nguyen.

Deputy Commander for Cyber Engineering, NAVSEA; integrates cybersecurity and digital engineering into ship programs[121].

RDML Jonathan J. Jett-Parmer.

Deputy Commander, Supervisor of Shipbuilding, Conversion and Repair (NAVSEA); leads SUPSHIP technical authority for new construction and overhauls[122].

RESERVE EDO FLAGS**RDML Robert J. Dodson (USNR).**

Deputy Commander, Supervision of Shipbuilding (NAVSEA); aligns reserve expertise to construction oversight[123].

RDML Michael S. Richman (USNR).

Deputy Director, Regional Strike Systems, Strategic Systems Program (SSP); supports strategic weapons sustainment[124].

B.2 Key Roles in Acquisition and Modernization

Department acquisition governance.

Sources: SECNAVINST 5400.15D; SECNAVINST 5000.2G; DoDI 5000.85.

- ASN(RD&A), the SAE and CAE for the DON, sets acquisition policy, charters PEOs / DRPMs, and adjudicates milestone decisions when delegated by the Milestone Decision Authority.
- PEOs and DRPMs hold delegated program authority, establish baselines, and are accountable for translating capability needs into executable acquisition strategies.
- SYSCOM Commanders provide the workforce, infrastructure, and TA warrants that underpin PEO execution inside the Navy matrix construct.

Joint requirements governance.

Sources [62, 70].

- **JROC:** Validates joint capability requirements, prioritizes gaps, and issues JROCM direction that becomes binding input to Gate 2 and milestone documentation.
- **Joint Staff Gatekeeper/JCB:** Controls the KM/DS queue, assigns Joint Staffing Designators, and routes most Navy submissions for JCB-level validation to keep JROC time focused on the highest-risk issues.
- **FCBs:** Chair certification working groups (Net-Ready, Force Protection, System Survivability), monitor DOTMLPF-P implementation, and elevate Configuration Steering Board actions when requirement trades alter program baselines.
- **PAEs and CTCs:** The 2025 Acquisition Transformation Strategy empowers PAEs to convene Capability Trade Councils that record requirement/cost/schedule trades in real time so PEO teams keep the DAS synchronized with evolving demand signals.

Combatant command sponsors.

Sources [1, 70].

- **CCDRs** own warfighting problems, sign JUONs/JEONs, and direct lead Services or agencies to prosecute capability gaps that demand rapid solutions.
- **CCMD staffs** conduct mission-thread analysis and architecture sprints with Fleets, labs, and industry to translate operational risk into measurable capability requirements before the Gatekeeper accepts a package.
- Commanders remain voting participants during validation reviews and oversee fielding to ensure delivered capability solves the theater problem that generated the requirement.

Joint Chiefs leadership.

Sources [1, 70, 72].

- **CJCS:** Principal military adviser to the President/SECWAR, co-chairs the JROC, and leads joint concept and readiness development that frames every Capability-Based Assessment.
- **VCJCS:** Chairs the JCB, co-leads the DMAG, and arbitrates cross-Service requirement or resourcing disputes before they escalate to the Deputies or Secretary level.

OPNAV corporate leadership.

Source [1].

- **CNO:** Serves as Service Chief, chairs the Navy corporate board, and approves Navy positions for Gate 0/1 decisions before submissions move to the Joint Staff.
- **OPNAV resource sponsors (N8, N82, N9, N91):** Translate Fleet priorities into the POM/FYDP, enforce two-pass/seven-gate rigor, and synchronize requirements, budgets, and PEO execution.

Acquisition decision authorities.

Sources [3, 32, 49].

- **DAE / USW(A&S):** Issues acquisition policy, chairs the Defense Acquisition Board, and retains MDA for ACAT ID/IAM programs unless delegated.
- **SAE/ASN(RD&A):** Assigns Navy MDA for ACAT II and below, approves Service-tailored governance (e.g., two-pass/seven-gate), and charters PEOs/DRPMs.
- **Delegated MDA (PEO/DRPM):** Signs Acquisition Decision Memoranda, manages Configuration Steering Boards, and keeps KPP/APB trades synchronized with requirements direction and statutory certifications.

Program Authority chain.

Sources: SECNAVINST 5000.2G; DoDI 5000.85.

- **PEO/PA:** Owns program outcomes, charters PMs, approves acquisition strategies and baselines.
- **DRPM:** Direct-report program leads for special access or priority portfolios with the same milestone authority as PEOs.
- **Ship Program Manager (SPM)/PM:** Accountable for cost/schedule/performance, leads risk management, orchestrates IPTs.
- **Ship Acquisition Program Manager (SHAPM):** Delivers new hulls; coordinates design/build/activation sequencing.
- **Ship Lifecycle Manager (SLM):** Drives sustainment and modernization packages for in-service assets.
- **PARM/Ship Project Director (SPD):** Sponsors platform-level upgrades and alteration packages.

Technical Authority chain.

Source: SECNAVINST 5400.15D.

- **UTA / Chief Engineer:** Issues enterprise technical policy, owns warrants.
- **TWHs:** Certifies design compliance, adjudicates departures/concurrence packages.
- **Engineering Agents (ISEA, DA, AEA, SIA, TDA):** Execute lifecycle engineering; provide in-service engineering decisions under delegated authority.

Operational test oversight.

Sources [32, 51].

- **DOT&E:** Independent OSD authority that concurs on TEMPs, monitors operational test realism, and reports suitability/effectiveness outcomes to Congress.
- **Service Operational Test Agencies.** COMOPTEVFOR (COTF), AFOTEC, ATEC, and MCOTEA plan and execute IOT&E/FOT&E, render the OTRR recommendation, and provide Service suitability findings to the MDA.

Operational test readiness and fleet release.

Sources [53, 77].

- **COTF/COMOPTEVFOR:** Chairs OTRR events, certifies whether OT&E adequacy and logistics/training enablers support Fleet introduction, and can recommend restricted Fleet release when Cat I deficiencies remain.
- **MDA/PEO:** Disposition Cat I/II findings, resource fixes, and may defer FRPDR until OT&E adequacy and deficiency burn-downs meet Title 10 standards.
- **TYCOM:** Accepts Fleet risk, sets operating restrictions, and updates tactics and training based on OT&E and LFT&E findings.

Developmental test governance.

Sources [53].

- **DASW(T&E):** Designates Chief Developmental Testers and Lead Developmental Test and Evaluation Organizations for ACAT I programs, adjudicates test resource shortfalls, and co-chairs the Defense T&E Working-level Integrated Product Team (WIPT).
- **CDT:** Synchronizes the integrated test schedule, deficiency management cadence, and readiness evidence required for each TRR/milestone.
- **LDTO:** Provides instrumentation, modeling & simulation accreditation, range execution, and authoritative data packages for developmental tests.

Live-fire statutory leads.

Sources [53, 54].

- **DOT&E Live Fire Test Director:** Approves plans, monitors execution realism, and delivers the statutory lethality/vulnerability report to Congress.
- **Program Manager/Chief Engineer:** Funds threat-representative articles, integrates survivability design fixes, and aligns LFT&E data with the Acquisition Program Baseline and sustainment plans.
- **Service Safety Centers and Warfare Centers:** Provide casualty data, threat surrogates, and post-test forensic analysis that feed both live-fire reports and Fleet tactics updates.

NAVSEA headquarters roles.

Source: EDO Coursebook Module 2.1.2 (NAVSEA Organization, 2025 edition).

- **SEA 01 Comptroller:** BSO lead for NAVSEA financial governance and funds control.
- **SEA 02 Contracts:** Enterprise contracting authority, policy, and warrant management.
- **SEA 03 Cyber Engineering and Digital:** Drives cyber resiliency, digital engineering, and data transformation initiatives.

- **SEA 04 Industrial Operations:** Oversees public shipyards, maintenance execution, and quality assurance.
- **SEA 05 Chief Engineer:** Chief TA; issues warrants, certifies designs, and maintains technical standards.
- **SEA 06 Sustainment:** Manages product support strategies and lifecycle logistics integration.
- **SEA 07 Undersea Warfare:** Dual-hatted as PEO UWS; oversees undersea combat systems sustainment.
- **SEA 08 Nuclear Propulsion:** Three-hatted nuclear propulsion authority across DON and DOE roles.
- **SEA 09 Safety and Regulatory Compliance:** Aligns enterprise safety governance and reporting.
- **SEA 10 Total Force and Corporate Ops:** Manages workforce planning, corporate services, and governance.
- **SEA 21 In-Service Ships/CNRM:** Leads surface-ship sustainment and modernization (PMS 321/326/339/421/443/451, SEA 21I, SURFMEPP).

NAVWAR enterprise directorates.

Source: EDO Coursebook Module 2.1.3 (NAVWAR Enterprise, 2024 edition).

- **Code 1.0 Comptroller:** Budget formulation/execution, BSO duties, and funds certification.
- **2.0 Contracts:** Contracting policy, strategy reviews, and award/administration oversight.
- **3.0 Counsel:** Acquisition law, ethics, protests, and claim resolution.
- **4.0 Logistics and Fleet Support:** Lifecycle logistics, technical data, and fleet distance support integration.
- **5.0 Chief Engineer/TA:** Enterprise architectures, interoperability, and cyber certification authority.
- **6.0 Program Management:** Portfolio governance, milestone preparation, and PEO integration.
- **7.0 Science and Technology:** S&T portfolio management, prototyping, and technology transition.
- **8.0 Corporate Operations:** Workforce, Chief Information Officer (CIO) services, facilities, security, and public affairs.
- **FRD-100 Fleet Support:** Deployed engineering assistance and sustainment response.
- **FRD-200 Installations:** Shore/afloat C4I installation planning and cutover execution.

Program Executive Offices (Navy portfolios).

Sources: SECNAVINST 5400.15D; EDO Coursebook Modules 2.1.4 and 3.1.5.

- **PEO CVN:** Designs, builds, and sustains nuclear-powered aircraft carriers (e.g., PMS 312/378/379).
- **PEO IWS:** Develops and sustains ship/submarine combat systems; mission-aligned IWS directorates cover sensors, weapons, C2, and allied integration.

- **PEO Ships:** Oversees surface combatant and amphibious ship construction and modernization.
- **PEO USC:** Leads Littoral Combat Ship (LCS), Guided-Missile Frigate (FFG) 62, expeditionary, and unmanned surface/undersea portfolios.
- **Team Submarines (PEO SSN, PEO SSBN, Program Executive Office, Columbia-Class Submarines (PEO Columbia), PEO UWS):** Manages attack/strategic submarine acquisition, in-service support, and undersea combat systems.
- **PEO C4I:** Delivers fleet C4I; PMW 1XX focus on capability development, PMW 7XX on platform integration.
- **PEO Digital:** Provides enterprise digital services (e.g., Flank Speed) across the DON.
- **PEO MLB:** Modernizes manpower, logistics, and business IT systems.

Contracting authority (KO family).

Sources: FAR; DFARS; NMCARS; EDO Coursebook Module 3.2.1.

- **PCO:** Plans the acquisition, synchs with PMs before solicitation, awards and signs contracts/mods.
- **ACO:** Oversees post-award performance, surveillance, and payment/contract administration (often DCMA / NAVSEA field activities).
- **TCO:** Leads partial/full terminations, settlement negotiations, and equitable adjustments.
- **KO warrant:** Defines the dollar/authority limits; only the warranted KO can bind or obligate the Government.
- **Contracting Officer's Representative (COR) / assistant PM / engineering support:** Provide technical surveillance and acceptance recommendations; cannot direct work or obligate funds (FAR Parts 1 and 42).
- **HCA:** Approves actions such as letter contracts and high-value single-award IDIQs; may delegate no lower than flag/Senior Executive levels within the DON per FAR 1.601 and DFARS/NMCARS supplements.
- **SAE, ASN(RD&A):** Signs D&Fs for multiyear contracting, extraordinary relief, and other actions reserved to the Service Acquisition Executive under FAR Subparts 1.7 and 17.1.

Program Manager / KO partnership.

Sources: DoDI 5000.85; EDO Coursebook Module 3.2.1.

- PM integrates warfighter need, technical baseline, and budget; KO CICA, incentives, and change management before RFP release or modification execution.

Source selection governance.

Sources: FAR Parts 1, 5, and 15; NAVSEA Source Selection Guide (2022); EDO Coursebook Module 3.2.2.

- **SSA:** Senior official who approves the Source Selection Plan, receives SSAC/SSEB recommendations, and signs the best-value decision memorandum.

- **SSAC:** Advisory council that synthesizes SSEB findings, compares proposals across factors, and briefs the SSA on trade-offs.
- **SSEB:** Multi-disciplinary evaluators (technical, management, past performance, cost/price) who rate proposals against Section M factors.
- **Small Business Professional / Competition Advocate:** Confirms set-aside decisions, reviews subcontracting plans, and endorses synopsis waivers.
- **Cost/Price Analyst & Legal Counsel:** Validate reasonableness determinations, alignment of Sections L/M, and clause sufficiency before release.
- **Award Fee Determining Official & Award Fee Board:** Plans award-fee periods, chairs performance reviews, and signs the determination memo authorizing or withholding fee payments (FAR Part 16; NAVSEA Source Selection Guide).

Fiscal control & certification.

Source: Department of War Financial Management Regulation (DoW FMR) 7000.14-R Volume 3.

- **Funds Certifying Official/Comptroller:** Verifies purpose/time/amount before obligation; ADA safeguard.
- **Program/Budget Analyst:** Tracks execution, monitors reprogramming thresholds, prepares obligation/expenditure burn-down.
- **Resource allocation chain:** OUSW(C) apportions to the Navy; OPNAV (N82/FMB) allocates to BSOs; SYSCOM comptrollers (e.g., SEA 01) issue suballocations to executing activities.

PPBE resource sponsors.

Source: EDO Coursebook Module 3.1.1 (PPBE, 2025 edition).

- **CAPE:** Provides independent cost assessments and program evaluation across the Department.
- **N8:** Builds the Navy POM, balancing capability and fiscal constraints across the FYDP.
- **N9:** Integrates warfare requirements and advocates mission-area investments.
- **N91:** Conducts cross-domain mission integration and orchestrates POM issue resolution.

Congressional resource chain.

Source: EDO Coursebook Module 3.1.2 (Congressional Enactment, 2025 edition).

- **HASC / SASC:** Authorize defense programs and policy in the annual NDAA.
- **HAC / SAC:** Produce appropriations bills that provide BA to execute Navy programs.
- **HBC / SBC:** Set topline guidance and 302 allocations through the budget resolution process.
- **CBO, CRS, GAO:** Supply independent scoring, research, and oversight that shape committee deliberations.

Cyber/Authorizations (when IT/C2 in scope).

Source: SECNAVINST 5000.2G.

- **Authorizing Official (AO):** Grants Authority to Operate; enforces Risk Management Framework (RMF) controls that drive design and integration requirements.

B.3 Combatant Commands (Reference)

AOR: The geographic (or astrographic) area assigned to a CDR for missions and forces. (Title 10)

Geographic (7):

- United States Africa Command (USAFRICOM)
- USCENTCOM
- USEUCOM
- United States Indo-Pacific Command (USINDOPACOM)
- United States Northern Command (USNORTHCOM)
- United States Southern Command (USSOUTHCOM)
- United States Space Command (USSPACECOM)¹

Functional (4):

- United States Cyber Command (USCYBERCOM)
- United States Special Operations Command (USSOCOM)
- United States Strategic Command (USSTRATCOM)
- United States Transportation Command (USTRANSCOM)

B.4 Current Events Relevant to EDOs and Acquisition

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¹USSPACECOM is treated by DoW as a **geographic (astrographic)** command with an AOR beginning at the Kármán line (~100 km)

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B.4.1 Project Overmatch and Digital Modernization

Five Eyes Project Arrangement.

In Oct 2024 Project Overmatch signed a Project Arrangement with Australia, Canada, New Zealand, and the United Kingdom, embedding Cooperative Project Personnel in San Diego to co-develop resilient coalition kill-chain networks ahead of RIMPAC 26; Capt. Remil Capili (PM), VADM Seiko Okano (steering chair/PMILDEP), Shayna Bond (international deputy), and David Flowers (international lead) direct the effort while a coalition lab stands up to support continuous testing[13].

Open DAGIR pilot.

By 18 Jun 2025 Project Overmatch, NAVWAR, the Chief Digital and AI Office, and Defense Innovation Unit (DIU) were onboarding commercial AI/ML applications through the Open DAGIR pipeline so Fleet commanders could visualize readiness data (“Maven Smart Systems”) across the hybrid fleet in weeks instead of years [14].

Leadership signal.

VADM Seiko Okano continues to serve as the direct reporting program manager and steering committee chair while serving as PMILDEP to ASN(RD&A), keeping Overmatch digital architecture tied to Program Decision Meetings and coalition governance[3, 13].

B.4.2 EDO Community Updates

NAVWAR change of command.

Rear Adm. (now VADM) Seiko Okano relieved Rear Adm. Doug Small on 9 Aug 2024, assuming command of NAVWAR and direct oversight of Project Overmatch; she emphasized information warfare dominance and continued alliance engagement with industry and academia [125].

Role of the PMILDEP.

As PMILDEP to ASN(RD&A), Okano now chairs Program Decision Meetings when delegated and synchronizes PAEs and SYSCOM technical authority, a critical linkage as Capability Trade Councils and portfolio scorecards begin replacing JCIDS artifacts [3, 62].

Tycom maintenance focus.

USFF and PACFLT maintenance flag officers continue to drive execution of regional maintenance center reforms and the Naval Sustainment System–Shipyards initiatives highlighted in the 2025 coursebook; these billets remain key board discussion topics because they link EDO leadership to depot performance [48].

B.4.3 Acquisition Policy and Industrial Base Highlights**WAS.**

The Acquisition Transformation Strategy replaces JCIDS with Capability Portfolio Management and Capability Trade Councils, empowering PAEs to adjudicate cost/schedule/performance trades in real time and document decisions in portfolio scorecards [62].

Industrial base actions.

SECWAR directed the Department to expand multi-year procurements, establish an Industrial Base Consortium, and reform bid protest incentives so the Munitions Acceleration Council and similar bodies can stabilize demand and attract private capital [62].

PPBE integration.

USW(C) and USW(A&S) are piloting portfolio budgets and flexible execution authorities so PAEs can move funds inside approved controls while maintaining transparency with OMB and Congress—a major shift boards will expect candidates to explain [62].

Security cooperation realignment.

DSCA and DTSA now fall under USW(A&S), aligning exportability, Foreign Military Sales, and industrial-base planning so allied orders and U.S. production trades can be brokered inside Capability Trade Councils [66].

National Security Capital Forum.

Section 1092 of the FY 2025 NDAA created the NSCF, chaired by the Office of Strategic Capital, to coordinate public/private financing decisions that affect defense materiel transactions—EDOs should highlight how their programs leverage or are scrutinized by the forum [67].

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List of Acronyms

Acronym	Definition
AAF	Adaptive Acquisition Framework
AC	Actual Cost
ACAT	Acquisition Category
ACO	Administrative Contracting Officer
ACWP	Actual Cost of Work Performed
ADA	Anti-Deficiency Act
ADA	Acquisition Data and Analytics
ADM	Acquisition Decision Memorandum
AEA	Alteration Engineering Agent
AER	Alteration Equivalent to Repair
AFOTEC	Air Force Operational Test and Evaluation Center
AI/ML	Artificial Intelligence and Machine Learning
AIM	Advanced Industrial Management
AIT	Alteration Installation Team
ANSI	American National Standards Institute
AO	Authorizing Official
AoA	Analysis of Alternatives
AOR	Accumulated Operating Result
APA	Additional Performance Attribute
APB	Acquisition Program Baseline
API	Application Programming Interface
APN	Aircraft Procurement, Navy
APUC	Average Procurement Unit Cost
ARC	Acquisition Research Center
ASN(FM&C)	Assistant Secretary of the Navy (Financial Management and Comptroller)
ASN(RD&A)	Assistant Secretary of the Navy for Research, Development and Acquisition
ASR	Alternative Systems Review
ASW	Anti-Submarine Warfare
ATEC	U.S. Army Test and Evaluation Command
AV	All Viewpoint

Acronym	Definition
AWI	Availability Work Integration
AWOL	Absent Without Leave
AWP	Availability Work Package
AWS	Aegis Weapon System
BA	Budget Authority
BAC	Budget at Completion
BAWP	Baseline Availability Work Package
BB	Block Buy
BCA	Business Case Analysis
BCAC	Business Capability Acquisition Cycle
BCWP	Budgeted Cost of Work Performed
BCWS	Budgeted Cost of Work Scheduled
BDAR	Battle Damage Assessment and Repair
BoM	Bill of Material
BSO	Budget Submitting Office
BTR	Below-Threshold Reprogramming
C4I	Command, Control, Communications, Computers, and Intelligence
CAE	Component Acquisition Executive
CAM	Control Account Manager
CANES	Consolidated Afloat Networks and Enterprise Services
CAPE	Cost Assessment and Program Evaluation
CARD	Cost Analysis Requirements Description
CAS	Cost Accounting Standards
CASREP	Casualty Report
CBA	Capability-Based Assessment
CBM	Condition-Based Maintenance
CBM+	Condition-Based Maintenance Plus
CBO	Congressional Budget Office
CCDR	Combatant Commander
CCMD	Combatant Command
CDD	Capabilities Development Document
CDD-V	Capabilities Development Document Validation
CDR	Critical Design Review
CDRL	Contract Data Requirements List
CDT	Chief Developmental Tester
CER	Cost Estimating Relationship
CFO	Chief Financial Officer
CHENG	Chief Engineer

Acronym	Definition
CICA	Competition in Contracting Act
CIO	Chief Information Officer
CIP	Capital Investment Program
CIVPERS	Civilian Personnel
CJADC2	Combined Joint All-Domain Command and Control
CJCS	Chairman of the Joint Chiefs of Staff
CMP	Class Maintenance Plan
CNO	Chief of Naval Operations
CNRMC	Commander, Navy Regional Maintenance Center
CO	Commanding Officer
COFC	Court of Federal Claims
COI	Critical Operational Issue
COMNAVAIRFOR	Commander, Naval Air Forces
COMNAVSEA	Commander, Naval Sea Systems Command
COMNAVSURFOR	Commander, Naval Surface Forces
COMNAVWAR	Commander, Naval Information Warfare Systems Command
COMOPTEVFOR	Commander, Operational Test and Evaluation Force
COMSUBFOR	Commander, Submarine Forces
CONOPS	Concept of Operations
COR	Contracting Officer's Representative
COTF	Commander, Operational Test Force
COTS	Commercial Off-the-Shelf
CPAF	Cost-Plus-Award-Fee
CPD	Capability Production Document
CPFF	Cost-Plus-Fixed-Fee
CPI	Cost Performance Index
CPIF	Cost-Plus-Incentive-Fee
CPM	Capability Portfolio Management
CR	Continuing Resolution
CRS	Congressional Research Service
CSB	Configuration Steering Board
CSDR	Cost and Software Data
CSFR	Contract Funds Status Report
CSMP	Current Ship's Maintenance Project
CSSQT	Combat System Ship Qualification Trial
CTC	Cost-to-Complete
CTC	Capability Trade Council
CTP	Critical Technical Parameter

Acronym	Definition
CV	Cost Variance
CV	Capability Viewpoint
CVN	Nuclear-Powered Aircraft Carrier
CWS	Compressed Work Schedule
D&F	Determination and Findings
DA	Design Agent
DAB	Defense Acquisition Board
DAE	Defense Acquisition Executive
DAES	Defense Acquisition Executive Summary
DAG	Defense Acquisition Guidebook
DAGIR	Data and Applications Government-owned Interoperable Repositories
DAS	Defense Acquisition System
DASN (APB)	Deputy Assistant Secretary of the Navy (Acquisition Policy and Budget)
DASW(T&E)	Deputy Assistant Secretary of War for Test and Evaluation
DBS	Defense Business System
DCAA	Defense Contract Audit Agency
DCAPE	Director, CAPE
DCMA	Defense Contract Management Agency
DCR	DOTMLPF-P Change Recommendation
DCS	Direct Commercial Sales
DDG	Guided Missile Destroyer
DevSecOps	Development, Security, and Operations
DFARS	Defense Federal Acquisition Regulation Supplement
DFAS	Defense Finance and Accounting Service
DFS	Departure from Specification
DHA	Direct Hire Authority
DIU	Defense Innovation Unit
DMAG	Deputy's Management Action Group
DMT	Directed Maintenance Task
DoDAF	Department of War Architecture Framework
DOE	Department of Energy
DON	Department of the Navy
DOT&E	Director, Operational Test and Evaluation
DOTMLPF-P	Doctrine, Organization, Training, Materiel, Leadership and Education, Personnel, Facilities, and Policy
DoW	Department of War
DoW	Department of War
DoW FMR	Department of War Financial Management Regulation

Acronym	Definition
DPRK	Democratic People's Republic of Korea
DRFPRD	Development Request for Proposal Release Decision
DRPM	Direct Reporting Program Manager
DSCA	Defense Security Cooperation Agency
DT	Developmental Test
DT&E	Developmental Test and Evaluation
DTSA	Defense Technology Security Administration
EA	Engineering Agent
EAC	Estimate at Completion
EDO	Engineering Duty Officer
EEO	Equal Employment Opportunity
EEOC	Equal Employment Opportunity Commission
EFT	Electronic Funds Transfer
EIA	Electronic Industries Alliance
EMD	Engineering and Manufacturing Development
EMRF	Expeditionary Maintenance and Repair Facility
EOA	Early Operational Assessment
EOQ	Economic Order Quantity
ERP	Enterprise Resource Planning
EV	Earned Value
EVM	Earned Value Management
EVMS	Earned Value Management System
FAR	Federal Acquisition Regulation
FCB	Functional Capabilities Board
FDA	Food and Drug Administration
FDSRT	Forward Deployed Ship Repair Team
FFG	Guided-Missile Frigate
FFP	Firm-Fixed-Price
FLRA	Federal Labor Relations Authority
FLSA	Fair Labor Standards Act
FMB	Financial Management Budget (OPNAV)
FMR	Financial Management Regulation
FMS	Foreign Military Sales
FOC	Full Operational Capability
FOT&E	Follow-on Operational Test and Evaluation
FPEPA	Fixed-Price with Economic Price Adjustment
FPIF	Fixed-Price Incentive (Firm Target)
FPR	Fixed-Price Redetermination

Acronym	Definition
FPRA	Forward Pricing Rate Agreement
FPRP	Forward Pricing Rate Proposal
FRD	Fleet Readiness Directorate
FRP	Full-Rate Production
FRPDR	Full-Rate Production Decision Review
FUNDS-ALT	Fleet Modernization Alteration funding
FUNDS-D	Fleet Modernization Design (engineering) funding
FUNDS-K	Fleet Modernization Procurement funding
FVEY	Five Eyes (Australia, Canada, New Zealand, United Kingdom, United States)
FY	Fiscal Year
FYDP	Future Years Defense Program
G&A	General and Administrative
GAAP	Generally Accepted Accounting Principles
GAO	Government Accountability Office
GPNTS	GPS-based Positioning, Navigation, and Timing Service
GPS	Global Positioning System
GS	General Schedule
HAC	House Appropriations Committee
HASC	House Armed Services Committee
HBC	House Budget Committee
HCA	Head Contracting Activity
HM&E	Hull, Mechanical, and Electrical
HRO	Human Resources Office
HUBZone	Historically Underutilized Business Zone
IA	Information Assurance
IAM	Major Automated Information System (legacy ACAT I equivalent)
IBR	Integrated Baseline Review
ICD	Initial Capabilities Document
ICE	Independent Cost Estimate
IDIQ	Indefinite-Delivery, Indefinite-Quantity
IGCE	Independent Government Cost Estimate
IGT	Intragovernmental Transaction
IMA	Intermediate Maintenance Activity
IMF	Intermediate Maintenance Facility
IMS	Integrated Master Schedule
INSURV	Board of Inspection and Survey
IOC	Initial Operational Capability
IOT&E	Initial Operational Test and Evaluation

Acronym	Definition
IOT&RE	Initial Operational Test and Readiness Evaluation
IPAC	Intra-Governmental Payment and Collection
IPMDAR	Integrated Program Manager's Data Analysis Report
IPP	Integrated Production Planning
IPT	Integrated Product Team
IR	Internal Reprogramming
IRB	Investment Review Board
IS	Information System
ISEA	In-Service Engineering Agent
IT	Information Technology
IWS	Integrated Warfare Systems
J&A	Justification and Approval
JAR	Joint Acceleration Reserve
JCB	Joint Capabilities Board
JCIDS	Joint Capabilities Integration and Development System
JEON	Joint Emergent Operational Need
JFMM	Joint Fleet Maintenance Manual
JITC	Joint Interoperability Test Command
JROC	Joint Requirements Oversight Council
JROCM	Joint Requirements Oversight Council Memorandum
JSD	Joint Staffing Designator
JSN	Job Sequence Number
JUON	Joint Urgent Operational Need
KM/DS	Knowledge Management and Decision Support
KO	Contracting Officer
KOP	Key Operational Problem
KPP	Key Performance Parameter
KSA	Key System Attribute
LCC	Life-Cycle Cost
LCS	Littoral Combat Ship
LCSP	Life-Cycle Sustainment Plan
LDTO	Lead Developmental Test and Evaluation Organization
LEGPROP	Legislative Proposal
LFT&E	Live Fire Test and Evaluation
LH	Labor-Hour
LHA	Landing Helicopter Assault
LHD	Landing Helicopter Dock
LLTM	Long-Lead Time Material

Acronym	Definition
LMA	Lead Maintenance Activity
LOA	Light-Off Assessment
LOE	Level of Effort
LPTA	Lowest Price Technically Acceptable
LRIP	Low-Rate Initial Production
LRMS	Long-Range Maintenance Schedule
LT	Letter Transfer
MAC-MO	Multiple Award Contract, Multi-Order
MCA	Major Capability Acquisition
MCOTEA	Marine Corps Operational Test and Evaluation Activity
MDA	Missile Defense Agency
MDA	Milestone Decision Authority
MDAP	Major Defense Acquisition Program
MDD	Materiel Development Decision
MDSU	Mobile Diving and Salvage Unit
MEIA	Mission Engineering and Integration Activity
MILCON	Military Construction
MILPERS	Military Personnel
MIPR	Military Interdepartmental Purchase Request
MOA	Memorandum of Agreement
MOE	Measure of Effectiveness
MOP	Measure of Performance
MOS	Measure of Suitability
MSA	Materiel Solution Analysis
MSC	Military Sealift Command
MSP	Maritime Support Plan
MTA	Middle Tier of Acquisition
MTBOMF	Mean Time Between Operational Mission Failure
MYP	Multiyear Procurement
N8	OPNAV Integration of Capabilities and Resources
N82	OPNAV Programming Division
N9	OPNAV Warfare Systems
N91	Warfare Integration Directorate
NAMTS	Navy Afloat Maintenance Training Strategy
NASA	National Aeronautics and Space Administration
NATO	North Atlantic Treaty Organization
NAVAIR	Naval Air Systems Command
Naval Reactors	Naval Nuclear Propulsion Program

Acronym	Definition
NAVSEA	Naval Sea Systems Command
NAVSEAINST	Naval Sea Systems Command Instruction
NAVSUP	Naval Supply Systems Command
NAVWAR	Naval Information Warfare Systems Command
NAWCAD	Naval Air Warfare Center Aircraft Division
NAWCWD	Naval Air Warfare Center Weapons Division
NDAA	National Defense Authorization Act
NDE	Navy Data Environment
NEC	Navy Enlisted Classification
NMCARS	Navy Marine Corps Acquisition Regulation Supplement
NMP	Navy Modernization Program
NOR	Net Operating Result
NR	Net-Ready
NRO	National Reconnaissance Office
NRRO	Naval Reactors Representative Office
NSA	Naval Supervising Activity
NSCF	National Security Capital Forum
NSS-SY	Naval Sustainment System—Shipyards
NSWC	Naval Surface Warfare Center
NSY	Naval Shipyard
NUWC	Naval Undersea Warfare Center
NWC	Naval Warfare Center
NWCF	Navy Working Capital Fund
O&S	Operations and Support Phase
OA	Operational Assessment
OC&MF	Operations and Maintenance, Fleet
OCI	Organizational Conflict of Interest
ODC	Other Direct Cost
OEM	Original Equipment Manufacturer
OFRP	Optimized Fleet Response Plan
OMB	Office of Management and Budget
OMN	Operation and Maintenance, Navy
OPCON	Operational Control
OPEVAL	Operational Evaluation
OPM	Office of Personnel Management
OPN	Other Procurement, Navy
OPNAV	Office of the Chief of Naval Operations
OPNAVINST	Office of the Chief of Naval Operations Instruction

Acronym	Definition
OPNAVNOTE	Office of the Chief of Naval Operations Notice
OQE	Objective Quality Evidence
OSD	Office of the Secretary of War
OSIC	On-Site Installation Coordinator
OT	Operational Test
OT	Other Transaction
OT&E	Operational Test and Evaluation
OTRR	Operational Test Readiness Review
OUSW(C)	Office of the Under Secretary of War (Comptroller)
OV	Operational Viewpoint
P&D	Production and Deployment Phase
PA	Program Authority
PA Reprog	Prior Approval Reprogramming
PACFLT	U.S. Pacific Fleet
PAE	Portfolio Acquisition Executive
PARM	Participating Acquisition Resource Manager
PATS	Pre-Availability Test
PAUC	Program Acquisition Unit Cost
PB	President's Budget
PBL	Performance-Based Logistics
PCD	Production Completion Date
PCO	Procuring Contracting Officer
PD	Position Description
PDM	Program Decision Memorandum
PDR	Preliminary Design Review
PEO	Program Executive Office
PEO C4I	Program Executive Office, Command, Control, Communications, Computers, and Intelligence
PEO Columbia	Program Executive Office, Columbia-Class Submarines
PEO CVN	Program Executive Office, Aircraft Carriers
PEO Digital	Program Executive Office, Digital and Enterprise Services
PEO IWS	Program Executive Office, Integrated Warfare Systems
PEO MLB	Program Executive Office, Manpower, Logistics and Business IT
PEO Ships	Program Executive Office, Ships
PEO SSBN	Program Executive Office, SSBN
PEO SSN	Program Executive Office, Attack Submarines
PEO USC	Program Executive Office, Unmanned and Small Combatants
PEO UWS	Program Executive Office, Undersea Warfare Systems

Acronym	Definition
PIP	Performance Improvement Plan
PM	Program Manager
PMB	Performance Measurement Baseline
PMILDEP	Principal Military Deputy
PMO	Program Management Office
PMS	Program Manager, Ships
PMW	Program Management Warfare (NAVWAR PEO C4I/NAVAIR PMS series)
POA&M	Plan of Action and Milestones
POE	Program Office Estimate
POM	Program Objective Memorandum
POTS	Pre-Overhaul Test
PPBE	Planning, Programming, Budgeting and Execution
PR	Purchase Request
PRC	People's Republic of China
PRR	Production Readiness Review
PTA	Point of Total Assumption
PV	Planned Value
R&A	Rescue and Assistance
RADM	Rear Admiral (Upper Half)
RCA	Request for Contract Action
RCM	Reliability-Centered Maintenance
RCOH	Refueling and Complex Overhaul
RDML	Rear Admiral (Lower Half)
RDT&E	Research, Development, Test, and Evaluation
RFI	Request for Information
RFP	Request for Proposal
RIF	Reduction in Force
RIMPAC	Rim of the Pacific Exercise
RMC	Regional Maintenance Center
RMD	Resource Management Decision
RMF	Risk Management Framework
RMMCO	Regional Maintenance and Modernization Coordination Office
ROA	Return on Assets
ROS	Return on Sales
RPCO	Reactor Plant Contractor's Office
RRAB	Requirements and Resourcing Alignment Board
SAC	Senate Appropriations Committee
SAE	Service Acquisition Executive

Acronym	Definition
SAM	System for Award Management
SAR	Selected Acquisition Report
SASC	Senate Armed Services Committee
SAT	Simplified Acquisition Threshold
SBA	Small Business Administration
SBC	Senate Budget Committee
SBIR	Small Business Innovation Research
SCN	Shipbuilding and Conversion, Navy
SDVOSB	Service-Disabled Veteran-Owned Small Business
SEA	NAVSEA headquarters staff code
SEA 05B	Office of the Chief Engineer
SEA 05C	Cost Engineering and Industrial Analysis
SEA 05E	Explosive Ordnance Engineering
SEA 05H	Integrated Warfare Systems Engineering
SEA 05M	Logistics and Maintenance Warfare Systems Engineering
SEA 05N	Undersea Warfare Systems Engineering
SEA 05P	Ship Integrity and Performance Engineering
SEA 05S	Industrial Engineering, Technical Policy and Standards
SEA 05T	Technology Office
SEA 05U	Submarine/Submersible Design and Systems Engineering
SEA 05V	Aircraft Carrier Design and Systems Engineering
SEA 05W	Surface Warfare Systems Engineering
SEA 05Z	Marine Engineering
SECNAV	Secretary of the Navy
SECNAVINST	Secretary of the Navy Instruction
SECWAR	Secretary of War
SEP	Systems Engineering Plan
SES	Senior Executive Service
SFR	System Functional Review
SFRM	Surface Force Readiness Manual
SHAPM	Ship Acquisition Program Manager
SHIPALT	Ship Alteration
SIA	Systems Integration Agent
SIB	Shop-in-a-Box
SIP	Separation Incentive Program
SLM	Ship Lifecycle Manager
SLOC	Source Lines of Code
SLR	Stabilized Labor Rate

Acronym	Definition
SMM	Ship's Maintenance Manager
SOM	SUPSHIP Operation Manual
SOVT	Shipboard Operational Verification Test
SOW	Statement of Work
SPD	Ship Project Director
SPI	Schedule Performance Index
SPM	Ship Program Manager
SRR	System Requirements Review
SS800	Submergence Systems Certification manual series
SSA	Source Selection Authority
SSAC	Source Selection Advisory Council
SSBN	Ballistic Missile Submarine, Nuclear
SSEB	Source Selection Evaluation Board
SSN	Attack Submarine, Nuclear
SSP	Strategic Systems Program
SUBMEPP	Submarine Maintenance Engineering Planning Program
SUBSAFE	Submarine Safety Program
SUBTECH	Submarine/Undersea Warfare Technology
SUPSALV	Supervisor of Salvage and Diving
SUPSHIP	Supervisor of Shipbuilding
SURFLANT	Naval Surface Force Atlantic
SURFMEPP	Surface Maintenance Engineering Planning Program
SURFPAC	Naval Surface Force Pacific
SV	Schedule Variance
SV	Systems Viewpoint
SVR	System Verification Review
SYSCOM	Systems Command
T&E	Test and Evaluation
T&M	Time-and-Materials
TA	Technical Authority
TCI	Total Cost Input
TCO	Termination Contracting Officer
TCPI	Total Cost Performance Index
TDA	Technical Direction Agent
TDP	Technical Data Package
TECHEVAL	Technical Evaluation
TEMP	Test and Evaluation Master Plan
TGI	Task Group Instruction

Acronym	Definition
TMRR	Technology Maturation and Risk Reduction
TOA	Total Obligational Authority
TOC	Total Ownership Cost
TPM	Technical Performance Measure
TRL	Technology Readiness Level
TRR	Test Readiness Review
TSO	Temporary Standing Order
TSRA	Total Ship Readiness Assessment
TUM	Tag-out Users Manual
TWD	Technical Work Document
TWH	Technical Warrant Holder
TYCOM	Type Commander
UCA	Undefinitized Contract Action
UCA	Urgent Capability Acquisition
UCF	Uniform Contract Format
ULP	Unfair Labor Practice
UON	Urgent Operational Need
USAFRICOM	United States Africa Command
USCENTCOM	United States Central Command
USCYBERCOM	United States Cyber Command
USD(P)	Under Secretary of Defense for Policy
USEUCOM	United States European Command
USFF	U.S. Fleet Forces
USINDOPACOM	United States Indo-Pacific Command
USNORTHCOM	United States Northern Command
USNR	United States Navy Reserve
USSOCOM	United States Special Operations Command
USSOUTHCOM	United States Southern Command
USSPACECOM	United States Space Command
USSTRATCOM	United States Strategic Command
USTRANSCOM	United States Transportation Command
USW(A&S)	Under Secretary of War for Acquisition and Sustainment
USW(C)	Under Secretary of War (Comptroller)
USW(R&E)	Under Secretary of War for Research and Engineering
UTA	Ultimate Technical Authority
V&V	Verification and Validation
VA	Department of Veterans Affairs
VAC	Variance at Complete

Acronym	Definition
VADM	Vice Admiral
VCJCS	Vice-Chairman of the Joint Chiefs of Staff
VERA	Voluntary Early Retirement Authority
VV&A	Verification, Validation, and Accreditation
WAF	Work Authorization Form
WARP	Wartime Acquisition Response Plan
WAS	Warfighting Acquisition System
WBS	Work Breakdown Structure
WCF	Working Capital Fund
WG	Wage Grade
WIPT	Working-level Integrated Product Team
WOSB	Women-Owned Small Business
WPI	Work Package Integration
WPN	Weapons Procurement, Navy
XO	Executive Officer